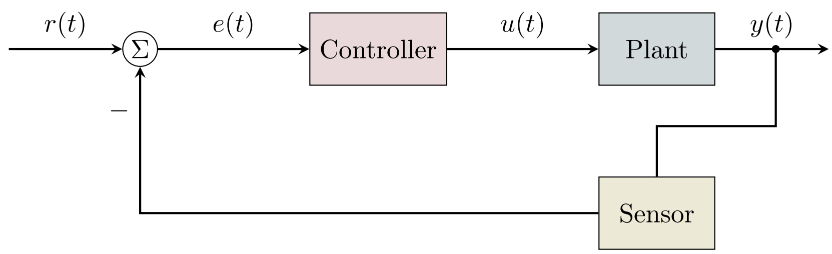
Modern Practical Control Systems

From Classical Theory to Machine Learning



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About the Author

J.C. Vaught is a graduate researcher in the Department of Mechanical Engineering at the University of South Carolina, where he is pursuing his M.S. under the supervision of Dr. Yi Wang. His research interests span control systems, optimization algorithms, and the intersection of classical control theory with modern machine learning techniques.

Throughout his academic career, Vaught has consistently demonstrated excellence, earning recognition on the President's List and Dean's List. His technical experience includes work on laser powder bed fusion additive manufacturing (LPBFAM) distortion prediction, CAD modeling, and finite element analysis.

Beyond research, Vaught has shown exceptional leadership in engineering organizations. He served as President of both IEEE and SAE student chapters and as an advisor to ASME. Notably, he co-founded the University of South Carolina's first university-affiliated NAR-insured Rocketry Club, growing its membership from zero to over 120 students. He also served as Student Body Treasurer, successfully managing and expanding the student government budget.

His diverse background in hands-on engineering, organizational leadership, and technical writing—including opinion contributions to *The Daily Gamecock*—informs this textbook's practical approach to control systems education. This text reflects his belief that control theory should be accessible to students from all engineering disciplines while maintaining mathematical rigor.

For correspondence regarding this textbook, please contact the Department of Mechanical Engineering at the University of South Carolina.

Preface

Control systems engineering stands at the intersection of mathematics, physics, and increasingly, computer science and artificial intelligence. From the thermostats in our homes to the autopilots in aircraft, from industrial robots to autonomous vehicles, control systems are ubiquitous in modern technology. This textbook aims to provide a comprehensive, practical introduction to control systems that bridges classical theory with contemporary machine learning approaches.

Philosophy of This Text

This book is written with several guiding principles:

1. **Broad Applicability**: While many control systems texts focus primarily on electrical or mechanical systems, this book deliberately covers applications across mechanical, electrical, chemical, aerospace, civil, biomedical, industrial, and computer engineering. Each chapter includes diverse examples to help students see the universality of control concepts.

2. **Mathematical Rigor with Intuition**: We begin with a comprehensive mathematical foundations chapter that reviews all prerequisite material—from complex numbers to linear algebra to probability theory. This ensures all students start from a common foundation while developing both computational skills and conceptual understanding.

3. **Practical Focus**: Every theoretical concept is accompanied by worked examples, MATLAB/Python code, and real-world applications. Theory without application breeds confusion; application without theory breeds limitation.

4. **Modern Integration**: This text seamlessly integrates classical control (Laplace transforms, root locus, Bode plots), modern control (state-space, optimal control), discrete-time control, and machine learning-based control (neural networks, reinforcement learning). Students learn when each approach is appropriate and how they complement each other.

5. **Progressive Complexity**: We begin with ideal assumptions—perfect sensors, instantaneous actuators, linear dynamics—and progressively relax these to address real-world challenges. This scaffolded approach builds confidence before introducing complexity.

How to Use This Book

The book is organized into several parts:

- **Part I: Foundations** covers mathematical prerequisites and system modeling with ideal components.

- **Part II: Classical Control** develops frequency-domain analysis and design techniques.
- **Part III: Modern Control** introduces state-space methods and optimal control.
- **Part IV: Discrete and Digital Control** addresses sampled-data systems and digital implementation.
- **Part V: Advanced Topics** covers nonlinear systems, robust control, and adaptive methods.
- **Part VI: Machine Learning in Control** explores neural networks, reinforcement learning, and hybrid approaches.

Each chapter includes learning objectives, worked examples, summary boxes, and end-of-chapter exercises ranging from computational to conceptual to design-oriented.

Acknowledgments

I am grateful to Dr. Yi Wang for his mentorship and guidance throughout my graduate studies. Thanks also to the faculty and students of the Molinaroli College of Engineering and Computing who provided feedback on early drafts. The vibrant engineering community at the University of South Carolina—through organizations like IEEE, SAE, ASME, and the Rocketry Club—has continually inspired my belief in hands-on, practical engineering education.

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Part I

Foundations

Chapter 1

Mathematical Foundations

Learning Objectives

After completing this chapter, you will be able to:

- Perform arithmetic and algebraic operations with complex numbers
- Apply calculus techniques including differentiation, integration, and differential equations
- Use Laplace and Fourier transforms to analyze systems
- Perform matrix operations and understand eigenvalue analysis
- Apply probability and statistics concepts relevant to stochastic control and machine learning
- Understand optimization fundamentals for control design
- Work with discrete-time mathematics for digital control systems

Control systems engineering draws upon a rich mathematical foundation spanning calculus, linear algebra, complex analysis, differential equations, and increasingly, probability theory and optimization. This chapter provides a comprehensive review of these mathematical tools, ensuring all readers have the prerequisite knowledge for the chapters that follow.

Whether you are approaching control from a mechanical, electrical, chemical, biomedical, aerospace, or computer science background, mastery of these fundamentals will enable you to understand system behavior, design controllers, and implement modern machine learning-based control strategies.

1.1 Complex Numbers and Imaginary Arithmetic

Complex numbers are fundamental to control systems analysis. They appear naturally in the solution of differential equations, in the Laplace transform, in frequency response analysis, and in stability analysis.

1.1.1 Definition and Representation

Definition 0.1: Complex Number

A **complex number** z is defined as:

$$z = a + \mathrm{j}b \tag{1.1}$$

where $a, b \in \mathbb{R}$, and $\mathrm{j} = \sqrt{-1}$ is the **imaginary unit**. The real part is $\mathrm{Re}(z) = a$ and the imaginary part is $\mathrm{Im}(z) = b$.

Note 1.1. In electrical engineering, j is often used instead of i for the imaginary unit to avoid confusion with current. We adopt this convention throughout.

Complex numbers can be represented in multiple forms:

Rectangular Form:

$$z = a + \mathrm{j}b \tag{1.2}$$

Polar Form:

$$z = r(\cos\theta + \mathrm{j}\sin\theta) = r\angle\theta \tag{1.3}$$

where $r = |z| = \sqrt{a^2 + b^2}$ is the **magnitude** (or modulus) and $\theta = \arg(z) = \arctan(b/a)$ is the **argument** (or phase angle).

Exponential Form (Euler's Formula):

$$z = r\mathrm{e}^{\mathrm{j}\theta} \tag{1.4}$$

Euler's formula, one of the most elegant results in mathematics, states:

$$\boxed{\mathrm{e}^{\mathrm{j}\theta} = \cos\theta + \mathrm{j}\sin\theta} \tag{1.5}$$

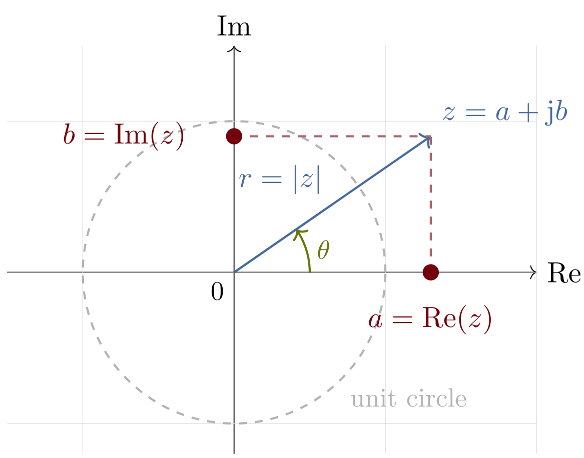


Figure 1.1: Geometric representation of a complex number in the complex plane (Argand diagram). The blue vector shows the complex number z, with dashed red projections onto the real and imaginary axes. The green angle θ represents the argument, and the unit circle (dashed gray) provides a reference. Colors: Atlantic (blue) for magnitude, Garnet (red) for projections, Horseshoe (green) for angle.

1.1.2 Arithmetic Operations

Addition and Subtraction:

$$(a + \mathrm{j}b) \pm (c + \mathrm{j}d) = (a \pm c) + \mathrm{j}(b \pm d) \tag{1.6}$$

Multiplication:

$$(a + \mathrm{j}b)(c + \mathrm{j}d) = (ac - bd) + \mathrm{j}(ad + bc) \tag{1.7}$$

In polar form: $r_1 \mathrm{e}^{\mathrm{j}\theta_1} \cdot r_2 \mathrm{e}^{\mathrm{j}\theta_2} = r_1 r_2 \mathrm{e}^{\mathrm{j}(\theta_1 + \theta_2)}$

Division:

$$\frac{a + \mathrm{j}b}{c + \mathrm{j}d} = \frac{(a + \mathrm{j}b)(c - \mathrm{j}d)}{(c + \mathrm{j}d)(c - \mathrm{j}d)} = \frac{(ac + bd) + \mathrm{j}(bc - ad)}{c^2 + d^2} \tag{1.8}$$

In polar form: $\frac{r_1 \mathrm{e}^{\mathrm{j}\theta_1}}{r_2 \mathrm{e}^{\mathrm{j}\theta_2}} = \frac{r_1}{r_2} \mathrm{e}^{\mathrm{j}(\theta_1 - \theta_2)}$

Complex Conjugate:

$$z^* = \overline{z} = a - \mathrm{j}b \tag{1.9}$$

Important property: $z \cdot z^* = |z|^2 = a^2 + b^2$

Powers (De Moivre's Theorem):

$$z^n = r^n \mathrm{e}^{\mathrm{j}n\theta} = r^n (\cos n\theta + \mathrm{j} \sin n\theta) \tag{1.10}$$

Roots: The nth roots of $z = r\mathrm{e}^{\mathrm{j}\theta}$ are:

$$z^{1/n} = r^{1/n} \mathrm{e}^{\mathrm{j}(\theta + 2\pi k)/n}, \quad k = 0, 1, \ldots, n - 1 \tag{1.11}$$

Example 0.1: Complex Number Operations

Find the polar form, magnitude, and phase of $z = 3 + 4\mathrm{j}$.

Solution:

$$|z| = \sqrt{3^2 + 4^2} = \sqrt{25} = 5 \tag{1.12}$$

$$\theta = \arctan\left(\frac{4}{3}\right) \approx 53.13 = 0.927 \text{ rad} \tag{1.13}$$

$$z = 5\mathrm{e}^{\mathrm{j}(0.927)} = 5\angle 53.13 \tag{1.14}$$

1.1.3 Complex Numbers in Control Systems

Complex numbers appear throughout control theory:

- **Poles and Zeros:** The roots of transfer function polynomials are generally complex.
- **Eigenvalues:** System matrices often have complex eigenvalues that determine stability and oscillatory behavior.
- **Frequency Response:** Evaluating transfer functions at $s = \mathrm{j}\omega$ yields complex numbers representing magnitude and phase.
- **Sinusoidal Steady-State:** Complex exponentials simplify analysis of sinusoidal inputs.

1.2 Calculus Review

1.2.1 Differential Calculus

Derivatives and Rules

The derivative of a function $f(t)$ represents its instantaneous rate of change:

$$\frac{\mathrm{d}f}{\mathrm{d}t} = f'(t) = \lim_{\Delta t \to 0} \frac{f(t+\Delta t) - f(t)}{\Delta t} \tag{1.15}$$

Key Differentiation Rules:

Rule	Formula
Power Rule	$\frac{\mathrm{d}}{\mathrm{d}t}(t^n) = nt^{n-1}$
Exponential	$\frac{\mathrm{d}}{\mathrm{d}t}(\mathrm{e}^{at}) = a\mathrm{e}^{at}$
Logarithm	$\frac{\mathrm{d}}{\mathrm{d}t}(\ln t) = \frac{1}{t}$
Sine	$\frac{\mathrm{d}}{\mathrm{d}t}(\sin \omega t) = \omega \cos \omega t$
Cosine	$\frac{\mathrm{d}}{\mathrm{d}t}(\cos \omega t) = -\omega \sin \omega t$
Product Rule	$\frac{\mathrm{d}}{\mathrm{d}t}(fg) = f'g + fg'$
Quotient Rule	$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{f}{g}\right) = \frac{f'g - fg'}{g^2}$
Chain Rule	$\frac{\mathrm{d}}{\mathrm{d}t} f(g(t)) = f'(g(t)) \cdot g'(t)$

Table 1.1: Essential differentiation rules for control systems.

Partial Derivatives

For functions of multiple variables $f(x_1, x_2, \ldots, x_n)$, partial derivatives hold all variables constant except one:

$$\frac{\partial f}{\partial x_i} = \lim_{\Delta x_i \to 0} \frac{f(x_1, \ldots, x_i + \Delta x_i, \ldots, x_n) - f(x_1, \ldots, x_n)}{\Delta x_i} \tag{1.16}$$

The **gradient** is the vector of all partial derivatives:

$$\nabla f = \begin{bmatrix} \frac{\partial f}{\partial x_1} & \frac{\partial f}{\partial x_2} & \cdots & \frac{\partial f}{\partial x_n} \end{bmatrix}^\top \tag{1.17}$$

The **Hessian** is the matrix of second partial derivatives:

$$\mathbf{H} = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix} \tag{1.18}$$

Taylor Series Expansion

Taylor series are essential for linearization of nonlinear systems:

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \cdots \tag{1.19}$$

For multivariate functions about point $\mathbf{x}_0$:

$$f(\mathbf{x}) \approx f(\mathbf{x}_0) + \nabla f(\mathbf{x}_0)^\top(\mathbf{x} - \mathbf{x}_0) + \frac{1}{2}(\mathbf{x} - \mathbf{x}_0)^\top \mathbf{H}(\mathbf{x}_0)(\mathbf{x} - \mathbf{x}_0) \tag{1.20}$$

Linearization in Control

For nonlinear systems $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u})$, we linearize about an equilibrium point $(\mathbf{x}_0, \mathbf{u}_0)$:

$$\Delta\dot{\mathbf{x}} \approx \mathbf{A}\Delta\mathbf{x} + \mathbf{B}\Delta\mathbf{u} \tag{1.21}$$

where $\mathbf{A} = \frac{\partial \mathbf{f}}{\partial \mathbf{x}}\big|_{(\mathbf{x}_0, \mathbf{u}_0)}$ and $\mathbf{B} = \frac{\partial \mathbf{f}}{\partial \mathbf{u}}\big|_{(\mathbf{x}_0, \mathbf{u}_0)}$

1.2.2 Integral Calculus

Definite and Indefinite Integrals

The indefinite integral (antiderivative):

$$\int f(t)\,\mathrm{d}t = F(t) + C \quad \text{where} \quad F'(t) = f(t) \tag{1.22}$$

The definite integral:

$$\int_a^b f(t)\,\mathrm{d}t = F(b) - F(a) \tag{1.23}$$

Key Integration Formulas:

Function	Integral
$t^n \ (n \neq -1)$	$\frac{t^{n+1}}{n+1} + C$
$\frac{1}{t}$	$\ln\lvert t\rvert + C$
e^{at}	$\frac{1}{a}\mathrm{e}^{at} + C$
$\sin\omega t$	$-\frac{1}{\omega}\cos\omega t + C$
$\cos\omega t$	$\frac{1}{\omega}\sin\omega t + C$
$\frac{1}{a^2+t^2}$	$\frac{1}{a}\arctan\frac{t}{a} + C$

Table 1.2: Essential integration formulas.

Integration by Parts

$$\int u\,\mathrm{d}v = uv - \int v\,\mathrm{d}u \tag{1.24}$$

This technique is essential for deriving Laplace transform properties.

Convolution Integral

The convolution of two functions $f(t)$ and $g(t)$ is:

$$(f * g)(t) = \int_{-\infty}^{\infty} f(\tau)g(t - \tau)\,\mathrm{d}\tau \tag{1.25}$$

For causal systems (signals zero for $t < 0$):

$$(f * g)(t) = \int_0^t f(\tau)g(t-\tau)\,\mathrm{d}\tau \tag{1.26}$$

Convolution in Control Systems

The output $y(t)$ of a linear time-invariant (LTI) system with impulse response $h(t)$ to input $u(t)$ is:

$$y(t) = (h * u)(t) = \int_0^t h(\tau)u(t-\tau)\,\mathrm{d}\tau \tag{1.27}$$

1.3 Differential Equations

Differential equations describe how systems evolve over time and are the mathematical language of dynamics.

1.3.1 Ordinary Differential Equations (ODEs)

First-Order Linear ODEs

The general first-order linear ODE:

$$\frac{\mathrm{d}y}{\mathrm{d}t} + p(t)y = q(t) \tag{1.28}$$

Solution using Integrating Factor: Multiply by $\mu(t) = \mathrm{e}^{\int p(t)\,\mathrm{d}t}$:

$$y(t) = \frac{1}{\mu(t)}\left[\int \mu(t)q(t)\,\mathrm{d}t + C\right] \tag{1.29}$$

For constant coefficients $\frac{\mathrm{d}y}{\mathrm{d}t} + ay = b$:

$$y(t) = \frac{b}{a} + \left(y_0 - \frac{b}{a}\right)\mathrm{e}^{-at} \tag{1.30}$$

Example 0.2: RC Circuit

An RC circuit with voltage source V_s satisfies:

$$RC\frac{\mathrm{d}v_c}{\mathrm{d}t} + v_c = V_s \tag{1.31}$$

With $v_c(0) = 0$, the capacitor voltage is:

$$v_c(t) = V_s\left(1 - \mathrm{e}^{-t/RC}\right) \tag{1.32}$$

The time constant $\tau = RC$ determines how quickly the system responds.

Second-Order Linear ODEs with Constant Coefficients

The standard form:

$$\frac{\mathrm{d}^2 y}{\mathrm{d}t^2} + 2\zeta\omega_n \frac{\mathrm{d}y}{\mathrm{d}t} + \omega_n^2 y = \omega_n^2 u(t) \tag{1.33}$$

where:

- ω_n = natural frequency (rad/s)
- ζ = damping ratio (dimensionless)

The **characteristic equation** is:

$$\lambda^2 + 2\zeta\omega_n\lambda + \omega_n^2 = 0 \tag{1.34}$$

with roots:

$$\lambda_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1} \tag{1.35}$$

Cases based on damping ratio:

1. **Overdamped** ($\zeta > 1$): Two distinct real roots. Response: $y(t) = C_1 \mathrm{e}^{\lambda_1 t} + C_2 \mathrm{e}^{\lambda_2 t}$
2. **Critically Damped** ($\zeta = 1$): Repeated real root $\lambda = -\omega_n$. Response: $y(t) = (C_1 + C_2 t)\mathrm{e}^{-\omega_n t}$
3. **Underdamped** ($0 < \zeta < 1$): Complex conjugate roots $\lambda = -\zeta\omega_n \pm \mathrm{j}\omega_d$ where $\omega_d = \omega_n\sqrt{1-\zeta^2}$. Response:
$$y(t) = \mathrm{e}^{-\zeta\omega_n t}(C_1 \cos\omega_d t + C_2 \sin\omega_d t) \tag{1.36}$$
4. **Undamped** ($\zeta = 0$): Pure imaginary roots. Response: $y(t) = C_1 \cos\omega_n t + C_2 \sin\omega_n t$

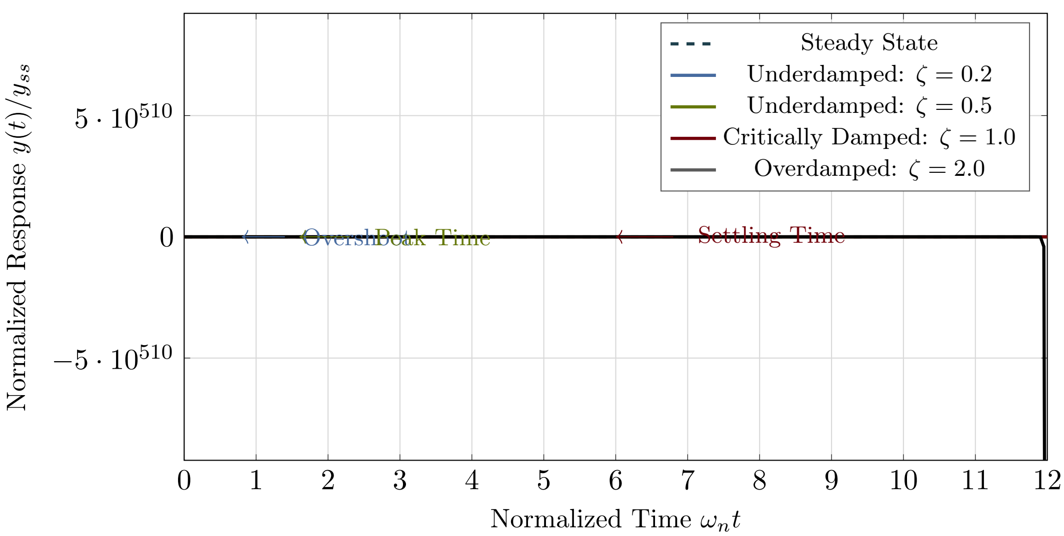


Figure 1.2: Step response of second-order systems for different damping ratios. The response shows how damping affects the transient behavior: underdamped ($\zeta < 1$) exhibits overshoot and oscillation, critically damped ($\zeta = 1$) provides fast response without overshoot, and overdamped ($\zeta > 1$) is slow but smooth. Key features include overshoot (Atlantic blue), peak time (Horseshoe green), and settling time (Garnet red) annotations. Dashed line shows steady-state value.

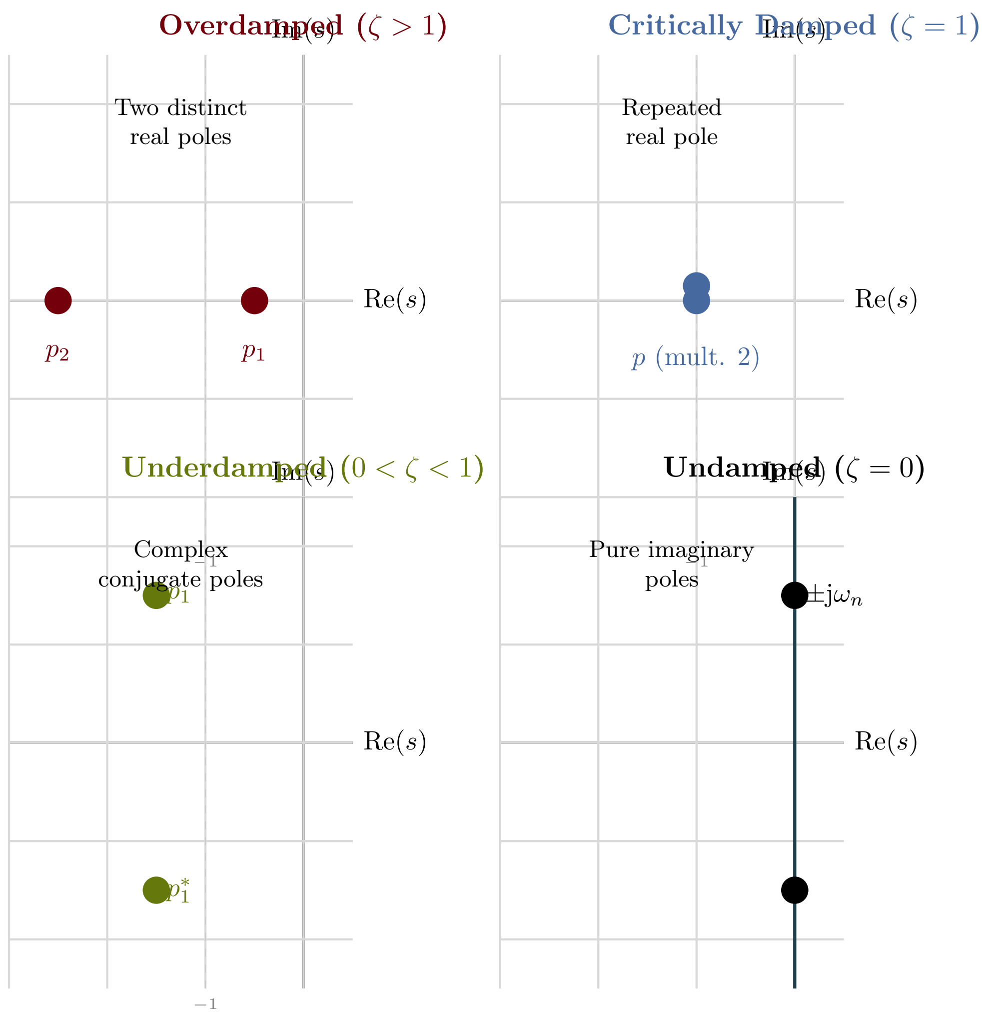


Figure 1.3: Pole locations in the s-plane for second-order systems under different damping conditions. Each subplot shows the characteristic pole configuration: overdamped (Garnet, two real poles on negative axis), critically damped (Atlantic, repeated real pole), underdamped (Horseshoe, complex conjugate pair), and undamped (Rose, pure imaginary pair on the imaginary axis). The stability region (left half-plane) ensures all poles are in the left half-plane for stable responses.

Higher-Order Linear ODEs

The general nth-order linear ODE with constant coefficients:

$$a_n \frac{\mathrm{d}^n y}{\mathrm{d}t^n} + a_{n-1}\frac{\mathrm{d}^{n-1} y}{\mathrm{d}t^{n-1}} + \cdots + a_1 \frac{\mathrm{d}y}{\mathrm{d}t} + a_0 y = f(t) \tag{1.37}$$

The solution consists of:

- **Homogeneous solution** $y_h(t)$: From the characteristic equation roots
- **Particular solution** $y_p(t)$: Depends on the forcing function $f(t)$

1.3.2 Systems of First-Order ODEs

Any higher-order ODE can be written as a system of first-order equations. This is the **state-space** representation:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} \tag{1.38}$$

where $\mathbf{x}$ is the state vector, $\mathbf{u}$ is the input vector, and $\mathbf{A}$, $\mathbf{B}$ are system matrices.
The solution is:

$$\mathbf{x}(t) = \mathrm{e}^{\mathbf{A}t}\mathbf{x}(0) + \int_0^t \mathrm{e}^{\mathbf{A}(t-\tau)}\mathbf{B}\mathbf{u}(\tau)\,\mathrm{d}\tau \tag{1.39}$$

where $\mathrm{e}^{\mathbf{A}t}$ is the **matrix exponential** (discussed in Section 1.6).

1.4 Laplace Transforms

The Laplace transform converts differential equations into algebraic equations, making system analysis much simpler.

1.4.1 Definition

Definition 0.2: Laplace Transform

The **Laplace transform** of a function $f(t)$ defined for $t \geq 0$ is:

$$F(s) = \mathcal{L}\{f(t)\} = \int_0^\infty f(t)\mathrm{e}^{-st}\,\mathrm{d}t \tag{1.40}$$

where $s = \sigma + \mathrm{j}\omega$ is a complex variable.

The **inverse Laplace transform** is:

$$f(t) = \mathcal{L}^{-1}\{F(s)\} = \frac{1}{2\pi\mathrm{j}}\int_{\sigma-\mathrm{j}\infty}^{\sigma+\mathrm{j}\infty} F(s)\mathrm{e}^{st}\,\mathrm{d}s \tag{1.41}$$

In practice, we use tables and partial fraction expansion rather than direct integration.

Signal Name	$f(t)$ for $t \geq 0$	$F(s)$
Impulse	$\delta(t)$	1
Unit step	$u(t) = 1$	$\frac{1}{s}$
Ramp	t	$\frac{1}{s^2}$
Parabolic	t^2	$\frac{2}{s^3}$
Power	t^n	$\frac{n!}{s^{n+1}}$
Exponential	e^{-at}	$\frac{1}{s+a}$
Decaying ramp	$t\mathrm{e}^{-at}$	$\frac{1}{(s+a)^2}$
Sine	$\sin \omega t$	$\frac{\omega}{s^2+\omega^2}$
Cosine	$\cos \omega t$	$\frac{s}{s^2+\omega^2}$
Damped sine	$\mathrm{e}^{-at} \sin \omega t$	$\frac{\omega}{(s+a)^2+\omega^2}$
Damped cosine	$\mathrm{e}^{-at} \cos \omega t$	$\frac{s+a}{(s+a)^2+\omega^2}$

Table 1.3: Common Laplace transform pairs.

Property	**Time Domain**	**s-Domain**
Linearity	$af(t) + bg(t)$	$aF(s) + bG(s)$
Differentiation	$\frac{\mathrm{d}f}{\mathrm{d}t}$	$sF(s) - f(0^-)$
Second derivative	$\frac{\mathrm{d}^2 f}{\mathrm{d}t^2}$	$s^2F(s) - sf(0^-) - f'(0^-)$
nth derivative	$\frac{\mathrm{d}^n f}{\mathrm{d}t^n}$	$s^nF(s) - \sum_{k=0}^{n-1} s^{n-1-k} f^{(k)}(0^-)$
Integration	$\int_0^t f(\tau)\,\mathrm{d}\tau$	$\frac{F(s)}{s}$
Time delay	$f(t-\tau)u(t-\tau)$	$\mathrm{e}^{-\tau s}F(s)$
Frequency shift	$\mathrm{e}^{-at}f(t)$	$F(s+a)$
Time scaling	$f(at)$	$\frac{1}{a}F\left(\frac{s}{a}\right)$
Convolution	$f(t) * g(t)$	$F(s)G(s)$
Initial value	$\lim_{t\to 0^+} f(t)$	$\lim_{s\to\infty} sF(s)$
Final value	$\lim_{t\to\infty} f(t)$	$\lim_{s\to 0} sF(s)$

Table 1.4: Laplace transform properties.

1.4.2 Common Laplace Transform Pairs

1.4.3 Laplace Transform Properties

Important Note

The **Final Value Theorem** only applies if all poles of $sF(s)$ are in the left half-plane (LHP). If any poles are on the imaginary axis or in the right half-plane, the theorem gives incorrect results.

1.4.4 Partial Fraction Expansion

To find the inverse Laplace transform, we decompose rational functions into simpler terms.

Distinct Real Poles:

$$\frac{N(s)}{(s+p_1)(s+p_2)\cdots(s+p_n)} = \frac{A_1}{s+p_1} + \frac{A_2}{s+p_2} + \cdots + \frac{A_n}{s+p_n} \tag{1.42}$$

The residue A_i is found by:

$$A_i = \lim_{s \to -p_i} (s+p_i)F(s) \tag{1.43}$$

Repeated Poles: For a pole of multiplicity m at $s = -p$:

$$\frac{N(s)}{(s+p)^m} = \frac{A_1}{s+p} + \frac{A_2}{(s+p)^2} + \cdots + \frac{A_m}{(s+p)^m} \tag{1.44}$$

Complex Conjugate Poles: For poles at $s = -\sigma \pm j\omega$:

$$\frac{N(s)}{(s+\sigma)^2+\omega^2} = \frac{As+B}{(s+\sigma)^2+\omega^2} \tag{1.45}$$

Example 0.3: Partial Fraction Expansion

Find the inverse Laplace transform of:

$$F(s) = \frac{2s+3}{(s+1)(s+2)} \tag{1.46}$$

Solution:

$$F(s) = \frac{A}{s+1} + \frac{B}{s+2} \tag{1.47}$$

Residues:

$$A = \lim_{s \to -1} (s+1) \cdot \frac{2s+3}{(s+1)(s+2)} = \frac{2(-1)+3}{-1+2} = 1 \tag{1.48}$$

$$B = \lim_{s \to -2} (s+2) \cdot \frac{2s+3}{(s+1)(s+2)} = \frac{2(-2)+3}{-2+1} = 1 \tag{1.49}$$

Therefore:

$$f(t) = \mathrm{e}^{-t} + \mathrm{e}^{-2t}, \quad t \geq 0 \tag{1.50}$$

1.4.5 Transfer Functions

Definition 0.3: Transfer Function

The **transfer function** $G(s)$ of a linear time-invariant system is the ratio of the Laplace transform of the output to the Laplace transform of the input, assuming zero initial conditions:

$$G(s) = \frac{Y(s)}{U(s)} \tag{1.51}$$

Transfer functions are typically written as:

$$G(s) = \frac{b_m s^m + b_{m-1}s^{m-1} + \cdots + b_1 s + b_0}{a_n s^n + a_{n-1}s^{n-1} + \cdots + a_1 s + a_0} = K\frac{\prod_{i=1}^{m}(s - z_i)}{\prod_{j=1}^{n}(s - p_j)} \tag{1.52}$$

where z_i are the **zeros** and p_j are the **poles** of the transfer function.

1.5 Fourier Transform and Frequency Analysis

1.5.1 Fourier Series

Any periodic function $f(t)$ with period T can be expressed as:

$$f(t) = a_0 + \sum_{n=1}^{\infty}\left[a_n \cos\left(\frac{2\pi nt}{T}\right) + b_n \sin\left(\frac{2\pi nt}{T}\right)\right] \tag{1.53}$$

where:

$$a_0 = \frac{1}{T}\int_0^T f(t)\,\mathrm{d}t \tag{1.54}$$

$$a_n = \frac{2}{T}\int_0^T f(t)\cos\left(\frac{2\pi nt}{T}\right)\mathrm{d}t \tag{1.55}$$

$$b_n = \frac{2}{T}\int_0^T f(t)\sin\left(\frac{2\pi nt}{T}\right)\mathrm{d}t \tag{1.56}$$

Complex exponential form:

$$f(t) = \sum_{n=-\infty}^{\infty} c_n \mathrm{e}^{\mathrm{j}n\omega_0 t}, \quad \omega_0 = \frac{2\pi}{T} \tag{1.57}$$

where $c_n = \frac{1}{T}\int_0^T f(t)\mathrm{e}^{-\mathrm{j}n\omega_0 t}\,\mathrm{d}t$

1.5.2 Fourier Transform

For non-periodic functions, the Fourier transform extends the series to continuous frequencies:

Definition 0.4: Fourier Transform

$$F(\omega) = \mathcal{F}\{f(t)\} = \int_{-\infty}^{\infty} f(t)\mathrm{e}^{-\mathrm{j}\omega t}\,\mathrm{d}t \tag{1.58}$$

$$f(t) = \mathcal{F}^{-1}\{F(\omega)\} = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(\omega)\mathrm{e}^{\mathrm{j}\omega t}\,\mathrm{d}\omega \tag{1.59}$$

Relationship to Laplace Transform: For causal signals, $F(\omega) = F(s)|_{s=\mathrm{j}\omega}$ when the Laplace transform converges on the imaginary axis.

1.5.3 Frequency Response

For a stable LTI system with transfer function $G(s)$, the **frequency response** is:

$$G(\mathrm{j}\omega) = |G(\mathrm{j}\omega)|\mathrm{e}^{\mathrm{j}\angle G(\mathrm{j}\omega)} \tag{1.60}$$

- $|G(\mathrm{j}\omega)|$ = magnitude (gain) at frequency ω
- $\angle G(\mathrm{j}\omega)$ = phase shift at frequency ω

For a sinusoidal input $u(t) = A\sin(\omega t)$, the steady-state output is:

$$y_{ss}(t) = A|G(\mathrm{j}\omega)|\sin(\omega t + \angle G(\mathrm{j}\omega)) \tag{1.61}$$

1.5.4 Bode Plots

Bode plots display magnitude (in dB) and phase versus frequency (on a log scale):

$$|G(\mathrm{j}\omega)|_{\mathrm{dB}} = 20\log_{10}|G(\mathrm{j}\omega)| \tag{1.62}$$

1.6 Linear Algebra

Linear algebra is essential for state-space control, multi-input multi-output (MIMO) systems, and machine learning.

1.6.1 Vectors and Matrices

Vector Operations:

- Addition: $\mathbf{x} + \mathbf{y} = [x_1 + y_1, x_2 + y_2, \ldots, x_n + y_n]^\top$
- Scalar multiplication: $\alpha\mathbf{x} = [\alpha x_1, \alpha x_2, \ldots, \alpha x_n]^\top$
- Dot product: $\mathbf{x} \cdot \mathbf{y} = \mathbf{x}^\top\mathbf{y} = \sum_{i=1}^{n} x_i y_i$
- Norm: $\|\mathbf{x}\| = \sqrt{\mathbf{x}^\top\mathbf{x}}$ (Euclidean norm)

Matrix Operations:

- Addition: $(\mathbf{A} + \mathbf{B})_{ij} = a_{ij} + b_{ij}$
- Multiplication: $(\mathbf{AB})_{ij} = \sum_k a_{ik}b_{kj}$
- Transpose: $(\mathbf{A}^\top)_{ij} = a_{ji}$
- Inverse: $\mathbf{AA}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$ (if exists)

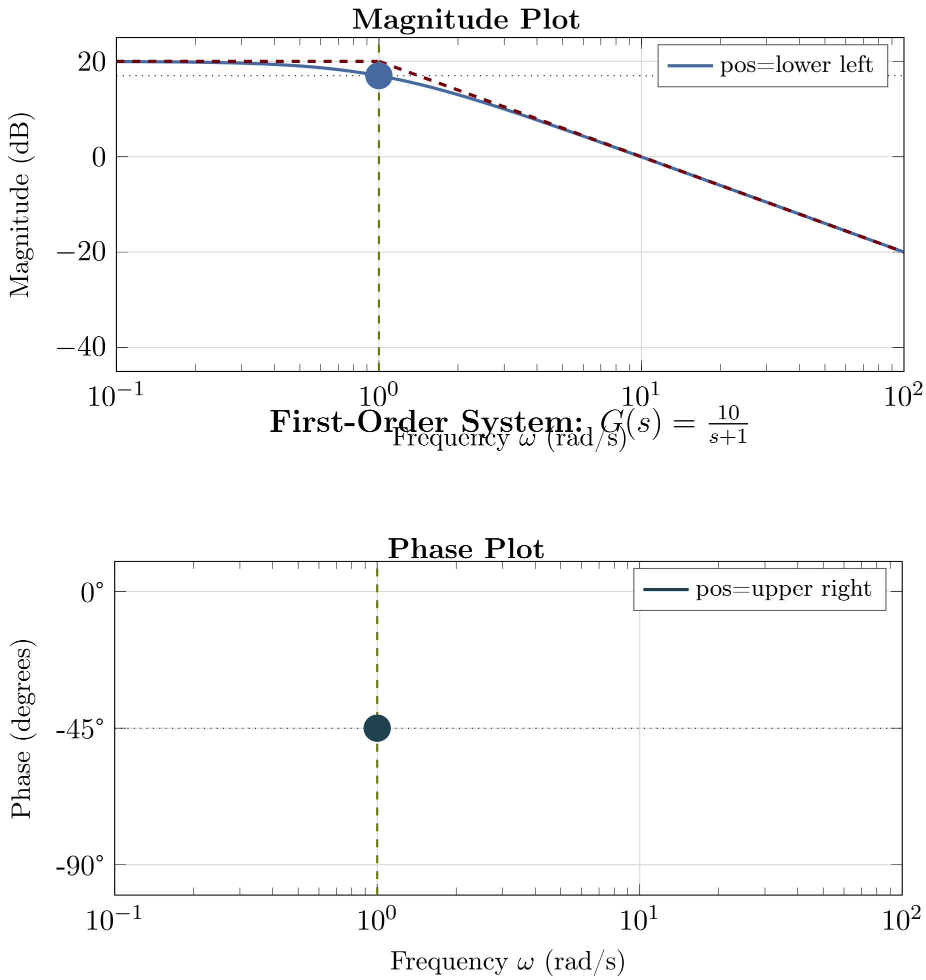


Figure 1.4: Bode plot for a first-order system $G(s) = \frac{10}{s+1}$. The magnitude plot (top, Atlantic blue) shows the steady-state gain of 20 dB below the corner frequency $\omega_c = 1$ rad/s, with a -20 dB/decade slope afterward. The phase plot (bottom, Congaree teal) shows -45° phase lag at the corner frequency. Dashed Garnet lines show asymptotic approximations. The Rose accent color highlights key features: the -3 dB magnitude point and -45° phase point at the corner frequency.

1.6.2 Special Matrices

- **Identity matrix:** $\mathbf{I}$ with ones on diagonal, zeros elsewhere
- **Diagonal matrix:** $\mathbf{D} = \text{diag}(d_1, d_2, \ldots, d_n)$
- **Symmetric matrix:** $\mathbf{A} = \mathbf{A}^\top$
- **Positive definite:** $\mathbf{x}^\top \mathbf{A}\mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$
- **Orthogonal matrix:** $\mathbf{Q}^\top \mathbf{Q} = \mathbf{Q}\mathbf{Q}^\top = \mathbf{I}$

1.6.3 Determinant and Rank

The **determinant** of a square matrix:

$$\det(\mathbf{A}) = |\mathbf{A}| = \sum_{j=1}^{n} (-1)^{1+j} a_{1j} M_{1j} \tag{1.63}$$

where M_{1j} is the minor (determinant of submatrix).

Properties:

- $\det(\mathbf{AB}) = \det(\mathbf{A})\det(\mathbf{B})$
- $\det(\mathbf{A}^{-1}) = 1/\det(\mathbf{A})$
- $\det(\mathbf{A}^\top) = \det(\mathbf{A})$
- $\mathbf{A}$ is invertible if and only if $\det(\mathbf{A}) \neq 0$

The **rank** is the number of linearly independent rows (or columns):

$$\text{rank}(\mathbf{A}) \leq \min(m, n) \text{ for } \mathbf{A} \in \mathbb{R}^{m\times n} \tag{1.64}$$

1.6.4 Eigenvalues and Eigenvectors

Definition 0.5: Eigenvalue Problem

For a square matrix $\mathbf{A}$, scalar λ is an **eigenvalue** and non-zero vector $\mathbf{v}$ is the corresponding **eigenvector** if:

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v} \tag{1.65}$$

Eigenvalues are roots of the **characteristic equation**:

$$\det(\mathbf{A} - \lambda\mathbf{I}) = 0 \tag{1.66}$$

Properties:

- Sum of eigenvalues = $\text{tr}(\mathbf{A}) = \sum_i a_{ii}$
- Product of eigenvalues = $\det(\mathbf{A})$
- Eigenvalues of $\mathbf{A}^{-1}$ are $1/\lambda_i$
- Eigenvalues of $\mathbf{A}^n$ are λ_i^n

Stability and Eigenvalues

A continuous-time system $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ is:

- **Stable** if all eigenvalues of $\mathbf{A}$ have negative real parts
- **Marginally stable** if eigenvalues have non-positive real parts with simple imaginary-axis eigenvalues
- **Unstable** if any eigenvalue has positive real part

1.6.5 Matrix Decompositions

Eigendecomposition: For diagonalizable matrix $\mathbf{A}$:

$$\mathbf{A} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^{-1} \tag{1.67}$$

where $\mathbf{V}$ contains eigenvectors and $\boldsymbol{\Lambda} = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$.

Singular Value Decomposition (SVD): Any matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$:

$$\mathbf{A} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top \tag{1.68}$$

where $\mathbf{U}, \mathbf{V}$ are orthogonal and $\boldsymbol{\Sigma}$ contains singular values.

QR Decomposition:

$$\mathbf{A} = \mathbf{Q}\mathbf{R} \tag{1.69}$$

where $\mathbf{Q}$ is orthogonal and $\mathbf{R}$ is upper triangular.

LU Decomposition:

$$\mathbf{A} = \mathbf{L}\mathbf{U} \tag{1.70}$$

where $\mathbf{L}$ is lower triangular and $\mathbf{U}$ is upper triangular.

1.6.6 Matrix Exponential

The matrix exponential is crucial for solving state-space equations:

Definition 0.6: Matrix Exponential

$$\mathrm{e}^{\mathbf{A}t} = \mathbf{I} + \mathbf{A}t + \frac{(\mathbf{A}t)^2}{2!} + \frac{(\mathbf{A}t)^3}{3!} + \cdots = \sum_{k=0}^{\infty} \frac{(\mathbf{A}t)^k}{k!} \tag{1.71}$$

Properties:

- $\mathrm{e}^{\mathbf{0}} = \mathbf{I}$
- $\frac{\mathrm{d}}{\mathrm{d}t}\mathrm{e}^{\mathbf{A}t} = \mathbf{A}\mathrm{e}^{\mathbf{A}t}$
- $\mathrm{e}^{\mathbf{A}(t_1+t_2)} = \mathrm{e}^{\mathbf{A}t_1}\mathrm{e}^{\mathbf{A}t_2}$
- $(\mathrm{e}^{\mathbf{A}t})^{-1} = \mathrm{e}^{-\mathbf{A}t}$

Computation via Laplace:

$$\mathrm{e}^{\mathbf{A}t} = \mathcal{L}^{-1}\{(s\mathbf{I} - \mathbf{A})^{-1}\} \tag{1.72}$$

Computation via eigendecomposition: If $\mathbf{A} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^{-1}$:

$$\mathrm{e}^{\mathbf{A}t} = \mathbf{V}\mathrm{e}^{\boldsymbol{\Lambda}t}\mathbf{V}^{-1} = \mathbf{V}\,\mathrm{diag}(\mathrm{e}^{\lambda_1 t}, \ldots, \mathrm{e}^{\lambda_n t})\mathbf{V}^{-1} \tag{1.73}$$

1.6.7 Linear Systems of Equations

Solving $\mathbf{A}\mathbf{x} = \mathbf{b}$:

- **Unique solution** if $\mathbf{A}$ is square and invertible: $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$
- **No solution** if $\mathbf{b} \notin \text{range}(\mathbf{A})$
- **Infinite solutions** if $\text{null}(\mathbf{A}) \neq \{\mathbf{0}\}$

Least Squares: When $\mathbf{A}\mathbf{x} = \mathbf{b}$ has no exact solution:

$$\hat{\mathbf{x}} = (\mathbf{A}^\top \mathbf{A})^{-1}\mathbf{A}^\top \mathbf{b} = \mathbf{A}^\dagger \mathbf{b} \tag{1.74}$$

where $\mathbf{A}^\dagger$ is the Moore-Penrose pseudoinverse.

1.7 Probability and Statistics

Probability theory is essential for stochastic control, Kalman filtering, and machine learning.

1.7.1 Probability Fundamentals

Axioms of Probability:

1. $P(A) \geq 0$ for all events A
2. $P(\Omega) = 1$ where Ω is the sample space
3. For mutually exclusive events: $P(A \cup B) = P(A) + P(B)$

Conditional Probability:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \tag{1.75}$$

Bayes' Theorem:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)} \tag{1.76}$$

1.7.2 Random Variables

Discrete Random Variables:

- Probability mass function: $p(x) = P(X = x)$
- Expected value: $E[X] = \sum_x x \cdot p(x)$
- Variance: $\text{Var}(X) = E[(X - E[X])^2] = E[X^2] - (E[X])^2$

Continuous Random Variables:

- Probability density function: $f(x)$ where $P(a \leq X \leq b) = \int_a^b f(x)\,\mathrm{d}x$
- Expected value: $E[X] = \int_{-\infty}^{\infty} x f(x)\,\mathrm{d}x$
- Cumulative distribution: $F(x) = P(X \leq x) = \int_{-\infty}^{x} f(t)\,\mathrm{d}t$

1.7.3 Common Distributions

Gaussian (Normal) Distribution:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \tag{1.77}$$

Notation: $X \sim \mathcal{N}(\mu, \sigma^2)$

Multivariate Gaussian:

$$f(\mathbf{x}) = \frac{1}{(2\pi)^{n/2}|\mathbf{\Sigma}|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^\top \mathbf{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right) \tag{1.78}$$

where $\boldsymbol{\mu}$ is the mean vector and $\mathbf{\Sigma}$ is the covariance matrix.

Other Important Distributions:

- **Uniform:** $f(x) = \frac{1}{b-a}$ for $x \in [a, b]$
- **Exponential:** $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$
- **Bernoulli:** $P(X = 1) = p$, $P(X = 0) = 1 - p$
- **Poisson:** $P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$

1.7.4 Stochastic Processes

A **stochastic process** is a collection of random variables indexed by time: $\{X(t) : t \in T\}$

White Noise: Uncorrelated random sequence with:

- $E[w(t)] = 0$
- $E[w(t)w(\tau)] = \sigma^2\delta(t - \tau)$

Markov Process: Future states depend only on present state:

$$P(X_{n+1}|X_n, X_{n-1}, \ldots, X_0) = P(X_{n+1}|X_n) \tag{1.79}$$

1.7.5 Estimation Theory

Maximum Likelihood Estimation (MLE):

$$\hat{\theta}_{ML} = \arg\max_\theta \prod_{i=1}^{n} f(x_i|\theta) = \arg\max_\theta \sum_{i=1}^{n} \log f(x_i|\theta) \tag{1.80}$$

Bayesian Estimation:

$$p(\theta|\mathbf{x}) \propto p(\mathbf{x}|\theta)p(\theta) \tag{1.81}$$

(posterior $\propto$ likelihood $\times$ prior)

Minimum Mean Square Error (MMSE):

$$\hat{\theta}_{MMSE} = E[\theta|\mathbf{x}] \tag{1.82}$$

1.8 Optimization

Optimization is central to optimal control, parameter estimation, and machine learning.

1.8.1 Unconstrained Optimization

Necessary Condition: At a local minimum $\mathbf{x}^*$:

$$\nabla f(\mathbf{x}^*) = \mathbf{0} \tag{1.83}$$

Sufficient Condition: Also $\mathbf{H}(\mathbf{x}^*) \succ 0$ (positive definite Hessian).
Gradient Descent:

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha \nabla f(\mathbf{x}_k) \tag{1.84}$$

where $\alpha > 0$ is the step size (learning rate).
Newton's Method:

$$\mathbf{x}_{k+1} = \mathbf{x}_k - [\mathbf{H}(\mathbf{x}_k)]^{-1} \nabla f(\mathbf{x}_k) \tag{1.85}$$

1.8.2 Constrained Optimization

Equality Constraints: Minimize $f(\mathbf{x})$ subject to $\mathbf{g}(\mathbf{x}) = \mathbf{0}$
The **Lagrangian**:

$$L(\mathbf{x}, \boldsymbol{\lambda}) = f(\mathbf{x}) + \boldsymbol{\lambda}^\top \mathbf{g}(\mathbf{x}) \tag{1.86}$$

Necessary Conditions (KKT):

$$\nabla_{\mathbf{x}} L = \mathbf{0} \tag{1.87}$$

$$\nabla_{\boldsymbol{\lambda}} L = \mathbf{g}(\mathbf{x}) = \mathbf{0} \tag{1.88}$$

Inequality Constraints: Minimize $f(\mathbf{x})$ subject to $\mathbf{h}(\mathbf{x}) \le \mathbf{0}$
The **Karush-Kuhn-Tucker (KKT) conditions**:

$$\nabla f(\mathbf{x}^*) + \sum_i \mu_i \nabla h_i(\mathbf{x}^*) = \mathbf{0} \tag{1.89}$$

$$h_i(\mathbf{x}^*) \le 0 \tag{1.90}$$

$$\mu_i \ge 0 \tag{1.91}$$

$$\mu_i h_i(\mathbf{x}^*) = 0 \quad \text{(complementary slackness)} \tag{1.92}$$

1.8.3 Convex Optimization

Definition 0.7: Convex Function

A function $f : \mathbb{R}^n \to \mathbb{R}$ is **convex** if for all $\mathbf{x}, \mathbf{y}$ and $\theta \in [0, 1]$:

$$f(\theta\mathbf{x} + (1-\theta)\mathbf{y}) \le \theta f(\mathbf{x}) + (1-\theta) f(\mathbf{y}) \tag{1.93}$$

Key property: Any local minimum of a convex function is a global minimum.
Examples of Convex Problems:

- Linear programming: $\min \mathbf{c}^\top \mathbf{x}$ s.t. $\mathbf{A}\mathbf{x} \le \mathbf{b}$
- Quadratic programming: $\min \frac{1}{2}\mathbf{x}^\top \mathbf{Q}\mathbf{x} + \mathbf{c}^\top \mathbf{x}$ s.t. $\mathbf{A}\mathbf{x} \le \mathbf{b}$ (if $\mathbf{Q} \succeq 0$)
- Least squares: $\min \|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2$

1.8.4 Dynamic Programming

For sequential decision problems, **Bellman's Principle of Optimality**:

> An optimal policy has the property that whatever the initial state and initial decision are, the remaining decisions must constitute an optimal policy with regard to the state resulting from the first decision.

The **Bellman equation** for discrete-time optimal control:

$$V^*(x_k) = \min_{u_k} [L(x_k, u_k) + V^*(f(x_k, u_k))] \tag{1.94}$$

where V^* is the optimal value function (cost-to-go).

1.9 Discrete-Time Mathematics

Digital control systems operate on sampled data, requiring discrete-time analysis.

1.9.1 Sequences and Difference Equations

A **sequence** is a function defined on integers: $x[n]$ or x_n for $n \in \mathbb{Z}$.

First-order difference equation:

$$x[n+1] = ax[n] + bu[n] \tag{1.95}$$

Solution:

$$x[n] = a^n x[0] + \sum_{k=0}^{n-1} a^{n-1-k} bu[k] \tag{1.96}$$

1.9.2 The Z-Transform

Definition 0.8: Z-Transform

The **Z-transform** of a sequence $x[n]$ is:

$$X(z) = \mathcal{Z}\{x[n]\} = \sum_{n=0}^{\infty} x[n]z^{-n} \tag{1.97}$$

Common Z-Transform Pairs:

Z-Transform Properties:

1.9.3 Relationship Between s-Domain and z-Domain

For a sampled signal with sampling period T:

$$z = \mathrm{e}^{sT} \tag{1.98}$$

Therefore:

$$s = \frac{1}{T} \ln z \tag{1.99}$$

Sequence	$x[n]$ for $n \geq 0$	$X(z)$
Unit impulse	$\delta[n]$	1
Unit step	$u[n] = 1$	$\frac{z}{z-1}$
Ramp	n	$\frac{z}{(z-1)^2}$
Exponential	a^n	$\frac{z}{z-a}$
Exponential ramp	na^n	$\frac{az}{(z-a)^2}$
Sine	$\sin(\omega n)$	$\frac{z \sin \omega}{z^2-2z\cos\omega+1}$
Cosine	$\cos(\omega n)$	$\frac{z(z-\cos\omega)}{z^2-2z\cos\omega+1}$

Table 1.5: Common Z-transform pairs.

Property	**Sequence**	**Z-Domain**
Linearity	$ax[n] + by[n]$	$aX(z) + bY(z)$
Time shift (delay)	$x[n-k]$	$z^{-k}X(z)$
Time advance	$x[n+1]$	$zX(z) - zx[0]$
Convolution	$x[n] * y[n]$	$X(z)Y(z)$
Initial value	$x[0]$	$\lim_{z\to\infty} X(z)$
Final value	$\lim_{n\to\infty} x[n]$	$\lim_{z\to 1}(z-1)X(z)$

Table 1.6: Z-transform properties.

Stability in z-Domain

A discrete-time system is stable if and only if all poles of its transfer function $G(z)$ lie **inside the unit circle** $|z| < 1$.

The imaginary axis in the s-plane maps to the unit circle in the z-plane:

- Left half-plane ($\mathrm{Re}(s) < 0$) $\to$ inside unit circle ($|z| < 1$)
- Right half-plane ($\mathrm{Re}(s) > 0$) $\to$ outside unit circle ($|z| > 1$)
- $s = 0 \to z = 1$
- $s = \pm \mathrm{j}\omega_s/2 \to z = -1$ (Nyquist frequency)

1.9.4 Discrete-Time State-Space

The discrete-time state-space representation:

$$\mathbf{x}[k+1] = \mathbf{A}_d\mathbf{x}[k] + \mathbf{B}_d\mathbf{u}[k] \tag{1.100}$$

$$\mathbf{y}[k] = \mathbf{C}_d\mathbf{x}[k] + \mathbf{D}_d\mathbf{u}[k] \tag{1.101}$$

Conversion from continuous-time (exact discretization with zero-order hold):

$$\mathbf{A}_d = \mathrm{e}^{\mathbf{A}T} \tag{1.102}$$

$$\mathbf{B}_d = \int_0^T \mathrm{e}^{\mathbf{A}\tau}\,\mathrm{d}\tau \cdot \mathbf{B} = \mathbf{A}^{-1}(\mathrm{e}^{\mathbf{A}T} - \mathbf{I})\mathbf{B} \tag{1.103}$$

1.10 Special Functions and Signals

1.10.1 Unit Step Function

$$u(t) = \begin{cases} 0 & t < 0 \\ 1 & t \geq 0 \end{cases} \tag{1.104}$$

The discrete-time unit step:

$$u[n] = \begin{cases} 0 & n < 0 \\ 1 & n \geq 0 \end{cases} \tag{1.105}$$

1.10.2 Impulse (Dirac Delta) Function

The **Dirac delta function** is defined by:

$$\delta(t) = 0 \quad \text{for } t \neq 0 \tag{1.106}$$

$$\int_{-\infty}^{\infty} \delta(t)\,\mathrm{d}t = 1 \tag{1.107}$$

Sifting property:

$$\int_{-\infty}^{\infty} f(t)\delta(t - t_0)\,\mathrm{d}t = f(t_0) \tag{1.108}$$

Relationship to step function:

$$\delta(t) = \frac{\mathrm{d}u(t)}{\mathrm{d}t} \tag{1.109}$$

1.10.3 Ramp Function

$$r(t) = t \cdot u(t) = \begin{cases} 0 & t < 0 \\ t & t \geq 0 \end{cases} \tag{1.110}$$

1.10.4 Sinusoidal Functions

General sinusoid:

$$x(t) = A\sin(\omega t + \phi) = A\cos(\omega t + \phi - \pi/2) \tag{1.111}$$

where:

- A = amplitude
- $\omega = 2\pi f$ = angular frequency (rad/s)
- f = frequency (Hz)
- ϕ = phase (rad)
- $T = 1/f = 2\pi/\omega$ = period (s)

Complex exponential representation:

$$A\cos(\omega t + \phi) = \mathrm{Re}\{A\mathrm{e}^{\mathrm{j}(\omega t + \phi)}\} \tag{1.112}$$

1.10.5 Exponential Functions

Real exponential:

$$x(t) = Ae^{\sigma t} \tag{1.113}$$

- $\sigma < 0$: decaying
- $\sigma > 0$: growing
- $\sigma = 0$: constant

Complex exponential (damped sinusoid):

$$x(t) = Ae^{\sigma t} \cos(\omega t + \phi) \tag{1.114}$$

1.11 Numerical Methods

1.11.1 Numerical Integration of ODEs

Euler's Method (Forward):

$$x_{n+1} = x_n + hf(t_n, x_n) \tag{1.115}$$

Runge-Kutta 4th Order (RK4):

$$k_1 = f(t_n, x_n) \tag{1.116}$$

$$k_2 = f(t_n + h/2, x_n + hk_1/2) \tag{1.117}$$

$$k_3 = f(t_n + h/2, x_n + hk_2/2) \tag{1.118}$$

$$k_4 = f(t_n + h, x_n + hk_3) \tag{1.119}$$

$$x_{n+1} = x_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4) \tag{1.120}$$

1.11.2 Root Finding

Newton-Raphson:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \tag{1.121}$$

Bisection Method: For continuous f with $f(a)f(b) < 0$, repeatedly halve the interval containing the root.

1.11.3 Numerical Linear Algebra

Key algorithms:

- **Gaussian elimination**: $O(n^3)$ for solving $\mathbf{Ax} = \mathbf{b}$
- **QR iteration**: Computing eigenvalues
- **SVD algorithms**: Golub-Kahan bidiagonalization
- **Iterative methods**: Conjugate gradient, GMRES for large sparse systems

1.12 Chapter Summary

Key Mathematical Concepts for Control Systems

1. **Complex Numbers**: Foundation for frequency analysis and pole-zero locations
2. **Calculus**: Describes system dynamics through derivatives and integrals
3. **Differential Equations**: Mathematical models of physical systems
4. **Laplace Transform**: Converts differential equations to algebraic equations
5. **Fourier Analysis**: Frequency-domain representation and analysis
6. **Linear Algebra**: Essential for state-space methods and MIMO systems
7. **Probability/Statistics**: Stochastic systems, estimation, and machine learning
8. **Optimization**: Controller design and parameter tuning
9. **Discrete Mathematics**: Digital control and sampled systems

1.13 Exercises

Exercise 1.1. Convert the following complex numbers to polar form:

(a) $z_1 = -3 + 4\mathrm{j}$

(b) $z_2 = 2 - 2\mathrm{j}$

(c) $z_3 = -5$

Exercise 1.2. Find the Laplace transform of:

(a) $f(t) = 3\mathrm{e}^{-2t}\sin(4t)$

(b) $f(t) = t^2\mathrm{e}^{-t}$

(c) $f(t) = (1 - \mathrm{e}^{-t})u(t)$

Exercise 1.3. Find the inverse Laplace transform of:

(a) $F(s) = \frac{5}{s(s+3)}$

(b) $F(s) = \frac{s+1}{s^2+4s+13}$

(c) $F(s) = \frac{1}{(s+1)^2(s+2)}$

Exercise 1.4. For the matrix $\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$:

(a) Find the eigenvalues

(b) Find the eigenvectors

(c) Compute $\mathrm{e}^{\mathbf{A}t}$

(d) Is the system $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ stable?

Exercise 1.5. Linearize the nonlinear system $\dot{x} = x^2 - u$ about the equilibrium point where $x_0 = 1$, $u_0 = 1$.

Exercise 1.6. Find the Z-transform and region of convergence for:

(a) $x[n] = (0.5)^n u[n]$

(b) $x[n] = n(0.8)^n u[n]$

(c) $x[n] = \cos(0.2\pi n)u[n]$

Exercise 1.7. Using gradient descent, find the minimum of $f(x, y) = x^2 + 2y^2 - 2xy - 2x$ starting from $(0, 0)$ with step size $\alpha = 0.1$. Perform 5 iterations.

Exercise 1.8. For a random variable $X \sim \mathcal{N}(5, 4)$:

(a) Find $P(X > 7)$

(b) Find $P(3 < X < 7)$

(c) Find the value x such that $P(X < x) = 0.95$

Chapter 2

System Modeling with Ideal Components

Learning Objectives

After completing this chapter, you will be able to:

- Understand the purpose and process of mathematical modeling
- Apply ideal sensor and actuator assumptions appropriately
- Derive differential equations from physical principles
- Convert between different model representations
- Recognize analogies between different physical domains

The foundation of control systems engineering is the mathematical model—a set of equations that describe how a system behaves. Before we can analyze stability, design controllers, or implement feedback, we must first capture the essential dynamics of the system we wish to control.

This chapter introduces systematic approaches to modeling physical systems across diverse engineering domains. We begin with ideal assumptions: perfect sensors that measure exactly what we want, and ideal actuators that respond instantly with unlimited capability. These assumptions let us focus on the fundamental physics before adding real-world complications in later chapters.

2.1 The Art and Science of Modeling

2.1.1 Why Model?

Mathematical models serve several critical purposes in control engineering:

1. **Prediction**: Models let us simulate system behavior before building hardware, saving time and money.
2. **Analysis**: We can mathematically determine stability, performance limits, and sensitivities.
3. **Design**: Controllers are designed based on models—better models lead to better controllers.
4. **Understanding**: The modeling process itself reveals what physical parameters matter most.

> *"All models are wrong, but some are useful."* — George E.P. Box

A model is always an approximation. The goal is not perfect fidelity but rather capturing the dynamics relevant to control at the frequencies and amplitudes of interest.

Historical Context: Watt's Flyball Governor and Early Feedback Control

The first automatic feedback controller was James Watt's flyball governor (1788), a mechanical device that regulated the speed of steam engines by adjusting fuel flow based on centrifugal force from rotating weights. When engine speed increased, the weights flew outward, reducing fuel flow; when speed decreased, they fell inward, increasing flow. This elegant mechanical feedback system enabled the Industrial Revolution by allowing steam engines to operate autonomously without constant human adjustment. Remarkably, no mathematical model existed for analyzing this system's stability—early engineers relied on trial-and-error design.

The mathematical analysis of feedback control began over 80 years later when James Clerk Maxwell, famous for electromagnetism, published a seminal paper in 1868 analyzing the stability of the flyball governor using differential equations. Maxwell showed that the governor's stability depended on system parameters and introduced the concept that a feedback system could become unstable if parameters were chosen incorrectly. His 1868 paper laid the mathematical foundation for all subsequent control theory, proving that governing equations could predict whether a system would oscillate or diverge—a revolutionary insight that transformed control from an art into a science.

2.1.2 The Modeling Process

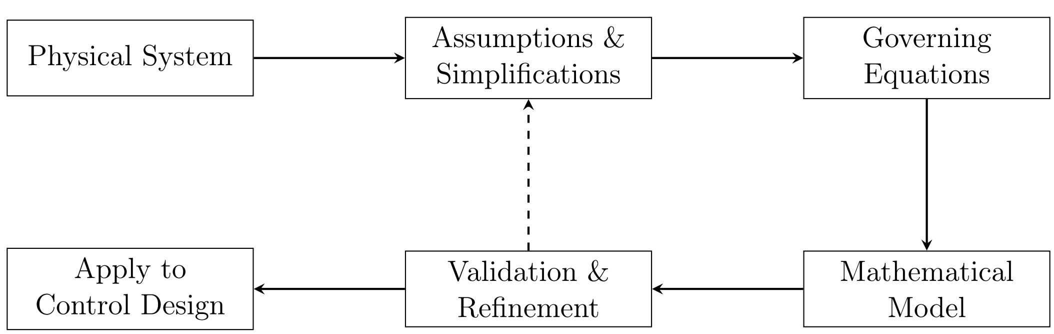


Figure 2.1: The iterative modeling process.

The modeling process typically involves:

1. **Define the system boundary**: What is included? What is external?
2. **Identify inputs and outputs**: What can we manipulate? What do we measure?
3. **List assumptions**: What simplifications are appropriate?
4. **Apply physical laws**: Conservation of energy, mass, momentum, charge, etc.
5. **Derive equations**: Combine physical laws into mathematical form.
6. **Validate**: Compare model predictions with experimental data.
7. **Refine**: Add complexity only where needed.

2.2 Ideal Assumptions

Throughout this chapter, we make two fundamental idealizations:

2.2.1 Ideal Sensors

An **ideal sensor** provides a perfect measurement of the quantity of interest:

- **No delay**: The measurement is instantaneous
- **No noise**: The measurement has no random errors
- **No bias**: The measurement has no systematic offset
- **Infinite bandwidth**: All frequencies are captured equally
- **Infinite resolution**: Any value can be measured exactly
- **No loading**: The sensor does not affect the measured quantity

Mathematically, if $x(t)$ is the true value, the ideal sensor output is:

$$y_{\text{measured}}(t) = x(t) \tag{2.1}$$

2.2.2 Ideal Actuators

An **ideal actuator** produces exactly the commanded output:

- **No delay**: Response is instantaneous
- **No dynamics**: No transient behavior in the actuator itself
- **Infinite bandwidth**: Can track any commanded signal perfectly
- **Unlimited force/torque/power**: No saturation limits
- **Infinite acceleration**: Can change output instantly
- **No backlash or friction**: Perfect mechanical transmission

Mathematically, if $u(t)$ is the commanded input, the ideal actuator produces:

$$F_{\text{actual}}(t) = u(t) \quad \text{(or voltage, torque, flow rate, etc.)} \tag{2.2}$$

Important Note

Real sensors and actuators violate all of these assumptions to varying degrees. Chapters on sensor modeling and actuator dynamics will address these non-idealities. For now, ideal assumptions let us focus on the plant dynamics.

2.3 Fundamental Physical Laws

Most dynamic systems can be modeled using a small set of conservation principles:

2.3.1 Newton's Laws (Mechanical)

Newton's Second Law for translation:

$$\sum F = ma = m\frac{\mathrm{d}^2 x}{\mathrm{d}t^2} \tag{2.3}$$

Rotational analog:

$$\sum \tau = J\alpha = J\frac{\mathrm{d}^2 \theta}{\mathrm{d}t^2} \tag{2.4}$$

where J is the moment of inertia and α is angular acceleration.

2.3.2 Kirchhoff's Laws (Electrical)

Kirchhoff's Voltage Law (KVL): The sum of voltages around any closed loop is zero:

$$\sum_{\text{loop}} V = 0 \tag{2.5}$$

Kirchhoff's Current Law (KCL): The sum of currents at any node is zero:

$$\sum_{\text{node}} i = 0 \tag{2.6}$$

2.3.3 Conservation of Energy (Thermal)

$$\frac{\mathrm{d}E}{\mathrm{d}t} = \dot{Q}_{\text{in}} - \dot{Q}_{\text{out}} + \dot{W} \tag{2.7}$$

For thermal systems:

$$mC_p\frac{\mathrm{d}T}{\mathrm{d}t} = \dot{Q}_{\text{in}} - \dot{Q}_{\text{out}} \tag{2.8}$$

2.3.4 Conservation of Mass (Fluid/Chemical)

$$\frac{\mathrm{d}m}{\mathrm{d}t} = \dot{m}_{\text{in}} - \dot{m}_{\text{out}} \tag{2.9}$$

For incompressible fluids with density ρ and volume V:

$$\rho\frac{\mathrm{d}V}{\mathrm{d}t} = \rho Q_{\text{in}} - \rho Q_{\text{out}} \tag{2.10}$$

where Q is volumetric flow rate.

2.4 Model Representations

The same dynamic system can be represented in several equivalent forms:

2.4.1 Differential Equation Form

The most direct form, derived from physical laws:

$$a_n\frac{\mathrm{d}^n y}{\mathrm{d}t^n} + a_{n-1}\frac{\mathrm{d}^{n-1} y}{\mathrm{d}t^{n-1}} + \cdots + a_1\frac{\mathrm{d}y}{\mathrm{d}t} + a_0 y = b_m\frac{\mathrm{d}^m u}{\mathrm{d}t^m} + \cdots + b_0 u \tag{2.11}$$

2.4.2 Transfer Function Form

Taking the Laplace transform (with zero initial conditions):

$$G(s) = \frac{Y(s)}{U(s)} = \frac{b_m s^m + b_{m-1}s^{m-1} + \cdots + b_0}{a_n s^n + a_{n-1}s^{n-1} + \cdots + a_0} \tag{2.12}$$

2.4.3 State-Space Form

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} \tag{2.13}$$

$$\mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u} \tag{2.14}$$

where $\mathbf{x}$ is the state vector containing the minimum information needed to predict future behavior.

2.4.4 Block Diagram Form

A graphical representation showing signal flow and interconnections.

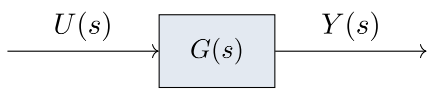


Figure 2.2: Simple block diagram representation.

2.5 System Analogies

Remarkably, systems from different physical domains often share the same mathematical structure. This allows us to transfer intuition and analysis techniques across domains.

Concept	**Mechanical (Trans.)**	**Electrical**	**Thermal**
Effort (across)	Force F	Voltage V	Temperature T
Flow (through)	Velocity v	Current i	Heat flow $\dot{Q}$
Resistance	Damper b	Resistor R	Thermal res. R_t
Capacitance	Spring $1/k$	Capacitor C	Thermal cap. C_t
Inductance	Mass m	Inductor L	—

Table 2.1: Analogous quantities across physical domains (force-voltage analogy).

Example 1.1: Mass-Spring-Damper ≡ RLC Circuit

The mechanical system:

$$m\ddot{x} + b\dot{x} + kx = F(t) \tag{2.15}$$

is mathematically identical to the series RLC circuit:

$$L\ddot{q} + R\dot{q} + \frac{1}{C}q = V(t) \tag{2.16}$$

Both have the same transfer function form:

$$G(s) = \frac{1}{ms^2 + bs + k} \equiv \frac{1}{Ls^2 + Rs + 1/C} \tag{2.17}$$

2.6 Standard Test Inputs

To characterize system behavior, we use standard test inputs:

2.6.1 Impulse Input

$$u(t) = \delta(t), \quad U(s) = 1 \tag{2.18}$$

The impulse response $h(t) = \mathcal{L}^{-1}\{G(s)\}$ completely characterizes an LTI system.

2.6.2 Step Input

$$u(t) = A \cdot u(t), \quad U(s) = \frac{A}{s} \tag{2.19}$$

Step response reveals settling time, overshoot, and steady-state error.

2.6.3 Ramp Input

$$u(t) = At \cdot u(t), \quad U(s) = \frac{A}{s^2} \tag{2.20}$$

Ramp response reveals tracking ability for constantly changing references.

2.6.4 Sinusoidal Input

$$u(t) = A\sin(\omega t), \quad U(s) = \frac{A\omega}{s^2 + \omega^2} \tag{2.21}$$

Sinusoidal response at various frequencies reveals the frequency response.

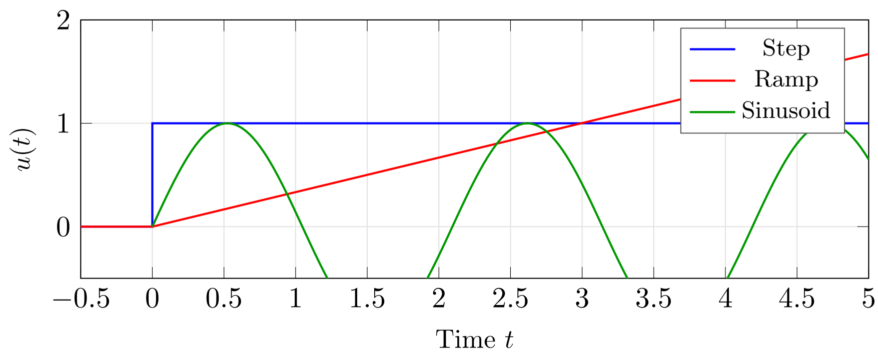


Figure 2.3: Standard test inputs for system characterization.

2.7 First and Second Order System Responses

2.7.1 First-Order Systems

The standard first-order transfer function:

$$G(s) = \frac{K}{\tau s + 1} \tag{2.22}$$

where:

- K = DC gain (steady-state gain)
- τ = time constant

Step Response:

$$y(t) = K\left(1 - \mathrm{e}^{-t/\tau}\right) \tag{2.23}$$

Key characteristics:

- At $t = \tau$: reaches 63.2% of final value
- At $t = 4\tau$: reaches 98.2% of final value (settling time)
- Initial slope: K/τ

2.7.2 Second-Order Systems

The standard second-order transfer function:

$$G(s) = \frac{K\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \tag{2.24}$$

where:

- K = DC gain
- ω_n = natural frequency (rad/s)
- ζ = damping ratio

Poles:

$$s_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1} \tag{2.25}$$

Step Response (underdamped, $0 < \zeta < 1$):

$$y(t) = K\left[1 - \frac{\mathrm{e}^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\sin(\omega_d t + \phi)\right] \tag{2.26}$$

where $\omega_d = \omega_n\sqrt{1-\zeta^2}$ and $\phi = \arccos(\zeta)$.

Time-Domain Specifications:

$$\text{Rise time (10-90\%):}\quad t_r \approx \frac{1.8}{\omega_n} \tag{2.27}$$

$$\text{Peak time:}\quad t_p = \frac{\pi}{\omega_d} \tag{2.28}$$

$$\text{Percent overshoot:}\quad \%OS = 100 \cdot \mathrm{e}^{-\pi\zeta/\sqrt{1-\zeta^2}} \tag{2.29}$$

$$\text{Settling time (2\%):}\quad t_s \approx \frac{4}{\zeta\omega_n} \tag{2.30}$$

2.8 Chapter Organization

The remainder of this chapter presents detailed modeling examples across engineering domains:

- **Section 1B**: Mechanical Systems—translational and rotational dynamics, mass-spring-damper systems, vehicle suspensions, multi-body systems
- **Section 1C**: Electrical Systems—passive circuits, operational amplifiers, filters, power electronics
- **Section 1D**: Electromechanical Systems—DC motors, generators, voice coils, solenoids
- **Section 1E**: Thermal and Fluid Systems—heat transfer, hydraulics, pneumatics, coupled thermal-fluid
- **Section 1F**: Chemical Systems—reaction kinetics, mixing, continuous stirred-tank reactors
- **Section 1G**: Aerospace Systems—aircraft dynamics, spacecraft attitude, rocket propulsion
- **Section 1H**: Biomedical Systems—pharmacokinetics, physiological models, medical devices
- **Section 1I**: Industrial and Computer Systems—manufacturing, queueing, computing

Each section follows a consistent pattern: physical description, governing equations, transfer function derivation, example analysis with step/sinusoidal inputs, and practical considerations.

2.9 Chapter Summary

Key Takeaways: System Modeling

- **Models are approximations, not truth**: All mathematical models are simplified representations of physical reality. The art of modeling lies in capturing the essential dynamics relevant to control at the frequencies and amplitudes of interest, while neglecting higher-order effects that do not significantly impact system behavior.
- **Conservation laws are universal tools**: Physical systems across all engineering domains—mechanical, electrical, thermal, fluid, and chemical—can be modeled by systematically applying conservation principles (energy, mass, momentum, charge). This universal framework allows deep transfer of knowledge between diverse application areas.
- **Multiple representations, same system**: A dynamic system can be expressed equivalently as a differential equation, transfer function, state-space model, or block diagram. Different representations are optimal for different analysis and design tasks; skilled engineers fluently translate between them.
- **Cross-domain analogies accelerate understanding**: Mathematical isomorphisms exist between seemingly unrelated physical systems. For example, a mass-spring-damper system is mathematically identical to an RLC circuit. Recognizing these analogies allows intuition and analysis techniques developed in one domain to directly inform engineering in another.

- **Ideal assumptions simplify without losing insight**: Idealizing sensors (perfect, noise-free, instantaneous) and actuators (perfectly responsive, unlimited power) allows us to focus on fundamental plant dynamics without getting distracted by the significant but secondary effects of real hardware. Nonidealities are addressed in later chapters after mastering the ideal case.

- **System order determines complexity and behavior**: The order of the differential equations governing a system determines its fundamental response characteristics. First-order systems exhibit exponential approach to steady state; second-order systems can oscillate with overshoot; higher-order systems exhibit more complex dynamics. Understanding order is essential for predicting qualitative behavior before detailed analysis.

- **The modeling-validation cycle is iterative**: Practical modeling is not a one-way linear process from assumptions to equations. Rather, it is an iterative cycle where model predictions are compared to experimental data, discrepancies drive refinement of assumptions, and additional complexity is added only where validation reveals it is necessary.

2.10 Mechanical Systems

Mechanical systems form the intuitive foundation for understanding dynamics. The physical concepts of mass, springs, and dampers have direct analogs in other domains and provide excellent intuition for system behavior.

2.10.1 Translational Mechanical Elements

Mass (Inertia)

Mass resists changes in velocity according to Newton's second law:

$$F = m\frac{\mathrm{d}v}{\mathrm{d}t} = m\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = m\ddot{x} \tag{2.31}$$

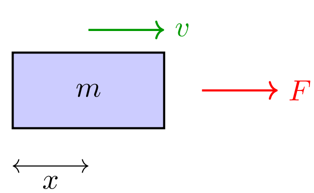


Figure 2.4: Mass element with applied force, coordinate system, and ground reference.

In the Laplace domain:

$$F(s) = ms^2X(s) \quad \Rightarrow \quad \frac{X(s)}{F(s)} = \frac{1}{ms^2} \tag{2.32}$$

Spring (Stiffness)

A linear spring produces a force proportional to displacement:

$$F = kx \quad \text{or} \quad F = k(x_1 - x_2) \tag{2.33}$$

where k is the spring constant (N/m or lb/in).

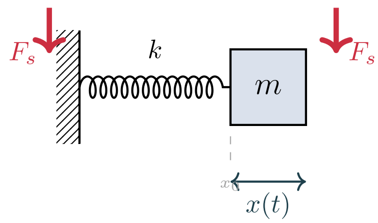


Figure 2.5: Spring element with force arrows on both ends and displacement annotation.

In the Laplace domain:

$$F(s) = kX(s) \quad \Rightarrow \quad \frac{X(s)}{F(s)} = \frac{1}{k} \tag{2.34}$$

Damper (Viscous Friction)

A viscous damper produces a force proportional to velocity:

$$F = b\dot{x} = b\frac{\mathrm{d}x}{\mathrm{d}t} \quad \text{or} \quad F = b(\dot{x}_1 - \dot{x}_2) \tag{2.35}$$

where b is the damping coefficient (N·s/m or lb·s/in).

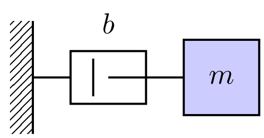


Figure 2.6: Viscous damper element.

In the Laplace domain:

$$F(s) = bsX(s) \quad \Rightarrow \quad \frac{X(s)}{F(s)} = \frac{1}{bs} \tag{2.36}$$

2.10.2 Mass-Spring-Damper System

The canonical mechanical system combines all three elements.

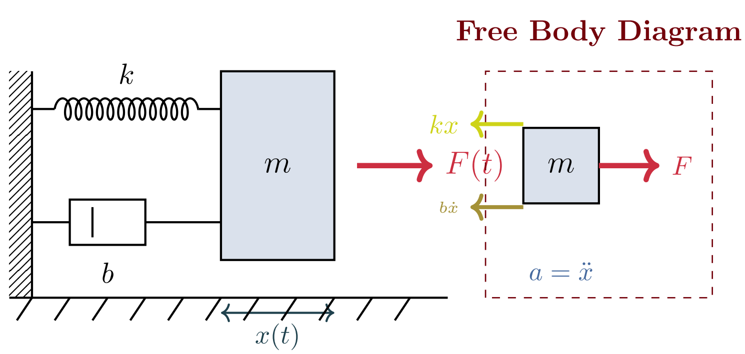


Figure 2.7: Mass-spring-damper system with free body diagram showing all forces.

Equation of Motion

Applying Newton's second law:

$$m\ddot{x} = F(t) - kx - b\dot{x} \tag{2.37}$$

Rearranging:

$$\boxed{m\ddot{x} + b\dot{x} + kx = F(t)} \tag{2.38}$$

Transfer Function

Taking the Laplace transform with zero initial conditions:

$$ms^2X(s) + bsX(s) + kX(s) = F(s) \tag{2.39}$$

$$G(s) = \frac{X(s)}{F(s)} = \frac{1}{ms^2 + bs + k} \tag{2.40}$$

In standard second-order form:

$$G(s) = \frac{1/k}{(m/k)s^2 + (b/k)s + 1} = \frac{1/k}{\frac{s^2}{\omega_n^2} + \frac{2\zeta}{\omega_n}s + 1} \tag{2.41}$$

where:

$$\omega_n = \sqrt{\frac{k}{m}} \quad \text{(natural frequency)} \tag{2.42}$$

$$\zeta = \frac{b}{2\sqrt{km}} \quad \text{(damping ratio)} \tag{2.43}$$

Pole-Zero Map for Different Damping Ratios

The location of poles in the complex s-plane determines the system's response characteristics. For a second-order system, the poles are located at:

$$s = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2} \tag{2.44}$$

The poles move along a circular locus as the damping ratio changes, creating three distinct response regions.

Key observations from the pole-zero map:

- **Underdamped** ($\zeta < 1$): Complex conjugate poles create oscillatory response with overshoot.
- **Critically damped** ($\zeta = 1$): Repeated real pole gives fastest non-oscillatory response.
- **Overdamped** ($\zeta > 1$): Two distinct real poles result in sluggish response without oscillation.

Example 1.2: Automobile Shock Absorber Test

A shock absorber test rig has $m = 50$ kg, $k = 20{,}000$ N/m, and $b = 1{,}000$ N·s/m. Find:

(a) Natural frequency and damping ratio

(b) The response to a 100 N step force

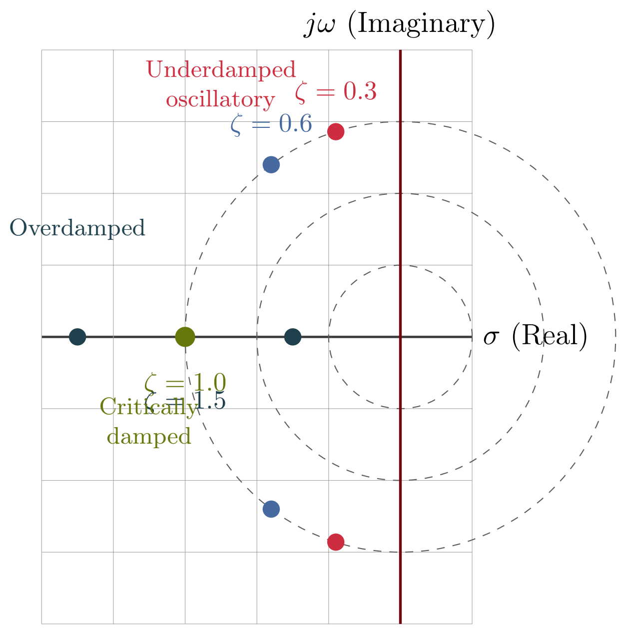


Figure 2.8: Pole-zero map showing pole locations for different damping ratios ($\omega_n = 3$ rad/s fixed). As ζ increases from underdamped to overdamped, poles move from the complex plane onto the real axis.

Solution:
(a) Natural frequency and damping ratio:

$$\omega_n = \sqrt{\frac{20{,}000}{50}} = \sqrt{400} = 20 \text{ rad/s} = 3.18 \text{ Hz} \tag{2.45}$$

$$\zeta = \frac{1{,}000}{2\sqrt{20{,}000 \times 50}} = \frac{1{,}000}{2{,}000} = 0.5 \tag{2.46}$$

The system is underdamped.
(b) Transfer function:

$$G(s) = \frac{1}{50s^2 + 1000s + 20000} = \frac{0.00005}{s^2 + 20s + 400} \tag{2.47}$$

For a 100 N step: $Y(s) = \frac{100}{s} \cdot G(s)$
Using partial fractions and inverse Laplace:

$$x(t) = \frac{100}{20000}\left[1 - \frac{\mathrm{e}^{-10t}}{\sqrt{1 - 0.25}} \sin(17.32t + 1.047)\right] \tag{2.48}$$

Steady-state displacement: $x_{ss} = \frac{F}{k} = \frac{100}{20000} = 5$ mm

Historical Context: Hooke's Law

The spring, one of the most fundamental control components, was mathematically described by Robert Hooke in 1660 when he published his law of elasticity: the force in a spring is proportional to its extension (Hooke stated it in Latin: "Ut tensio, sic vis"—as the extension, so the force). This simple linear relationship, $F = kx$, became the foundation for modeling elastic systems for centuries. Combined with Isaac Newton's laws of motion (published in 1687), Hooke's law enabled the prediction of oscillatory behavior in mechanical systems. The mass-spring system's equation of motion, $m\ddot{x} + kx = F$, is perhaps the most important differential equation in all of engineering.

The systematic analysis of mechanical vibrations emerged in the 18th and 19th centuries as engineers faced resonance problems in machinery, bridges, and vehicles. Understanding that vibrations depended on mass, stiffness, and damping—and that underdamped systems could produce dangerous oscillations—drove the development of stability analysis methods. Vibration analysis, grounded in Hooke's and Newton's laws, ultimately led to the Nyquist and Bode analysis techniques that dominate modern control design, making the humble spring-mass-damper system the pedagogical gateway to control theory.

2.10.3 Vehicle Suspension System

A quarter-car model represents one corner of a vehicle suspension.

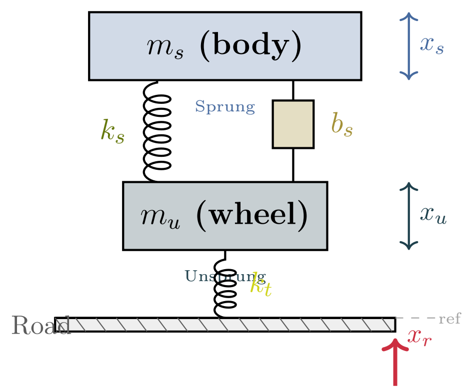


Figure 2.9: Quarter-car suspension model with distinct colors for sprung (body) and unsprung (wheel) masses, and road input visualization.

Equations of Motion

For the sprung mass (body):

$$m_s\ddot{x}_s = -k_s(x_s - x_u) - b_s(\dot{x}_s - \dot{x}_u) \tag{2.49}$$

For the unsprung mass (wheel):

$$m_u\ddot{x}_u = k_s(x_s - x_u) + b_s(\dot{x}_s - \dot{x}_u) - k_t(x_u - x_r) \tag{2.50}$$

State-Space Representation

Defining state variables: $\mathbf{x} = [x_s - x_u, \dot{x}_s, x_u - x_r, \dot{x}_u]^\top$

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 & -1 \\ -k_s/m_s & -b_s/m_s & 0 & b_s/m_s \\ 0 & 0 & 0 & 1 \\ k_s/m_u & b_s/m_u & -k_t/m_u & -b_s/m_u \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 0 \\ -1 \\ 0 \end{bmatrix} \dot{x}_r \tag{2.51}$$

Transfer Functions

From road input $X_r(s)$ to body displacement $X_s(s)$:

$$\frac{X_s(s)}{X_r(s)} = \frac{(b_s s + k_s)k_t}{m_s m_u s^4 + (m_s + m_u)b_s s^3 + [m_s k_t + (m_s + m_u)k_s]s^2 + b_s k_t s + k_s k_t} \tag{2.52}$$

Example 1.3: Quarter-Car Parameters

Typical values for a passenger car:

- Sprung mass: $m_s = 250$ kg (quarter of body mass)
- Unsprung mass: $m_u = 35$ kg (wheel, brake, suspension links)
- Suspension stiffness: $k_s = 16{,}000$ N/m
- Suspension damping: $b_s = 1{,}000$ N·s/m
- Tire stiffness: $k_t = 160{,}000$ N/m

The body bounce frequency: $f_s = \frac{1}{2\pi}\sqrt{\frac{k_s}{m_s}} = 1.27$ Hz

The wheel hop frequency: $f_u = \frac{1}{2\pi}\sqrt{\frac{k_t}{m_u}} = 10.8$ Hz

2.10.4 Rotational Mechanical Systems

Rotational systems follow analogous laws with angular quantities.

Translational	**Rotational**	**Unit**
Displacement x	Angle θ	rad
Velocity $v = \dot{x}$	Angular velocity $\omega = \dot{\theta}$	rad/s
Acceleration $a = \ddot{x}$	Angular acceleration $\alpha = \ddot{\theta}$	rad/s^2
Force F	Torque τ	N·m
Mass m	Moment of inertia J	kg·m^2
Spring constant k	Torsional stiffness κ	N·m/rad
Damping b	Rotational damping B	N·m·s/rad

Table 2.2: Translational-rotational analogies.

Rotational Inertia

$$\tau = J\ddot{\theta} = J\dot{\omega} \tag{2.53}$$

Torsional Spring

$$\tau = \kappa\theta \quad \text{or} \quad \tau = \kappa(\theta_1 - \theta_2) \tag{2.54}$$

Rotational Damper

$$\tau = B\dot{\theta} = B\omega \tag{2.55}$$

Example 1.4: Rotating Shaft System

A motor drives a load through a flexible shaft. The motor inertia is $J_m = 0.01$ kg·m^2, load inertia is $J_L = 0.1$ kg·m^2, shaft stiffness is $\kappa = 100$ N·m/rad, and bearing friction is $B = 0.5$ N·m·s/rad.
Equations of motion:

$$J_m\ddot{\theta}_m = \tau_m - \kappa(\theta_m - \theta_L) \tag{2.56}$$
$$J_L\ddot{\theta}_L = \kappa(\theta_m - \theta_L) - B\dot{\theta}_L \tag{2.57}$$

Transfer function from motor torque to load angle:

$$\frac{\Theta_L(s)}{\tau_m(s)} = \frac{\kappa}{J_m J_L s^4 + J_m B s^3 + (J_m + J_L)\kappa s^2 + B\kappa s} \tag{2.58}$$

2.10.5 Gear Trains

Gears transform speed and torque between shafts.

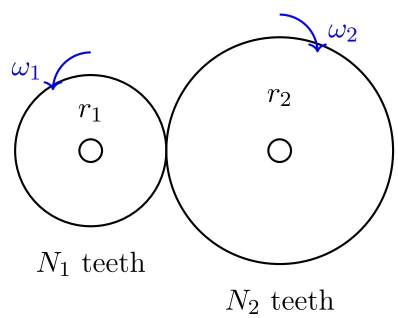


Figure 2.11: Gear pair.

Gear Ratio:

$$n = \frac{N_2}{N_1} = \frac{r_2}{r_1} = \frac{\omega_1}{\omega_2} = \frac{\tau_2}{\tau_1} \tag{2.59}$$

Reflected Inertia: When reflecting load inertia J_L to the motor side:

$$J_{\text{reflected}} = \frac{J_L}{n^2} \tag{2.60}$$

Reflected Damping:

$$B_{\text{reflected}} = \frac{B_L}{n^2} \tag{2.61}$$

Gear Ratio Selection

The gear ratio affects both speed and torque:

- High gear ratio ($n > 1$): Reduces speed, increases torque, reduces reflected inertia
- Low gear ratio ($n < 1$): Increases speed, reduces torque, increases reflected inertia

Optimal gear ratio often maximizes acceleration: $n_{\text{opt}} = \sqrt{J_L/J_m}$

2.10.6 Multi-Body Mechanical Systems

Example 1.5: Two-Mass System

Consider two masses connected by springs and dampers:
Equations of motion:

$$m_1\ddot{x}_1 = -k_1x_1 - b_1\dot{x}_1 + k_2(x_2 - x_1) + b_2(\dot{x}_2 - \dot{x}_1) \tag{2.62}$$
$$m_2\ddot{x}_2 = F - k_2(x_2 - x_1) - b_2(\dot{x}_2 - \dot{x}_1) \tag{2.63}$$

State-space form with $\mathbf{x} = [x_1, \dot{x}_1, x_2, \dot{x}_2]^\top$:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -(k_1+k_2)/m_1 & -(b_1+b_2)/m_1 & k_2/m_1 & b_2/m_1 \\ 0 & 0 & 0 & 1 \\ k_2/m_2 & b_2/m_2 & -k_2/m_2 & -b_2/m_2 \end{bmatrix} \tag{2.64}$$

2.10.7 Pendulum Systems

Simple Pendulum

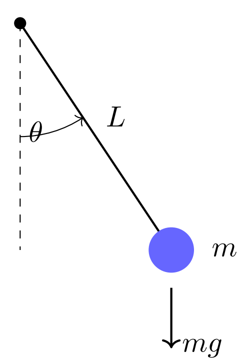


Figure 2.13: Simple pendulum.

The equation of motion (nonlinear):

$$mL^2\ddot{\theta} + mgL\sin\theta = \tau \tag{2.65}$$

For small angles ($\sin\theta \approx \theta$), the linearized equation:

$$\ddot{\theta} + \frac{g}{L}\theta = \frac{\tau}{mL^2} \tag{2.66}$$

Natural frequency: $\omega_n = \sqrt{g/L}$

Inverted Pendulum on Cart

The classic control benchmark problem:

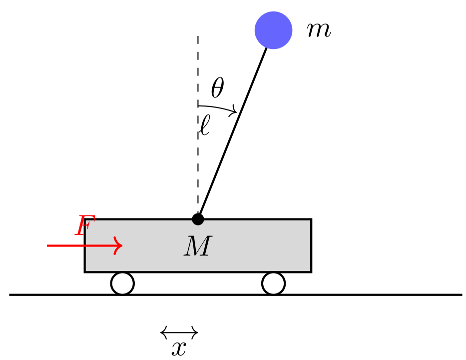


Figure 2.14: Inverted pendulum on cart.

Linearized equations about the upright position ($\theta = 0$):

$$(M + m)\ddot{x} + m\ell\ddot{\theta} = F \tag{2.67}$$

$$m\ell\ddot{x} + m\ell^2\ddot{\theta} - mg\ell\theta = 0 \tag{2.68}$$

Transfer function from force to angle:

$$\frac{\Theta(s)}{F(s)} = \frac{1}{(M\ell)s^2 - (M+m)g} \tag{2.69}$$

This system is **open-loop unstable** (pole in the right half-plane) and requires feedback control for stabilization.

2.10.8 Exercises: Mechanical Systems

Exercise 2.1. A spring-mass system has $m = 2$ kg and $k = 200$ N/m. Find:

(a) The natural frequency in rad/s and Hz

(b) The transfer function $X(s)/F(s)$

(c) The response to a 10 N step force

Exercise 2.2. Add a damper with $b = 8$ N·s/m to the system in Exercise 1.

(a) Calculate the damping ratio

(b) Find the damped natural frequency

(c) Determine if the system is underdamped, critically damped, or overdamped

Exercise 2.3. A motor with $J_m = 0.005$ kg·m^2 drives a load with $J_L = 0.5$ kg·m^2 through a 10:1 gear ratio.

(a) Calculate the reflected load inertia at the motor

(b) Find the total effective inertia seen by the motor

(c) Derive the transfer function from motor torque to load angle

Exercise 2.4. For the quarter-car model, derive the transfer function from road velocity $\dot{X}_r(s)$ to body acceleration $s^2 X_s(s)$. This represents the acceleration felt by passengers.

Exercise 2.5. Linearize the simple pendulum equation about $\theta_0 = \pi/6$ (30 degrees). Find the equilibrium torque and the linearized dynamics.

Exercise 2.6. A tuned mass damper for building vibration control has the following parameters:

- Building mass: $M = 1000$ kg
- Damper mass: $m = 20$ kg
- Damper spring constant: $k = 5000$ N/m
- Damper coefficient: $b = 500$ N·s/m

(a) Calculate the natural frequency of the tuned damper

(b) If the building is excited at 2 Hz, find the phase lag of the damper response

(c) Design the spring constant to match the building frequency of 1.5 Hz

Exercise 2.7. Consider a flexible robot arm modeled as a two-inertia system: motor inertia $J_m = 0.01$ kg·m^2, link inertia $J_L = 0.5$ kg·m^2, shaft torsional stiffness $\kappa = 50$ N·m/rad, and bearing friction $B = 0.1$ N·m·s/rad.

(a) Derive the transfer function from motor torque to link angle

(b) Find the natural frequencies (radians and Hz)

(c) Explain why this system exhibits vibration and how to suppress it

Exercise 2.8. Concept: Compare the responses of three second-order mechanical systems with the same natural frequency ($\omega_n = 10$ rad/s) but different damping ratios: $\zeta_1 = 0.3$, $\zeta_2 = 0.7$, $\zeta_3 = 1.0$.

(a) For each, describe the qualitative step response: overshoot, oscillations, settling time

(b) Rank them by settling time (2% criterion)

(c) Which would you choose for a robotics application and why?

Exercise 2.9. Design: A vibration isolator for precision equipment must satisfy:

- Support mass: $m = 50$ kg
- Isolation efficiency $> 80\%$ at 10 Hz (reduce vibration amplitude by factor of 5)
- Static deflection < 10 cm at rated load

(a) Calculate the required spring constant

(b) Find the natural frequency

(c) Verify the isolation efficiency at 10 Hz (calculate transmissibility)

(d) Specify a damping ratio for practical implementation

Exercise 2.10. Concept: The inverted pendulum on a cart is fundamentally unstable. Explain:

(a) Why feedback control is necessary (discuss the pole locations)

(b) What happens to the system response without control when perturbed

(c) Why a cart-based solution is simpler than a balance-wheel solution

2.11 Electrical Systems

Electrical circuits provide clean examples of dynamic systems with well-defined mathematical models. The concepts developed here—impedance, frequency response, filtering—form the historical foundation of control theory.

Historical Context: Black's Negative Feedback Amplifier and Frequency Response Methods

The foundation of modern control theory emerged not from mechanical engineers but from electrical engineers at Bell Telephone Laboratories. In 1927, Harold Black invented the negative feedback amplifier, recognizing that feeding a portion of the output back inverted to the input could stabilize amplifier gain against component variations and temperature changes. This elegant insight—that feedback could improve performance—was initially rejected by patent offices, who considered it obvious. Yet Black's feedback amplifier enabled long-distance telephone transmission by compensating for signal loss over cables, fundamentally changing telecommunications.

The mathematical analysis of feedback systems was subsequently developed by Harry Nyquist (1932) and Hendrik Bode (1945), who created graphical methods to analyze stability and performance from frequency response measurements. The Nyquist stability criterion and Bode plots transformed control design from an empirical art into a quantitative science. These frequency response techniques, born from electrical circuits and amplifiers, became universal tools applicable to all domains—mechanical, thermal, biological, and beyond—establishing electrical systems as the historical birthplace of modern control theory.

2.11.1 Passive Circuit Elements

Resistor

Ohm's law relates voltage and current:

$$v(t) = Ri(t) \tag{2.70}$$

In the Laplace domain: $V(s) = RI(s)$
Impedance: $Z_R(s) = R$

Figure 2.15: Resistor element.

Capacitor

The capacitor stores charge:

$$i(t) = C\frac{\mathrm{d}v}{\mathrm{d}t} \quad \Leftrightarrow \quad v(t) = \frac{1}{C}\int i(\tau)\,\mathrm{d}\tau \tag{2.71}$$

In the Laplace domain: $I(s) = CsV(s)$ or $V(s) = \frac{1}{Cs}I(s)$
Impedance: $Z_C(s) = \frac{1}{Cs}$

Figure 2.16: Capacitor element.

Inductor

The inductor stores magnetic energy:

$$v(t) = L\frac{\mathrm{d}i}{\mathrm{d}t} \quad \Leftrightarrow \quad i(t) = \frac{1}{L}\int v(\tau)\,\mathrm{d}\tau \tag{2.72}$$

In the Laplace domain: $V(s) = LsI(s)$ or $I(s) = \frac{1}{Ls}V(s)$
Impedance: $Z_L(s) = Ls$

Figure 2.17: Inductor element.

2.11.2 RC Circuit (First-Order System)

Time-Domain Analysis

Applying KVL:

$$v_{\text{in}} = v_R + v_{\text{out}} = Ri + v_{\text{out}} \tag{2.73}$$

Since $i = C\frac{\mathrm{d}v_{\text{out}}}{\mathrm{d}t}$:

$$\boxed{RC\frac{\mathrm{d}v_{\text{out}}}{\mathrm{d}t} + v_{\text{out}} = v_{\text{in}}} \tag{2.74}$$

This is a first-order ODE with time constant $\tau = RC$.

Transfer Function

$$G(s) = \frac{V_{\text{out}}(s)}{V_{\text{in}}(s)} = \frac{1}{RCs + 1} = \frac{1}{\tau s + 1} \tag{2.75}$$

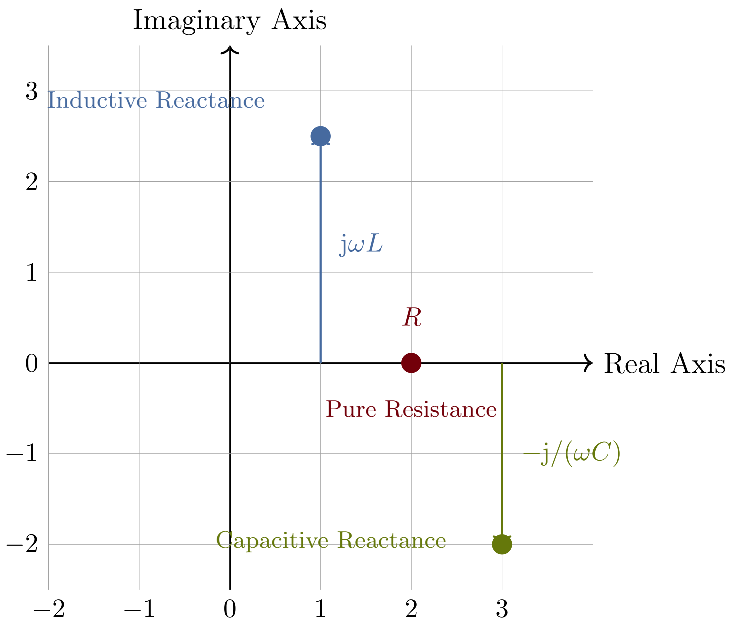


Figure 2.18: Impedance in the complex plane showing resistive, inductive, and capacitive elements.

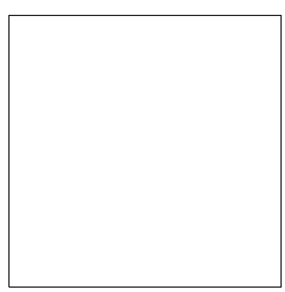


Figure 2.19: RC low-pass filter circuit.

Frequency Response

Substituting $s = \mathrm{j}\omega$:

$$G(\mathrm{j}\omega) = \frac{1}{1 + \mathrm{j}\omega RC} \tag{2.76}$$

Magnitude and phase:

$$|G(\mathrm{j}\omega)| = \frac{1}{\sqrt{1 + (\omega RC)^2}} \tag{2.77}$$

$$\angle G(\mathrm{j}\omega) = -\arctan(\omega RC) \tag{2.78}$$

Cutoff frequency (where magnitude is $1/\sqrt{2}$ or -3 dB):

$$\omega_c = \frac{1}{RC} = \frac{1}{\tau} \tag{2.79}$$

Example 1.6: RC Circuit Step Response

An RC circuit has $R = 10$ kΩ and $C = 1$ μF. Find:

(a) Time constant

(b) Cutoff frequency

(c) Response to a 5V step input

Solution:
(a) Time constant:

$$\tau = RC = 10{,}000 \times 10^{-6} = 0.01 \text{ s} = 10 \text{ ms} \tag{2.80}$$

(b) Cutoff frequency:

$$f_c = \frac{1}{2\pi RC} = \frac{1}{2\pi \times 0.01} = 15.9 \text{ Hz} \tag{2.81}$$

(c) Step response:

$$v_{\text{out}}(t) = 5(1 - \mathrm{e}^{-t/0.01}) = 5(1 - \mathrm{e}^{-100t}) \text{ V} \tag{2.82}$$

At $t = \tau = 10$ ms: $v_{\text{out}} = 5(1 - \mathrm{e}^{-1}) = 3.16$ V (63.2% of final)

2.11.3 RL Circuit

For current as output:

$$L\frac{\mathrm{d}i}{\mathrm{d}t} + Ri = v_{\text{in}} \tag{2.83}$$

Transfer function (voltage to current):

$$\frac{I(s)}{V_{\text{in}}(s)} = \frac{1}{Ls + R} = \frac{1/R}{(L/R)s + 1} \tag{2.84}$$

Time constant: $\tau = L/R$

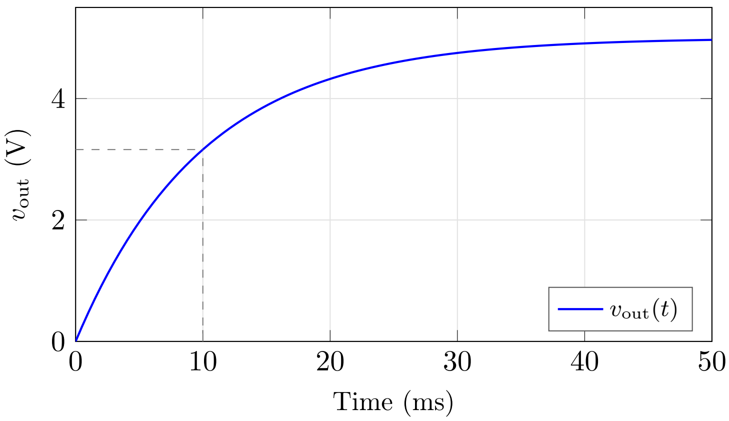
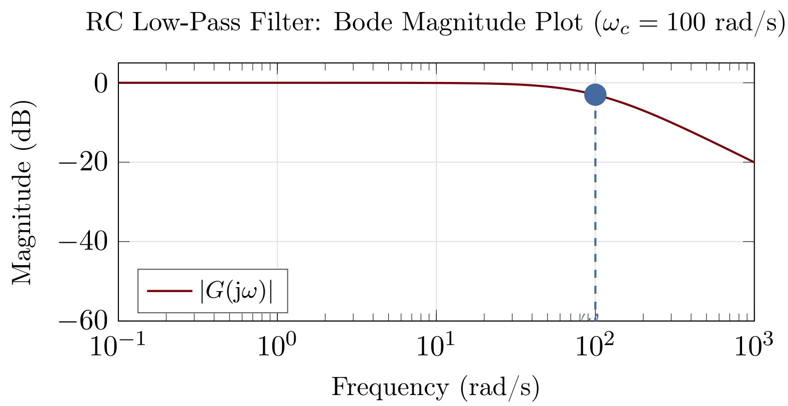


Figure 2.20: RC circuit step response.



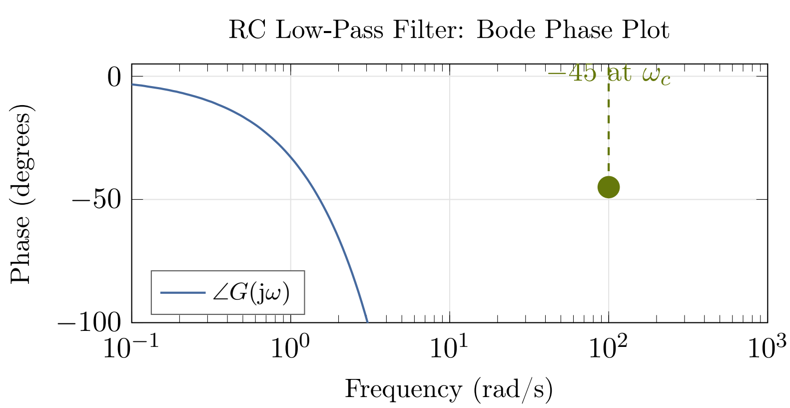


Figure 2.21: RC low-pass filter Bode plots (magnitude and phase).

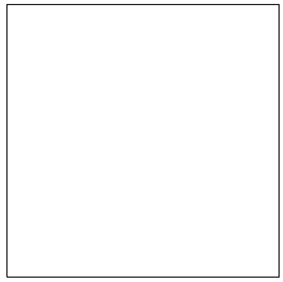


Figure 2.22: RL circuit.

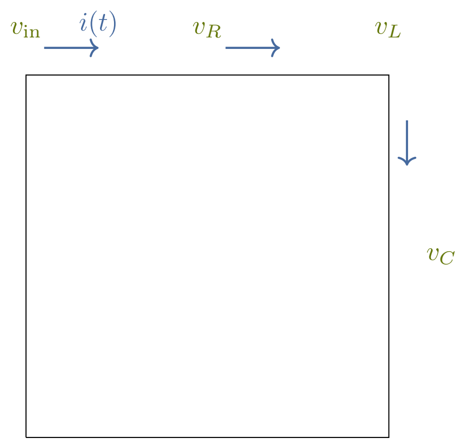


Figure 2.23: Series RLC circuit with current direction and node voltages labeled.

2.11.4 RLC Circuit (Second-Order System)

Differential Equation

Applying KVL:

$$v_{\text{in}} = v_R + v_L + v_C = Ri + L\frac{\mathrm{d}i}{\mathrm{d}t} + v_{\text{out}} \tag{2.85}$$

Since $i = C\frac{\mathrm{d}v_{\text{out}}}{\mathrm{d}t}$:

$$\boxed{LC\frac{\mathrm{d}^2 v_{\text{out}}}{\mathrm{d}t^2} + RC\frac{\mathrm{d}v_{\text{out}}}{\mathrm{d}t} + v_{\text{out}} = v_{\text{in}}} \tag{2.86}$$

Transfer Function

$$G(s) = \frac{V_{\text{out}}(s)}{V_{\text{in}}(s)} = \frac{1}{LCs^2 + RCs + 1} \tag{2.87}$$

In standard form:

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \tag{2.88}$$

where:

$$\omega_n = \frac{1}{\sqrt{LC}} \quad \text{(resonant frequency)} \tag{2.89}$$

$$\zeta = \frac{R}{2}\sqrt{\frac{C}{L}} \quad \text{(damping ratio)} \tag{2.90}$$

Quality Factor

The quality factor Q measures the sharpness of resonance:

$$Q = \frac{1}{2\zeta} = \frac{1}{R}\sqrt{\frac{L}{C}} \tag{2.91}$$

High Q means sharp resonance peak and low damping.

Example 1.7: RLC Circuit Design

Design an RLC circuit with resonant frequency $f_n = 1$ kHz and $Q = 10$.
Solution:
Choose $C = 0.1$ μF. Then:

$$L = \frac{1}{(2\pi f_n)^2 C} = \frac{1}{(2\pi \times 1000)^2 \times 10^{-7}} = 0.253 \text{ H} \tag{2.92}$$

For $Q = 10$:

$$R = \frac{1}{Q}\sqrt{\frac{L}{C}} = \frac{1}{10}\sqrt{\frac{0.253}{10^{-7}}} = 159\ \Omega \tag{2.93}$$

Damping ratio: $\zeta = 1/(2Q) = 0.05$ (highly underdamped)

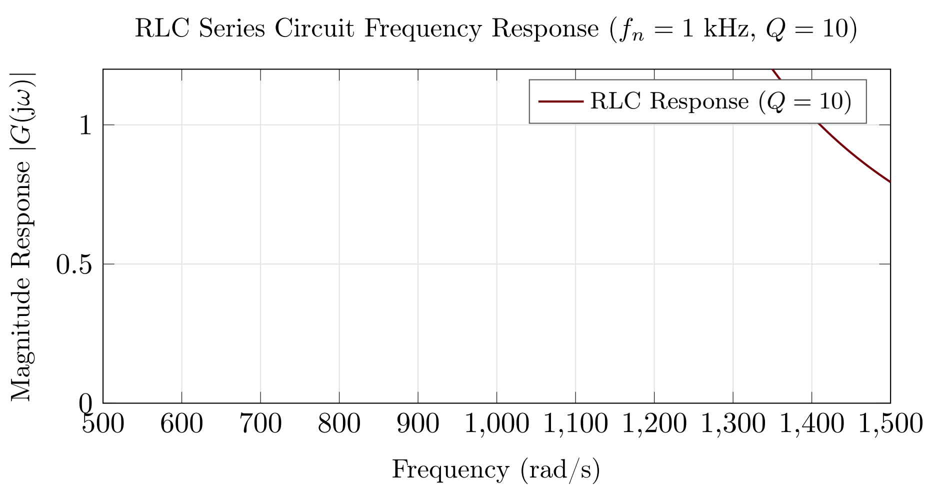


Figure 2.24: RLC series circuit frequency response showing resonance peak at ω_n and bandwidth.

2.11.5 Operational Amplifier Circuits

Operational amplifiers (op-amps) are fundamental building blocks for analog control systems.

Ideal Op-Amp Assumptions

- Infinite open-loop gain: $A \to \infty$
- Infinite input impedance: $Z_{\text{in}} \to \infty$ (no input current)
- Zero output impedance: $Z_{\text{out}} = 0$
- Infinite bandwidth

With negative feedback, these lead to the "golden rules":

1. $V_+ = V_-$ (virtual short)
2. $I_+ = I_- = 0$ (no input current)

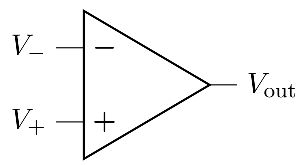


Figure 2.25: Op-amp symbol.

Inverting Amplifier

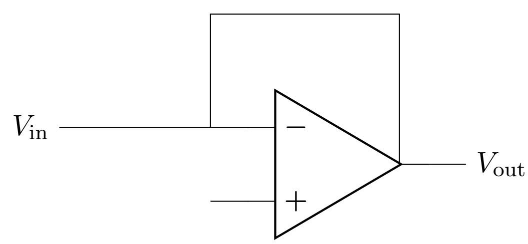


Figure 2.26: Inverting amplifier.

Transfer function:

$$\frac{V_{\text{out}}}{V_{\text{in}}} = -\frac{R_2}{R_1} \tag{2.94}$$

Non-Inverting Amplifier

Transfer function:

$$\frac{V_{\text{out}}}{V_{\text{in}}} = 1 + \frac{R_2}{R_1} \tag{2.95}$$

Integrator

Transfer function:

$$G(s) = \frac{V_{\text{out}}(s)}{V_{\text{in}}(s)} = -\frac{1}{RCs} \tag{2.96}$$

Time-domain:

$$v_{\text{out}}(t) = -\frac{1}{RC}\int_0^t v_{\text{in}}(\tau)\,\mathrm{d}\tau \tag{2.97}$$

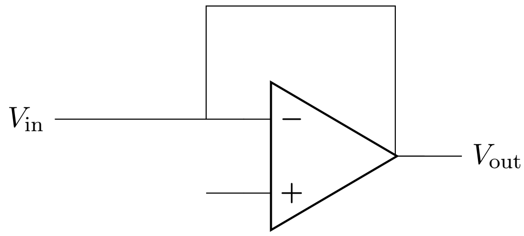


Figure 2.27: Op-amp integrator.

Differentiator

Transfer function:

$$G(s) = \frac{V_{\text{out}}(s)}{V_{\text{in}}(s)} = -RCs \tag{2.98}$$

Important Note

Pure differentiators amplify high-frequency noise. In practice, add a small resistor in series with C to limit high-frequency gain.

PID Controller Implementation

A PID controller can be implemented with op-amps:

$$G_{PID}(s) = K_p + \frac{K_i}{s} + K_d s \tag{2.99}$$

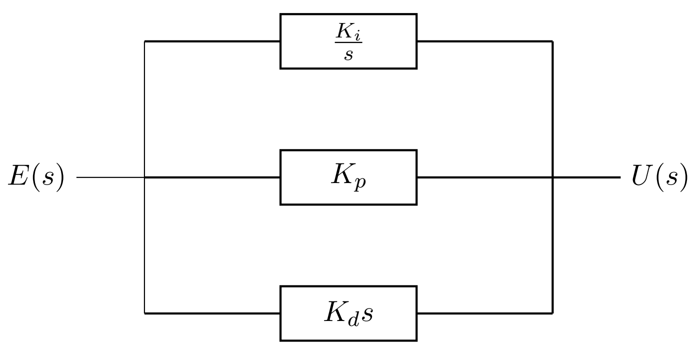


Figure 2.28: PID controller block diagram.

2.11.6 Active Filters

First-Order Low-Pass Filter

$$G(s) = \frac{K}{s/\omega_c + 1} \tag{2.100}$$

Passes low frequencies, attenuates high frequencies.

First-Order High-Pass Filter

$$G(s) = \frac{Ks/\omega_c}{s/\omega_c + 1} \tag{2.101}$$

Passes high frequencies, attenuates low frequencies (including DC).

Second-Order Butterworth Low-Pass

Maximally flat passband response:

$$G(s) = \frac{\omega_c^2}{s^2 + \sqrt{2}\omega_c s + \omega_c^2} \tag{2.102}$$

Damping ratio $\zeta = 1/\sqrt{2} \approx 0.707$, giving no overshoot in step response.

Band-Pass Filter

$$G(s) = \frac{K(\omega_0/Q)s}{s^2 + (\omega_0/Q)s + \omega_0^2} \tag{2.103}$$

Passes frequencies near ω_0, attenuates both lower and higher frequencies.

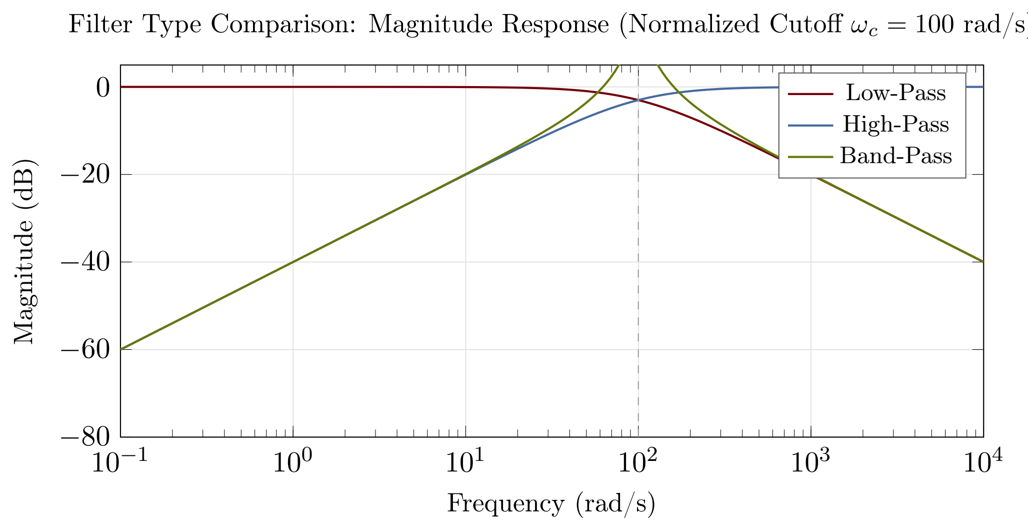


Figure 2.29: Comparison of filter types showing magnitude response of low-pass, high-pass, and band-pass filters with normalized cutoff frequency.

2.11.7 Power Electronics

Buck Converter (Step-Down)

Steady-state output (ideal):

$$V_{\text{out}} = D \cdot V_{\text{in}} \tag{2.104}$$

where D is the duty cycle ($0 \leq D \leq 1$).

Small-signal transfer function (control to output):

$$\frac{\hat{v}_{\text{out}}(s)}{\hat{d}(s)} = V_{\text{in}} \cdot \frac{1}{LCs^2 + (L/R_L)s + 1} \tag{2.105}$$

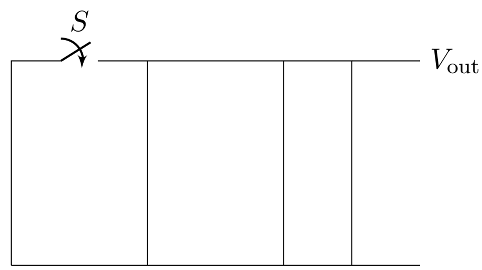


Figure 2.30: Buck converter circuit.

2.11.8 Exercises: Electrical Systems

Exercise 2.11. For an RC circuit with $R = 47$ kΩ and $C = 0.22$ μF:

(a) Find the time constant and cutoff frequency

(b) Sketch the Bode plot (magnitude and phase)

(c) Find the output voltage 5 ms after a 10V step input

Exercise 2.12. Design an RLC band-pass filter with center frequency 10 kHz and bandwidth 1 kHz. Calculate R, L, C, and Q.

Exercise 2.13. An op-amp integrator has $R = 100$ kΩ and $C = 1$ μF. If the input is $v_{\text{in}}(t) = 2\sin(100t)$ V:

(a) Find the transfer function

(b) Calculate the steady-state output amplitude and phase

(c) Write the time-domain output expression

Exercise 2.14. For the series RLC circuit, find the current $i(t)$ response to a unit step voltage input when $R = 100$ Ω, $L = 0.1$ H, and $C = 100$ μF. Is the circuit underdamped, critically damped, or overdamped?

Exercise 2.15. Design a second-order Sallen-Key low-pass filter with cutoff frequency 1 kHz and Butterworth response. Specify all component values.

Exercise 2.16. Concept: Compare the frequency response characteristics of three filters:

(a) A first-order low-pass filter with $\omega_c = 1000$ rad/s

(b) A second-order Butterworth low-pass filter with $\omega_c = 1000$ rad/s

(c) A first-order high-pass filter with $\omega_c = 1000$ rad/s

For each, sketch the Bode magnitude plot and describe:

(a) The slope (dB/decade) in the stopband

(b) The phase shift at the cutoff frequency

(c) Which frequencies are attenuated and which are passed

Exercise 2.17. A boost converter steps up a 12V input to 48V at a switching frequency of 200 kHz. Assume efficiency $\eta = 0.92$.

(a) Calculate the required duty cycle

(b) If the load draws 5A at 48V, find the input current

(c) Design the filter inductance to limit current ripple to $\pm 5\%$

Exercise 2.18. Design: An active low-pass filter must attenuate 60 Hz power line noise by at least 40 dB while preserving signals below 50 Hz.

(a) Determine the minimum filter order required

(b) Choose between Butterworth, Chebyshev, and Bessel responses and justify your choice

(c) If using a second-order topology, calculate the component values for unity-gain implementation

(d) Estimate the phase lag at 20 Hz

Exercise 2.19. Computational: For the series RLC circuit with $R = 10\ \Omega$, $L = 1$ H, and $C = 1$ μF:

(a) Find the resonant frequency

(b) Calculate the quality factor Q

(c) Find the magnitude of impedance at resonance, at $\omega = \omega_n/2$, and at $2\omega_n$

(d) If driven by $v_{\text{in}}(t) = 100\sin(\omega_n t)$ V, find the steady-state current amplitude at resonance

Exercise 2.20. Concept: The PID controller can be implemented using three op-amp stages (one proportional, one integrator, one differentiator).

(a) Draw a block diagram showing how three op-amp circuits combine to form a PID controller

(b) Explain the role of each stage in controlling a second-order system

(c) Discuss practical issues with the differentiator stage (e.g., noise) and propose solutions

2.12 Electromechanical Systems

Electromechanical systems convert between electrical and mechanical energy. They are ubiquitous in control applications—from servo motors in robots to voice coils in hard drives.

2.12.1 DC Motor Fundamentals

The DC motor is the workhorse of motion control, converting electrical power to rotational mechanical power.

Historical Context: Faraday's Electromagnetic Induction and the Motor-Generator Principle

The DC motor's operation is rooted in Michael Faraday's 1831 discovery of electromagnetic induction—that a changing magnetic flux induces an electric current. Faraday's insight that electricity and magnetism were fundamentally related led to the realization that the inverse was also true: a current-carrying conductor in a magnetic field experiences a force. This force-current relationship, quantified as $F = BIL$ (where B is magnetic flux density and L is conductor length), became the principle underlying the DC motor. Early motors (1830s-1890s) were crude devices driven by this principle, but lacked practical commutation until the development of split-ring commutators.

The fierce competition between Thomas Edison's direct current (DC) and Nikola Tesla's alternating current (AC) systems (the "War of Currents," 1880s-1890s) hinged largely on motor capabilities. DC motors' superior controllability made them ideal for variable-speed applications and industrial drives, while AC motors' simplicity eventually won for grid applications. The DC motor's ease of control—coupled with its simple mathematical model involving back-EMF and torque constants—made it the canonical example for teaching control theory, a role it maintains to this day even as brushless motors have replaced traditional commutators in many applications.

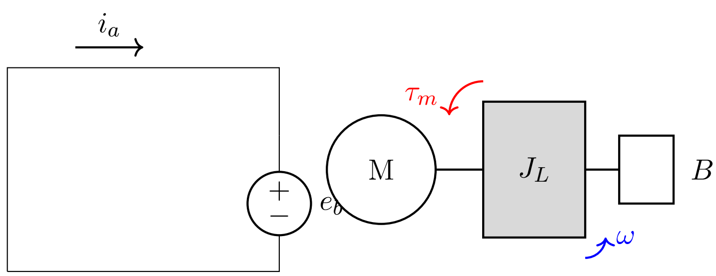


Figure 2.31: DC motor equivalent circuit and mechanical load.

Governing Equations

Electrical equation (armature circuit):

$$V_a = R_a i_a + L_a \frac{\mathrm{d}i_a}{\mathrm{d}t} + e_b \tag{2.106}$$

Back-EMF (proportional to speed):

$$e_b = K_e \omega = K_e \dot{\theta} \tag{2.107}$$

Motor torque (proportional to current):

$$\tau_m = K_t i_a \tag{2.108}$$

Mechanical equation:

$$J\ddot{\theta} = \tau_m - B\dot{\theta} - \tau_L = K_t i_a - B\omega - \tau_L \tag{2.109}$$

where:

- V_a = armature voltage (V)
- R_a = armature resistance (Ω)
- L_a = armature inductance (H)
- i_a = armature current (A)
- e_b = back-EMF (V)
- K_e = back-EMF constant (V·s/rad)
- K_t = torque constant (N·m/A)
- J = total inertia (kg·m^2)
- B = viscous friction (N·m·s/rad)
- $\omega = \dot{\theta}$ = angular velocity (rad/s)

Motor Constants

For an ideal DC motor: $K_t = K_e$ (in SI units). This follows from energy conservation—electrical power in equals mechanical power out.

Transfer Functions

Taking Laplace transforms:

$$V_a(s) = (R_a + L_a s)I_a(s) + K_e s\Theta(s) \tag{2.110}$$

$$Js^2\Theta(s) = K_t I_a(s) - Bs\Theta(s) - \tau_L(s) \tag{2.111}$$

Voltage to angular velocity (with $\tau_L = 0$):

$$\frac{\Omega(s)}{V_a(s)} = \frac{K_t}{(L_a s + R_a)(Js + B) + K_t K_e} \tag{2.112}$$

Voltage to angular position:

$$\frac{\Theta(s)}{V_a(s)} = \frac{K_t}{s[(L_a s + R_a)(Js + B) + K_t K_e]} \tag{2.113}$$

Simplified Model (Neglecting Inductance)

For many motors, the electrical time constant $\tau_e = L_a/R_a$ is much smaller than the mechanical time constant $\tau_m = J/B$. Setting $L_a \approx 0$:

$$\frac{\Omega(s)}{V_a(s)} = \frac{K_t/R_a}{Js + B + K_t K_e/R_a} = \frac{K_m}{\tau_m s + 1} \tag{2.114}$$

where:

$$K_m = \frac{K_t}{R_a B + K_t K_e} = \frac{K_t}{K_t K_e + R_a B} \quad \text{(motor gain, rad/s/V)} \tag{2.115}$$

$$\tau_m = \frac{R_a J}{R_a B + K_t K_e} \quad \text{(mechanical time constant)} \tag{2.116}$$

Example 1.8: DC Motor Analysis

A DC motor has the following parameters:

- $R_a = 2\ \Omega$, $L_a = 0.5$ mH
- $K_t = K_e = 0.1$ N·m/A $= 0.1$ V·s/rad
- $J = 0.01$ kg·m^2, $B = 0.001$ N·m·s/rad

Find the transfer function and steady-state speed for $V_a = 24$ V.
Solution:
Electrical time constant: $\tau_e = L_a/R_a = 0.0005/2 = 0.25$ ms
Since τ_e is small, use the simplified model:

$$K_m = \frac{0.1}{2 \times 0.001 + 0.1 \times 0.1} = \frac{0.1}{0.012} = 8.33 \text{ rad/s/V} \tag{2.117}$$

$$\tau_m = \frac{2 \times 0.01}{0.012} = 1.67 \text{ s} \tag{2.118}$$

Transfer function:

$$\frac{\Omega(s)}{V_a(s)} = \frac{8.33}{1.67s + 1} \tag{2.119}$$

Steady-state speed at 24V:

$$\omega_{ss} = K_m \times V_a = 8.33 \times 24 = 200 \text{ rad/s} = 1910 \text{ rpm} \tag{2.120}$$

2.12.2 DC Motor with Position Control

For position control applications, we need the voltage-to-position transfer function:

$$\frac{\Theta(s)}{V_a(s)} = \frac{K_m/s}{\tau_m s + 1} = \frac{K_m}{s(\tau_m s + 1)} \tag{2.121}$$

This is a Type 1 system (one free integrator), which can track step position commands with zero steady-state error.

State-Space Model

Choosing states $\mathbf{x} = [\theta, \omega, i_a]^\top$:

$$\begin{bmatrix} \dot{\theta} \\ \dot{\omega} \\ \dot{i}_a \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -B/J & K_t/J \\ 0 & -K_e/L_a & -R_a/L_a \end{bmatrix} \begin{bmatrix} \theta \\ \omega \\ i_a \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1/L_a \end{bmatrix} V_a \tag{2.122}$$

Output (position): $y = \theta = [1, 0, 0]\mathbf{x}$

2.12.3 Brushless DC Motors (BLDC)

Brushless DC motors use electronic commutation instead of mechanical brushes. The control-relevant model is similar to the brushed DC motor:

$$\frac{\Omega(s)}{V_q(s)} = \frac{K_t/R}{(L/R)Js^2 + Js + K_tK_e/R} \tag{2.123}$$

where V_q is the q-axis voltage in the field-oriented control frame.

2.12.4 Stepper Motors

Stepper motors move in discrete angular steps, typically 1.8° (200 steps/revolution).

Open-loop position (idealized):

$$\theta = N_{\text{steps}} \times \theta_{\text{step}} \tag{2.124}$$

Dynamic model (simplified single phase):

$$J\ddot{\theta} + B\dot{\theta} + K_d \sin(N_r\theta) = K_t i \tag{2.125}$$

where N_r is the number of rotor teeth and K_d is the detent torque constant.

2.12.5 Voice Coil Actuators

Voice coils provide linear motion and are used in hard drives, loudspeakers, and precision positioning.

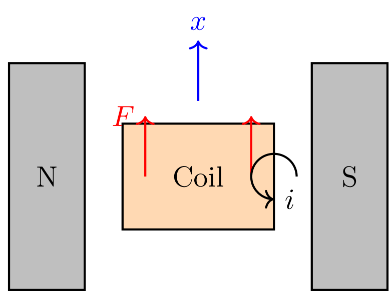


Figure 2.32: Voice coil actuator schematic.

Governing equations:

$$V = Ri + L\frac{\mathrm{d}i}{\mathrm{d}t} + K_e\dot{x} \tag{2.126}$$

$$F = K_f i = m\ddot{x} + b\dot{x} \tag{2.127}$$

Transfer function (voltage to position):

$$\frac{X(s)}{V(s)} = \frac{K_f}{s[(Ls + R)(ms + b) + K_fK_e]} \tag{2.128}$$

2.12.6 Solenoids

Solenoids provide on/off linear actuation. The force depends nonlinearly on position:

$$F = \frac{1}{2}i^2\frac{\mathrm{d}L(x)}{\mathrm{d}x} \tag{2.129}$$

where $L(x)$ is the position-dependent inductance.

Linearized about operating point (x_0, i_0):

$$\Delta F = K_i\Delta i + K_x\Delta x \tag{2.130}$$

where K_i and K_x are partial derivatives of force with respect to current and position.

2.12.7 Piezoelectric Actuators

Piezoelectric actuators provide very small displacements (nm to μm) with high force and fast response.

Constitutive equations:

$$x = d_{33}V + s_{33}F/A \tag{2.131}$$
$$Q = CV + d_{33}F \tag{2.132}$$

where:

- d_{33} = piezoelectric coefficient (m/V or C/N)
- s_{33} = elastic compliance (m^2/N)
- C = capacitance (F)

Transfer function (simplified, free displacement):

$$\frac{X(s)}{V(s)} = \frac{d_{33}\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \tag{2.133}$$

2.12.8 Generators and Regeneration

When a motor is driven faster than its no-load speed, or when decelerating a load, it acts as a generator:

$$P_{\text{regen}} = e_b \cdot i_a = K_e\omega \cdot i_a \tag{2.134}$$

This regenerated power can be:

- Dissipated in braking resistors
- Returned to the power supply (regenerative braking)
- Stored in capacitors or batteries

2.12.9 Motor Selection Example

Example 1.9: Robot Joint Motor Selection

A robot joint must:

- Move a load with inertia $J_L = 0.05$ kg·m^2
- Achieve maximum velocity $\omega_{\text{max}} = 10$ rad/s
- Achieve maximum acceleration $\alpha_{\text{max}} = 50$ rad/s^2

Select an appropriate motor and gear ratio.
Solution:
Required torque at load:

$$\tau_L = J_L\alpha_{\text{max}} = 0.05 \times 50 = 2.5 \text{ N·m} \tag{2.135}$$

Consider a motor with:

- Rated torque: $\tau_m = 0.3$ N·m
- Rated speed: 3000 rpm = 314 rad/s
- Rotor inertia: $J_m = 5 \times 10^{-5}$ kg·m^2

Required gear ratio (from torque):

$$n \geq \frac{\tau_L}{\tau_m} = \frac{2.5}{0.3} = 8.3 \tag{2.136}$$

Check speed (with $n = 10$):

$$\omega_m = n \cdot \omega_L = 10 \times 10 = 100 \text{ rad/s} < 314 \text{ rad/s} \quad \checkmark \tag{2.137}$$

Reflected inertia check:

$$J_{\text{reflected}} = \frac{J_L}{n^2} = \frac{0.05}{100} = 5 \times 10^{-4} \text{ kg·m}^2 \tag{2.138}$$

Total inertia at motor: $J_{\text{total}} = J_m + J_{\text{reflected}} = 5.5 \times 10^{-4}$ kg·m^2
Motor acceleration:

$$\alpha_m = \frac{\tau_m}{J_{\text{total}}} = \frac{0.3}{5.5 \times 10^{-4}} = 545 \text{ rad/s}^2 \tag{2.139}$$

Load acceleration achieved:

$$\alpha_L = \frac{\alpha_m}{n} = \frac{545}{10} = 54.5 \text{ rad/s}^2 > 50 \text{ rad/s}^2 \quad \checkmark \tag{2.140}$$

The motor with 10:1 gear ratio meets all requirements.

2.12.10 Exercises: Electromechanical Systems

Exercise 2.21. A DC motor has $R_a = 1\ \Omega$, $K_t = K_e = 0.05$ N·m/A, $J = 2 \times 10^{-4}$ kg·m^2, and $B = 10^{-4}$ N·m·s/rad. Neglect inductance.

(a) Derive the transfer function from voltage to velocity

(b) Find the steady-state speed for $V_a = 12$ V with no load

(c) Find the stall torque (torque at zero speed) for $V_a = 12$ V

Exercise 2.22. For the motor in Exercise 1, a 5:1 gear reducer is added, and the load inertia is $J_L = 0.01$ kg·m^2 with load damping $B_L = 0.01$ N·m·s/rad.

(a) Find the reflected load inertia and damping at the motor

(b) Derive the new transfer function from voltage to load velocity

(c) Compare the new time constant to the original

Exercise 2.23. A voice coil actuator has $R = 5\ \Omega$, $L = 1$ mH, $K_f = K_e = 10$ N/A, $m = 0.1$ kg, and $b = 2$ N·s/m.

(a) Derive the transfer function from voltage to position

(b) Find all poles and determine if the system is stable

(c) Calculate the DC gain

Exercise 2.24. A piezoelectric actuator has $d_{33} = 500$ pm/V, resonant frequency 30 kHz, and damping ratio 0.02.

(a) Write the transfer function

(b) Find the static displacement for 100V input

(c) Find the displacement amplitude at 20 kHz for 100V sinusoidal input

Exercise 2.25. Design: A servo motor must position a robotic arm joint with:

- Load inertia: $J_L = 0.02$ kg·m^2
- Load damping: $B_L = 0.05$ N·m·s/rad
- Desired bandwidth: 20 rad/s
- Maximum allowed steady-state error: 0.01 rad

(a) Select a motor with appropriate torque and speed ratings

(b) Choose a gear ratio to match the bandwidth requirement

(c) Design a proportional feedback controller to meet the steady-state error specification

(d) Estimate the settling time (2% criterion)

Exercise 2.26. Concept: Explain the role of back-EMF in DC motor speed regulation.

(a) Derive how back-EMF creates a speed-dependent resistance to applied voltage

(b) Show that steady-state speed increases linearly with applied voltage

(c) Explain why increased load torque causes the motor to slow down

(d) Discuss how feedback control can maintain constant speed despite load variations

Exercise 2.27. A stepper motor with 200 steps/revolution (1.8° per step) drives a load with moment of inertia $J = 10^{-4}$ kg·m^2 and friction damping $B = 10^{-5}$ N·m·s/rad.

(a) Calculate the motor's step angle in radians

(b) If stepped at 1000 Hz, what is the steady-state rotational velocity?

(c) Calculate the acceleration at a 1000 rad/s^2 acceleration specification

(d) Explain the hunting and resonance issues in stepper motor control

Exercise 2.28. Computational: For a brushless DC motor (BLDC) with field-oriented control:

(a) Derive the relationship between q-axis voltage and motor torque

(b) Show that the motor behaves like a first-order system in the current loop

(c) Calculate the current loop bandwidth for $L = 5$ mH, $R = 1\ \Omega$, and proportional gain $K_p = 10$

(d) Find the settling time for a 10A step reference current

Exercise 2.29. Concept: Compare direct drive motors versus geared motors for a positioning application.

(a) List advantages and disadvantages of each approach

(b) Show how gear ratio affects reflected inertia and bandwidth

(c) Explain the tradeoff between speed and torque

(d) In what applications would you choose each approach?

2.13 Thermal and Fluid Systems

Thermal and fluid systems are governed by energy and mass conservation. These systems often exhibit first-order dynamics with relatively slow time constants, making them excellent candidates for feedback control.

2.13.1 Thermal Systems Fundamentals

Heat Transfer Mechanisms

Conduction (through solids):

$$\dot{Q} = \frac{kA}{L}(T_1 - T_2) = \frac{T_1 - T_2}{R_{th}} \tag{2.141}$$

where $R_{th} = L/(kA)$ is the thermal resistance.
Convection (between solid and fluid):

$$\dot{Q} = hA(T_s - T_\infty) \tag{2.142}$$

where h is the convection coefficient (W/m^{2}K).
Radiation:

$$\dot{Q} = \epsilon\sigma A(T_1^4 - T_2^4) \tag{2.143}$$

For small temperature differences, linearized: $\dot{Q} \approx h_r A(T_1 - T_2)$ where $h_r = 4\epsilon\sigma T_{\text{avg}}^3$.

Thermal Capacitance

The energy stored in a body:

$$E = mC_pT = \rho V C_p T \tag{2.144}$$

Rate of temperature change:

$$mC_p\frac{\mathrm{d}T}{\mathrm{d}t} = \dot{Q}_{\text{in}} - \dot{Q}_{\text{out}} \tag{2.145}$$

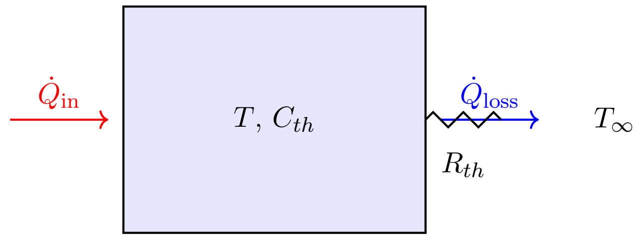


Figure 2.33: Single-capacitance thermal system.

2.13.2 Single-Capacitance Thermal System

Energy balance:

$$C_{th}\frac{\mathrm{d}T}{\mathrm{d}t} = \dot{Q}_{\text{in}} - \frac{T - T_\infty}{R_{th}} \tag{2.146}$$

where $C_{th} = mC_p$ is the thermal capacitance.
Rearranging:

$$\boxed{R_{th}C_{th}\frac{\mathrm{d}T}{\mathrm{d}t} + T = R_{th}\dot{Q}_{\text{in}} + T_\infty} \tag{2.147}$$

Transfer Function

Defining $\theta = T - T_\infty$ (temperature above ambient):

$$G(s) = \frac{\Theta(s)}{\dot{Q}_{\text{in}}(s)} = \frac{R_{th}}{\tau s + 1} \tag{2.148}$$

where $\tau = R_{th}C_{th}$ is the thermal time constant.

Example 1.10: Room Heating

A room has thermal capacitance $C_{th} = 500$ kJ/K and thermal resistance to outdoors $R_{th} = 0.01$ K/W. Find:

(a) The thermal time constant

(b) The heater power needed to maintain 20°C when outdoor temperature is 0°C

(c) The room temperature 1 hour after turning off the heater

Solution:
(a) Time constant:

$$\tau = R_{th}C_{th} = 0.01 \times 500{,}000 = 5000 \text{ s} = 83.3 \text{ min} \tag{2.149}$$

(b) Steady-state heat balance:

$$\dot{Q}_{\text{in}} = \frac{T - T_\infty}{R_{th}} = \frac{20 - 0}{0.01} = 2000 \text{ W} = 2 \text{ kW} \tag{2.150}$$

(c) After heater off (cooling from 20°C):

$$T(t) = T_\infty + (T_0 - T_\infty)\mathrm{e}^{-t/\tau} = 0 + 20\mathrm{e}^{-3600/5000} = 9.7\text{C} \tag{2.151}$$

2.13.3 Multi-Capacitance Thermal Systems

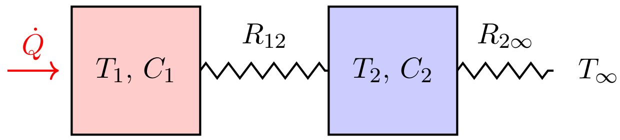


Figure 2.34: Two-capacitance thermal system.

Energy balances:

$$C_1 \frac{\mathrm{d}T_1}{\mathrm{d}t} = \dot{Q} - \frac{T_1 - T_2}{R_{12}} \tag{2.152}$$

$$C_2 \frac{\mathrm{d}T_2}{\mathrm{d}t} = \frac{T_1 - T_2}{R_{12}} - \frac{T_2 - T_\infty}{R_{2\infty}} \tag{2.153}$$

This yields a second-order system with two thermal time constants.

2.13.4 Fluid Systems: Tank Level

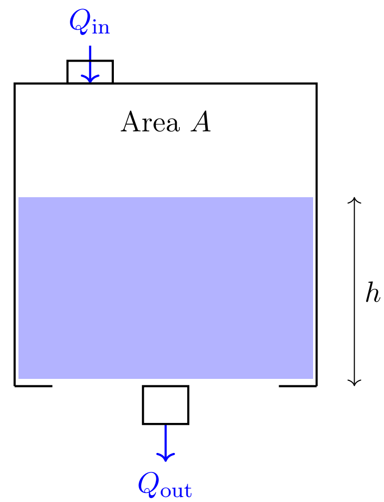


Figure 2.35: Single tank level system.

Mass Balance

For incompressible fluid:

$$A \frac{\mathrm{d}h}{\mathrm{d}t} = Q_{\text{in}} - Q_{\text{out}} \tag{2.154}$$

Outflow Models

Linear resistance (laminar flow through valve):

$$Q_{\text{out}} = \frac{h}{R_f} \tag{2.155}$$

where R_f is the fluid resistance.
Nonlinear resistance (turbulent flow, orifice):

$$Q_{\text{out}} = C_d A_o \sqrt{2gh} \tag{2.156}$$

This is nonlinear but can be linearized about an operating point.

Linear Tank Model

With linear resistance:

$$A\frac{\mathrm{d}h}{\mathrm{d}t} + \frac{h}{R_f} = Q_{\text{in}} \tag{2.157}$$

Transfer function:

$$G(s) = \frac{H(s)}{Q_{\text{in}}(s)} = \frac{R_f}{\tau s + 1} \tag{2.158}$$

where $\tau = AR_f$ is the hydraulic time constant.

Example 1.11: Water Tank

A cylindrical tank has diameter 2 m and drains through a valve with resistance $R_f = 100$ s/m^2. Find:

(a) The time constant

(b) Steady-state level for inflow of 0.01 m^3/s

(c) Time to reach 90% of steady-state from empty

Solution:
Tank area: $A = \pi(1)^2 = 3.14$ m^2
(a) Time constant:

$$\tau = AR_f = 3.14 \times 100 = 314 \text{ s} = 5.2 \text{ min} \tag{2.159}$$

(b) Steady-state level:

$$h_{ss} = R_f \times Q_{\text{in}} = 100 \times 0.01 = 1 \text{ m} \tag{2.160}$$

(c) Time to 90%:

$$0.9 = 1 - \mathrm{e}^{-t/\tau} \implies t = -\tau \ln(0.1) = 314 \times 2.3 = 723 \text{ s} = 12 \text{ min} \tag{2.161}$$

2.13.5 Interacting Tank Systems

Mass balances:

$$A_1\frac{\mathrm{d}h_1}{\mathrm{d}t} = Q_{\text{in}} - \frac{h_1 - h_2}{R_1} \tag{2.162}$$

$$A_2\frac{\mathrm{d}h_2}{\mathrm{d}t} = \frac{h_1 - h_2}{R_1} - \frac{h_2}{R_2} \tag{2.163}$$

This is a second-order system. The interaction between tanks creates coupling that affects the system dynamics.

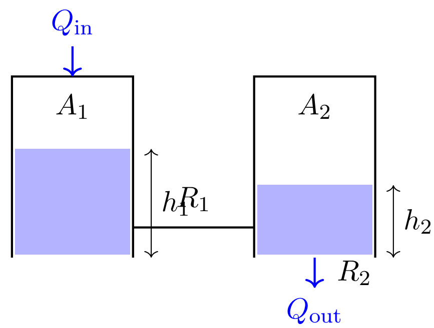


Figure 2.36: Two interacting tanks.

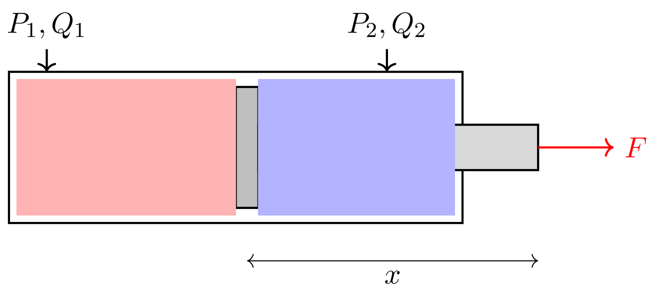


Figure 2.37: Double-acting hydraulic cylinder.

2.13.6 Hydraulic Actuators

Flow Continuity

For incompressible fluid:

$$Q_1 = A_1 \dot{x} + \frac{V_1}{\beta} \dot{P}_1 \tag{2.164}$$

where β is the bulk modulus.

Force Balance

$$M\ddot{x} = P_1 A_1 - P_2 A_2 - F_{\text{load}} - B\dot{x} \tag{2.165}$$

Servo Valve Model

Flow through a servo valve:

$$Q = C_d w x_v \sqrt{\frac{2(P_s - P_L)}{\rho}} \tag{2.166}$$

where x_v is the valve spool position.
Linearized about operating point:

$$Q = K_q x_v - K_c P_L \tag{2.167}$$

2.13.7 Pneumatic Systems

Pneumatic systems use compressible gas (usually air), adding complexity:

Ideal gas law:

$$PV = mRT \tag{2.168}$$

Mass flow through orifice:

For subsonic flow ($P_2/P_1 > 0.528$):

$$\dot{m} = C_d A P_1 \sqrt{\frac{2}{RT_1}} \sqrt{\frac{\gamma}{\gamma - 1}\left[\left(\frac{P_2}{P_1}\right)^{2/\gamma} - \left(\frac{P_2}{P_1}\right)^{(\gamma+1)/\gamma}\right]} \tag{2.169}$$

For choked flow ($P_2/P_1 \leq 0.528$):

$$\dot{m} = C_d A P_1 \sqrt{\frac{\gamma}{RT_1}} \left(\frac{2}{\gamma + 1}\right)^{(\gamma+1)/(2(\gamma-1))} \tag{2.170}$$

Pneumatic Cylinder

Pressure dynamics in chamber:

$$\frac{\mathrm{d}P}{\mathrm{d}t} = \frac{\gamma RT}{V}(\dot{m}_{\text{in}} - \dot{m}_{\text{out}}) - \frac{\gamma P}{V}\dot{V} \tag{2.171}$$

2.13.8 Heat Exchangers

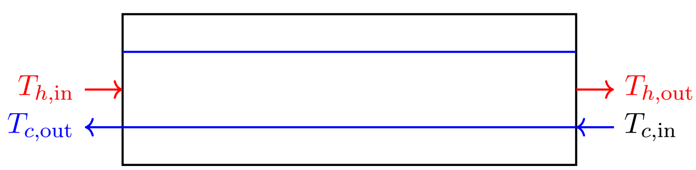


Figure 2.38: Counter-flow heat exchanger.

Simplified lumped model:

$$C_h \frac{\mathrm{d}T_h}{\mathrm{d}t} = \dot{m}_h c_{p,h}(T_{h,\text{in}} - T_h) - UA(T_h - T_c) \tag{2.172}$$

$$C_c \frac{\mathrm{d}T_c}{\mathrm{d}t} = \dot{m}_c c_{p,c}(T_{c,\text{in}} - T_c) + UA(T_h - T_c) \tag{2.173}$$

where UA is the overall heat transfer coefficient times area.

2.13.9 Thermal Resistance Networks

Network Analysis

For **series resistances**, the total thermal resistance equals the sum:

$$R_{\text{total}} = R_1 + R_2 + R_3 + \cdots \tag{2.174}$$

The heat flow through all resistances is identical (same current). The temperature difference across each resistance is proportional to its value.

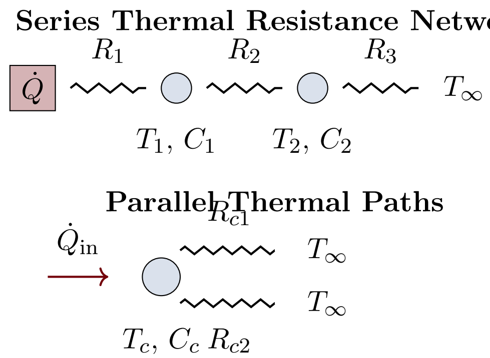


Figure 2.39: Thermal resistance networks showing series (top) and parallel (bottom) heat flow paths with thermal capacitances at nodes.

For **parallel resistances**, the equivalent resistance is:

$$\frac{1}{R_{\text{eq}}} = \frac{1}{R_{c1}} + \frac{1}{R_{c2}} + \cdots \tag{2.175}$$

Each path dissipates heat proportionally to the inverse of its resistance. Parallel paths reduce the effective resistance, increasing heat flow.

2.13.10 Two-Tank Fluid System Dynamics

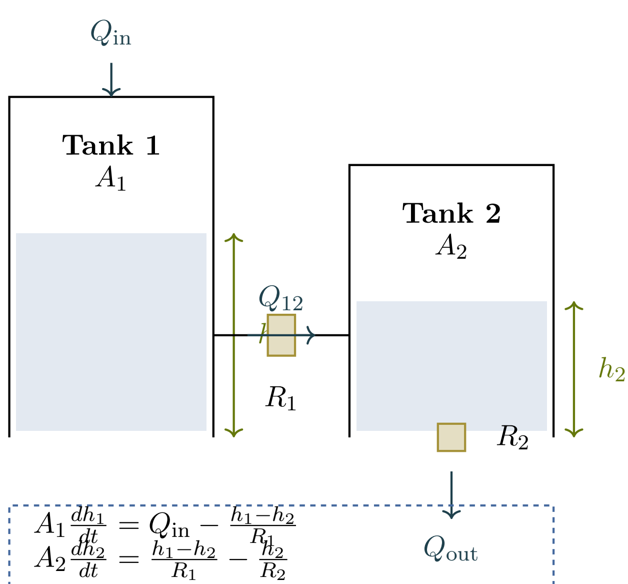


Figure 2.40: Two-tank fluid system showing mass conservation, inter-tank flow through resistance R_1, and outflow through R_2. Shaded regions indicate liquid level.

This configuration creates a second-order system where tank interactions cause non-minimum-phase behavior. The inter-tank resistance R_1 introduces coupling between the tanks.

2.13.11 Exercises: Thermal and Fluid Systems

Exercise 2.30. A metal block ($m = 5$ kg, $C_p = 500$ J/kg·K) is heated by a 100 W heater and loses heat to 25°C ambient through convection ($hA = 5$ W/K).

(a) Write the differential equation for block temperature

(b) Find the thermal time constant

(c) Find the steady-state temperature

(d) How long to reach 90% of steady-state from ambient?

Exercise 2.31. Thermal Resistance Network: A system has two thermal resistances in series: $R_1 = 0.5$ K/W and $R_2 = 0.8$ K/W, with a heat source $\dot{Q} = 500$ W and ambient temperature $T_\infty = 20$°C. The total thermal capacitance is $C_{th} = 10$ kJ/K.

(a) Find the total thermal resistance

(b) Find the steady-state temperature rise above ambient

(c) If the system starts at ambient, find the temperature after 100 seconds

Exercise 2.32. Two tanks in series each have $A = 1$ m^2. The first tank drains into the second through resistance $R_1 = 50$ s/m^2, and the second drains to atmosphere through $R_2 = 100$ s/m^2.

(a) Write the state-space model

(b) Find the transfer function from Q_{in} to h_2

(c) Find the system poles and time constants

(d) Explain how the inter-tank interaction affects the response

Exercise 2.33. Tank Level Dynamics: A conical tank has height-dependent area: $A(h) = \pi(rh/H)^2$ where $r = 0.5$ m is base radius and $H = 2$ m is total height. The outlet has resistance $R_f = 80$ s/m^2.

(a) Write the nonlinear ODE for level h

(b) For $h = 1$ m at steady-state with constant inflow, find Q_{in}

(c) Linearize the model about this operating point

Exercise 2.34. A hydraulic servo system has a 10 cm diameter piston, 1000 kg load mass, and bulk modulus $\beta = 1.5 \times 10^9$ Pa. The chamber volume is 0.5 L.

(a) Calculate the hydraulic natural frequency

(b) If damping ratio is 0.3, find the damping coefficient

(c) Sketch the expected step response

Exercise 2.35. Heat Exchanger Thermal Dynamics: A counter-flow heat exchanger receives hot fluid at 80°C (inlet) with flow rate 0.5 kg/s, and cold fluid at 20°C with flow rate 0.3 kg/s. The heat transfer coefficient is $UA = 2$ kW/K, and thermal capacitances are $C_h = C_c = 5$ kJ/K.

(a) Write the energy balance equations for both sides

(b) Find the steady-state outlet temperatures

(c) Determine the time constants of the hot and cold sides

Exercise 2.36. A hot water tank ($V = 200$ L) receives water at 15°C at rate 0.1 L/s and has a 3 kW heater. Heat loss to ambient is $UA = 10$ W/K. Find the steady-state temperature and time constant. What is the response time to reach 95% of steady-state?

2.14 Chemical Systems

Chemical processes involve mass and energy balances coupled with reaction kinetics. These systems often exhibit nonlinear behavior, time delays, and complex interactions, making them challenging but important control applications.

2.14.1 Continuous Stirred-Tank Reactor (CSTR)

The CSTR is a fundamental unit in chemical engineering and a classic control problem.

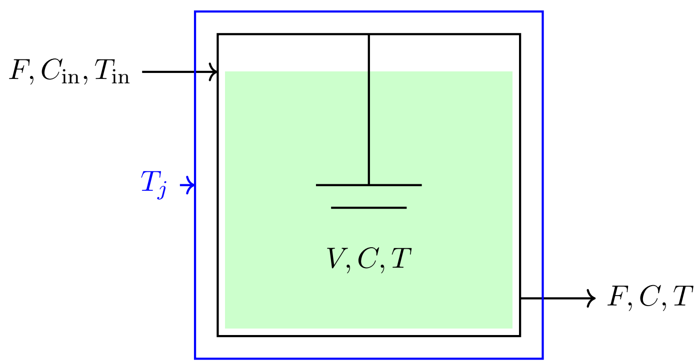


Figure 2.41: Continuous stirred-tank reactor with cooling jacket.

Assumptions

- Perfect mixing (uniform concentration and temperature)
- Constant volume V and density ρ
- Constant flow rate F (in = out)
- Single first-order irreversible reaction: $A \to B$

Component Mass Balance

$$V \frac{\mathrm{d}C_A}{\mathrm{d}t} = F(C_{A,\text{in}} - C_A) - Vk(T)C_A \tag{2.176}$$

where $k(T)$ is the temperature-dependent reaction rate constant.

Arrhenius Equation

$$k(T) = k_0 \exp\left(-\frac{E_a}{RT}\right) \tag{2.177}$$

where:

- k_0 = pre-exponential factor (1/s)
- E_a = activation energy (J/mol)
- R = universal gas constant (8.314 J/mol·K)

Energy Balance

$$\rho V C_p \frac{\mathrm{d}T}{\mathrm{d}t} = F\rho C_p(T_{\text{in}} - T) + (-\Delta H_r)Vk(T)C_A - UA(T - T_j) \tag{2.178}$$

where:

- ΔH_r = heat of reaction (J/mol, negative for exothermic)
- UA = heat transfer coefficient times area (W/K)
- T_j = jacket temperature (K)

Linearized Model

Linearizing about steady-state (C_{A0}, T_0):

$$\frac{\mathrm{d}\Delta C_A}{\mathrm{d}t} = a_{11}\Delta C_A + a_{12}\Delta T + b_1 \Delta C_{A,\text{in}} \tag{2.179}$$

$$\frac{\mathrm{d}\Delta T}{\mathrm{d}t} = a_{21}\Delta C_A + a_{22}\Delta T + b_2 \Delta T_j \tag{2.180}$$

State-space form:

$$\begin{bmatrix} \Delta \dot{C}_A \\ \Delta \dot{T} \end{bmatrix} = \mathbf{A} \begin{bmatrix} \Delta C_A \\ \Delta T \end{bmatrix} + \mathbf{B} \begin{bmatrix} \Delta C_{A,\text{in}} \\ \Delta T_j \end{bmatrix} \tag{2.181}$$

Example 1.12: CSTR Linearization

A CSTR operates at steady-state with:

- $V = 1\ \text{m}^3$, $F = 0.1\ \text{m}^3/\text{min}$
- $C_{A0} = 2\ \text{mol/L}$, $T_0 = 350\ \text{K}$
- $k_0 = 10^{10}\ \text{min}^{-1}$, $E_a/R = 8750\ \text{K}$

The linearized **A** matrix elements are:

$$a_{11} = -\frac{F}{V} - k(T_0) = -0.1 - k_0 \mathrm{e}^{-8750/350} = -0.1 - 1.2 = -1.3\ \text{min}^{-1} \tag{2.182}$$

$$a_{12} = -C_{A0} \left.\frac{\mathrm{d}k}{\mathrm{d}T}\right|_{T_0} = -C_{A0}k(T_0)\frac{E_a}{RT_0^2} = -0.17\ \text{mol/L·K·min} \tag{2.183}$$

The negative a_{12} indicates that increased temperature increases reaction rate, consuming more C_A.

2.14.2 Mixing Tank

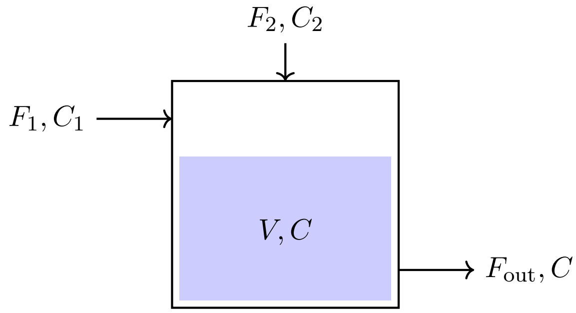


Figure 2.42: Mixing tank with two inlets.

Mass Balance (Total)

$$\frac{\mathrm{d}(\rho V)}{\mathrm{d}t} = \rho_1 F_1 + \rho_2 F_2 - \rho F_{\text{out}} \tag{2.184}$$

For constant density and volume: $F_{\text{out}} = F_1 + F_2$

Component Balance

$$V\frac{\mathrm{d}C}{\mathrm{d}t} = F_1 C_1 + F_2 C_2 - (F_1 + F_2)C \tag{2.185}$$

Transfer function (for constant F_1, F_2):

$$\frac{C(s)}{C_1(s)} = \frac{F_1/V}{s + (F_1 + F_2)/V} = \frac{F_1/(F_1 + F_2)}{\tau s + 1} \tag{2.186}$$

where $\tau = V/(F_1 + F_2)$ is the residence time.

2.14.3 pH Neutralization

pH control is notoriously difficult due to the highly nonlinear relationship between acid/base concentration and pH.

$$\text{pH} = -\log_{10}[H^+] \tag{2.187}$$

For a strong acid-strong base system:

$$[H^+] = \frac{C_a - C_b + \sqrt{(C_a - C_b)^2 + 4K_w}}{2} \tag{2.188}$$

where $K_w = 10^{-14}$ at 25°C.

Titration curve: The pH changes dramatically near the equivalence point, creating a high-gain nonlinearity.

2.14.4 Distillation Column

Distillation separates mixtures based on volatility differences.

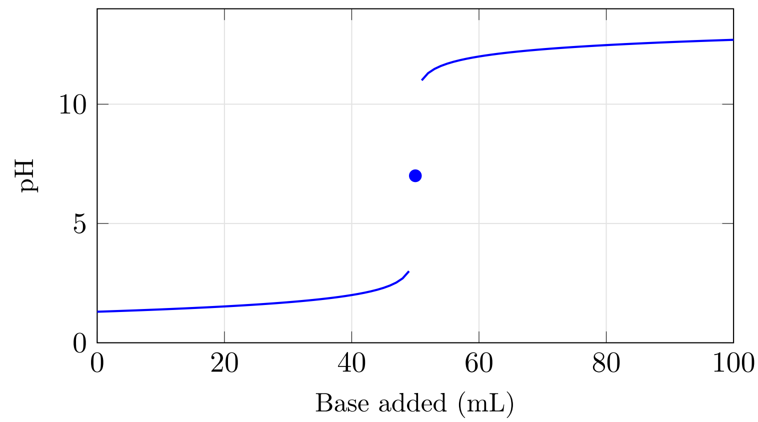


Figure 2.43: Typical pH titration curve showing high gain near equivalence.

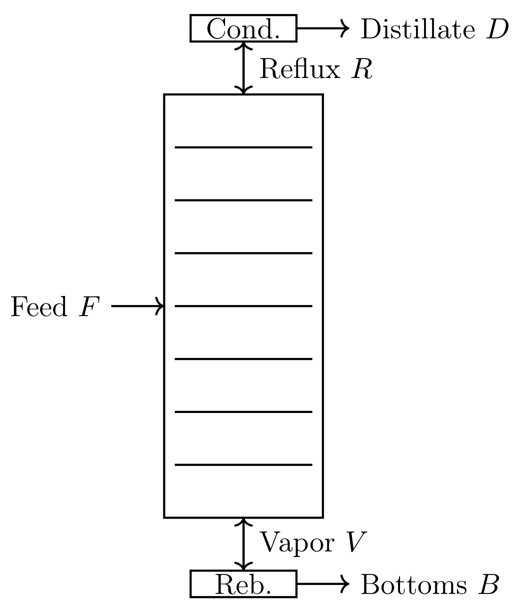


Figure 2.44: Distillation column schematic.

Simplified Model

For a binary mixture, the composition dynamics on tray n:

$$M_n \frac{\mathrm{d}x_n}{\mathrm{d}t} = L_{n-1}x_{n-1} + V_{n+1}y_{n+1} - L_n x_n - V_n y_n \tag{2.189}$$

where:

- M_n = liquid holdup on tray
- L_n = liquid flow rate
- V_n = vapor flow rate
- x_n = liquid composition (mole fraction)
- y_n = vapor composition

Vapor-liquid equilibrium:

$$y_n = \frac{\alpha x_n}{1 + (\alpha - 1)x_n} \tag{2.190}$$

where α is the relative volatility.

2.14.5 Batch Reactor

Unlike continuous reactors, batch reactors have no flow during reaction:

$$\frac{\mathrm{d}C_A}{\mathrm{d}t} = -k(T)C_A \tag{2.191}$$

$$\frac{\mathrm{d}T}{\mathrm{d}t} = \frac{(-\Delta H_r)k(T)C_A}{\rho C_p} - \frac{UA(T - T_j)}{\rho V C_p} \tag{2.192}$$

The concentration decreases exponentially (for first-order kinetics):

$$C_A(t) = C_{A0} \exp\left(-\int_0^t k(T(\tau))\,\mathrm{d}\tau\right) \tag{2.193}$$

2.14.6 Reaction Kinetics

Zero-Order Reaction

$$\frac{\mathrm{d}C}{\mathrm{d}t} = -k \tag{2.194}$$

Solution: $C(t) = C_0 - kt$

First-Order Reaction

$$\frac{\mathrm{d}C}{\mathrm{d}t} = -kC \tag{2.195}$$

Solution: $C(t) = C_0 \mathrm{e}^{-kt}$
Half-life: $t_{1/2} = \ln(2)/k$

Second-Order Reaction

$$\frac{\mathrm{d}C}{\mathrm{d}t} = -kC^2 \tag{2.196}$$

Solution: $\frac{1}{C(t)} = \frac{1}{C_0} + kt$

Enzyme Kinetics (Michaelis-Menten)

$$\frac{\mathrm{d}C}{\mathrm{d}t} = -\frac{V_{\max}C}{K_m + C} \tag{2.197}$$

where $V_{\max}$ is maximum rate and K_m is the Michaelis constant.

2.14.7 Process Dead Time

Many chemical processes exhibit significant time delays (dead time, transport lag):

$$y(t) = u(t - \theta) \tag{2.198}$$

Transfer function:

$$G(s) = \mathrm{e}^{-\theta s} \tag{2.199}$$

Dead time makes control more challenging. Common approximations:
First-order Padé:

$$\mathrm{e}^{-\theta s} \approx \frac{1 - \theta s/2}{1 + \theta s/2} \tag{2.200}$$

Second-order Padé:

$$\mathrm{e}^{-\theta s} \approx \frac{1 - \theta s/2 + (\theta s)^2/12}{1 + \theta s/2 + (\theta s)^2/12} \tag{2.201}$$

Example 1.13: First-Order Plus Dead Time (FOPDT)

Many chemical processes can be approximated as:

$$G(s) = \frac{K\mathrm{e}^{-\theta s}}{\tau s + 1} \tag{2.202}$$

For a heat exchanger with $K = 2$ °C/%, $\tau = 60$ s, $\theta = 10$ s:

$$G(s) = \frac{2\mathrm{e}^{-10s}}{60s + 1} \tag{2.203}$$

The dead-time-to-time-constant ratio $\theta/\tau = 10/60 = 0.17$ indicates this system is relatively easy to control.

2.14.8 CSTR Schematic and Inlet/Outlet Configuration

The inlet provides fresh reactant at concentration $C_{A,\text{in}}$ and temperature T_{in}, while the outlet removes mixture at the reactor conditions (C_A, T) since perfect mixing is assumed. The cooling jacket maintains temperature control through T_j.

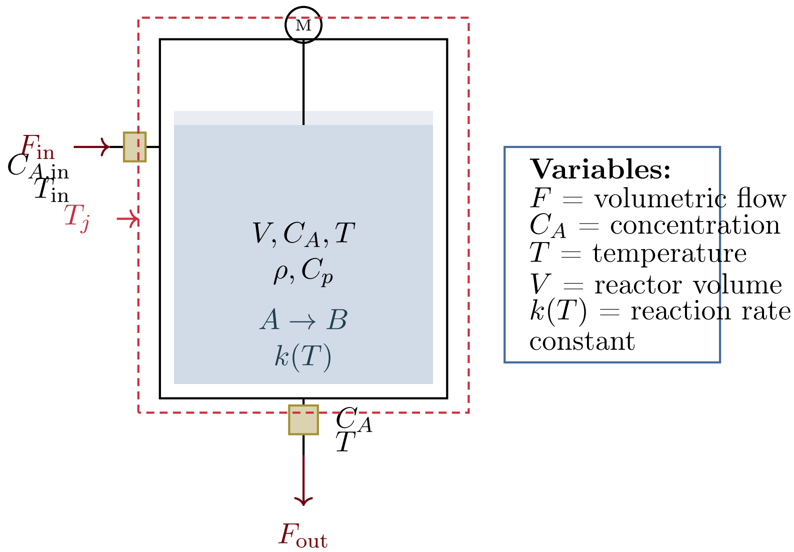


Figure 2.45: Continuous stirred-tank reactor (CSTR) with detailed inlet/outlet ports, cooling jacket, and labeled flows showing reactant concentration and temperature at inlet and outlet.

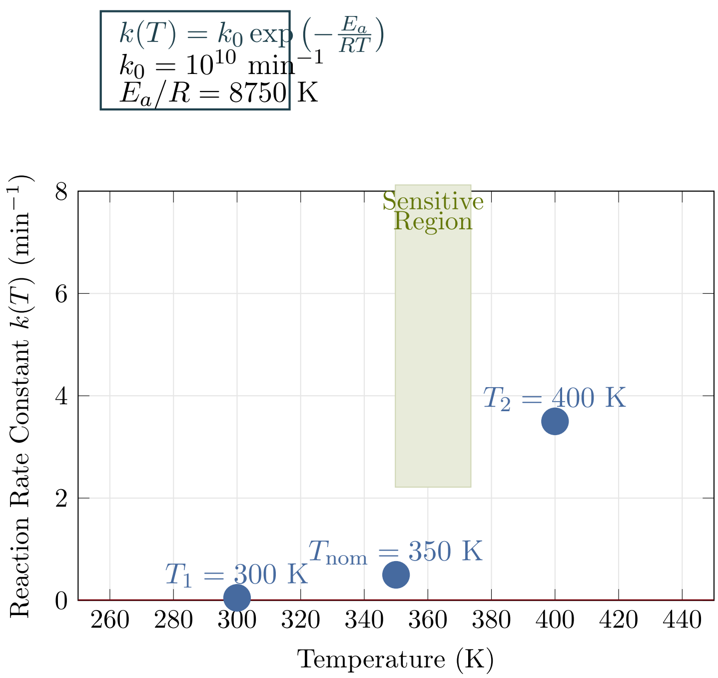


Figure 2.46: Arrhenius equation showing exponential increase in reaction rate with temperature. The nonlinear relationship creates control challenges, with the system becoming increasingly sensitive to temperature in the operating range.

2.14.9 Reaction Rate vs. Temperature: Arrhenius Behavior

The exponential dependence means small temperature changes can dramatically affect reaction rate. For a 50 K increase from 350 K to 400 K, the reaction rate increases approximately 7-fold. This nonlinear behavior is fundamental to exothermic reaction dynamics and runaway concerns.

2.14.10 Exercises: Chemical Systems

Exercise 2.37. A CSTR with volume 500 L and flow rate 50 L/min processes a first-order reaction with $k = 0.1$ min^{-1}. Feed concentration is 2 mol/L.

(a) Find the steady-state exit concentration

(b) Find the residence time

(c) Write the transfer function from C_{in} to C

(d) Find the time constant

Exercise 2.38. A mixing tank blends two streams: Stream 1 at 100 L/min with 10% salt, and Stream 2 at 50 L/min with 2% salt. Tank volume is 1000 L.

(a) Find the steady-state exit concentration

(b) If Stream 1 concentration suddenly changes to 15%, find the new steady-state and the time to reach 90% of the change

Exercise 2.39. An exothermic batch reactor starts at 25°C with 1000 mol of reactant. Heat of reaction is −50 kJ/mol, heat capacity is 4 kJ/kg·K, mass is 1000 kg, and reaction rate constant is $k = 0.01$ min^{-1} at 25°C with $E_a/R = 5000$ K.

(a) Write the coupled ODEs for concentration and temperature (adiabatic case)

(b) Estimate the adiabatic temperature rise if reaction goes to completion

Exercise 2.40. CSTR Reaction Kinetics: A CSTR operates with a second-order reaction $2A \to B$ having rate law $r = kC_A^2$ with $k = 0.1$ L/(mol·min). The reactor volume is 100 L and inlet concentration is 1 mol/L.

(a) For inlet flow 10 L/min, calculate the steady-state concentration

(b) If temperature increases causing k to double, find the new steady-state

(c) Compare reaction rates at both operating points

Exercise 2.41. CSTR Temperature Control: An exothermic reaction in a CSTR generates heat at rate $\dot{Q}_{\text{gen}} = (-\Delta H_r)k(T)C_A \cdot V$ where $-\Delta H_r = 50$ kJ/mol, $k(T)$ follows Arrhenius with $E_a/R = 8750$ K. The cooling jacket removes heat at $\dot{Q}_{\text{cool}} = UA(T - T_j)$ with $UA = 5$ kW/K.

(a) Write the energy balance ODE

(b) For $T_j = 350$ K (set at nominal operating point), find the steady-state temperature

(c) If T_j suddenly drops to 300 K, predict the qualitative response

Exercise 2.42. A process has FOPDT model with $K = 1.5$, $\tau = 120$ s, and $\theta = 30$ s.

(a) Write the transfer function

(b) Calculate θ/τ and comment on controllability

(c) Use first-order Padé approximation and find the system poles

Exercise 2.43. pH Control Challenge: A strong acid-strong base neutralization uses $[H^+] = \frac{C_a - C_b + \sqrt{(C_a - C_b)^2 + 4K_w}}{2}$ where $K_w = 10^{-14}$.

(a) With acid concentration 0.1 M flowing at 100 mL/min and base 0.1 M at 95 mL/min, find pH at steady-state

(b) Calculate the sensitivity $\frac{d(\text{pH})}{dC_b}$ at this point

(c) If base flow increases to 100.5 mL/min, estimate the new pH and explain the control difficulty

2.15 Aerospace Systems

Aerospace systems present unique control challenges: operation in three dimensions, coupling between axes, varying dynamics with flight conditions, and stringent safety requirements.

2.15.1 Aircraft Longitudinal Dynamics

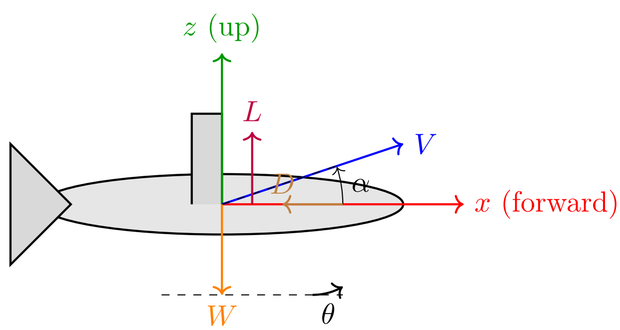


Figure 2.47: Aircraft longitudinal plane showing forces and angles.

State Variables

- u = forward velocity perturbation (m/s)
- w = vertical velocity perturbation (m/s)
- q = pitch rate (rad/s)
- θ = pitch angle (rad)

Linearized Equations of Motion

$$\begin{bmatrix} \dot{u} \\ \dot{w} \\ \dot{q} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} X_u & X_w & 0 & -g \\ Z_u & Z_w & U_0 & 0 \\ M_u & M_w & M_q & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} u \\ w \\ q \\ \theta \end{bmatrix} + \begin{bmatrix} X_{\delta_e} \\ Z_{\delta_e} \\ M_{\delta_e} \\ 0 \end{bmatrix} \delta_e \tag{2.204}$$

where δ_e is elevator deflection and X_u, Z_w, etc. are stability derivatives.

Characteristic Modes

The longitudinal dynamics typically have two modes:

Short Period Mode:

- Fast oscillation (1-3 seconds period)
- Primarily α and q motion
- Well-damped ($\zeta \approx 0.3 - 0.7$)

Approximate transfer function:

$$\frac{\alpha(s)}{\delta_e(s)} \approx \frac{Z_{\delta_e}/U_0}{s^2 - (M_q + M_{\dot{\alpha}} + Z_w/U_0)s + (Z_w M_q/U_0 - M_w)} \tag{2.205}$$

Phugoid Mode:

- Slow oscillation (30-100 seconds period)
- Exchange between kinetic and potential energy
- Lightly damped ($\zeta \approx 0.04 - 0.1$)

Approximate frequency and damping:

$$\omega_p \approx \sqrt{2}\frac{g}{U_0} \tag{2.206}$$

$$\zeta_p \approx \frac{1}{\sqrt{2}}\frac{C_D}{C_L} \tag{2.207}$$

Example 1.14: Boeing 747 Longitudinal Modes

At cruise ($U_0 = 250$ m/s, altitude 10 km):
Short period:

- $\omega_{sp} = 0.9$ rad/s, $\zeta_{sp} = 0.35$
- Period: $T_{sp} = 2\pi/\omega_{sp} = 7$ s

Phugoid:

- $\omega_p = 0.06$ rad/s, $\zeta_p = 0.05$
- Period: $T_p = 2\pi/\omega_p = 105$ s

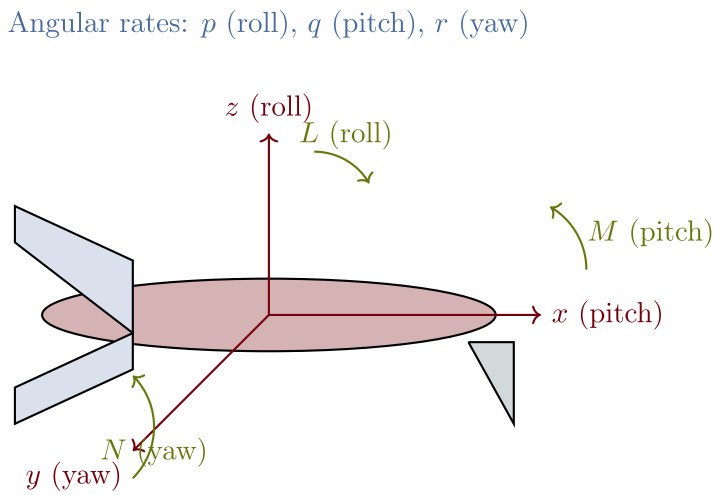


Figure 2.48: Aircraft body-fixed coordinate axes showing pitch, roll, and yaw moments with associated angular velocities.

2.15.2 Aircraft Body Axes and Rotational Dynamics

2.15.3 Aircraft Lateral-Directional Dynamics

State Variables

- β = sideslip angle (rad)
- p = roll rate (rad/s)
- r = yaw rate (rad/s)
- ϕ = bank angle (rad)

Characteristic Modes

Dutch Roll:

- Coupled yaw-roll oscillation
- Period: 3-10 seconds
- Damping varies with aircraft design

Roll Mode:

- Pure rolling motion
- First-order, typically fast (time constant 0.5-2 s)

Spiral Mode:

- Slow divergence or convergence
- Often slightly unstable (acceptable if slow)

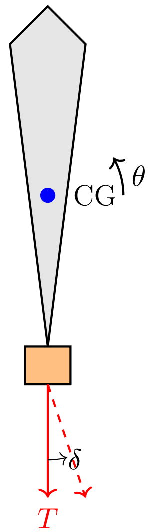


Figure 2.49: Rocket with thrust vector control.

2.15.4 Rocket Dynamics

Pitch Dynamics (Simplified)

For a rocket with thrust vector control:

$$I_{yy}\ddot{\theta} = T \cdot \ell_T \cdot \delta \tag{2.208}$$

where:

- I_{yy} = pitch moment of inertia
- T = thrust
- ℓ_T = distance from CG to gimbal point
- δ = gimbal angle

Transfer function:

$$\frac{\Theta(s)}{\delta(s)} = \frac{T\ell_T/I_{yy}}{s^2} = \frac{K_r}{s^2} \tag{2.209}$$

This is a double integrator—marginally stable and requires feedback for control.

Aerodynamic Effects

With aerodynamic forces, additional terms appear:

$$I_{yy}\ddot{\theta} = T\ell_T\delta + M_\alpha\alpha + M_q q \tag{2.210}$$

For a statically unstable rocket ($M_\alpha > 0$), the system has a pole in the right half-plane.

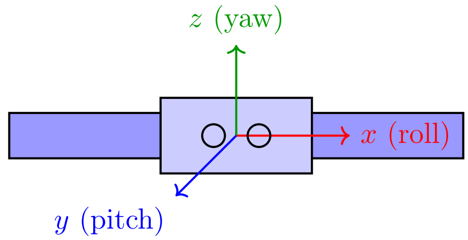


Figure 2.50: Satellite with body-fixed axes.

2.15.5 Satellite Attitude Dynamics

Euler's Equations

$$I_{xx}\dot{\omega}_x + (I_{zz} - I_{yy})\omega_y\omega_z = \tau_x \tag{2.211}$$
$$I_{yy}\dot{\omega}_y + (I_{xx} - I_{zz})\omega_z\omega_x = \tau_y \tag{2.212}$$
$$I_{zz}\dot{\omega}_z + (I_{yy} - I_{xx})\omega_x\omega_y = \tau_z \tag{2.213}$$

Linearized Single-Axis Model

For small angles about one axis:

$$I\ddot{\theta} = \tau \tag{2.214}$$

Transfer function:

$$\frac{\Theta(s)}{\tau(s)} = \frac{1}{Is^2} \tag{2.215}$$

Reaction Wheel Dynamics

A reaction wheel applies torque by changing its angular momentum:

$$\tau_{\text{satellite}} = -I_w\dot{\omega}_w \tag{2.216}$$

Combined system:

$$I_s\ddot{\theta} = -I_w\dot{\omega}_w = I_w\ddot{\theta}_w \tag{2.217}$$

2.15.6 Quadrotor Dynamics

Control Inputs

$$\text{Total thrust:} \quad T = F_1 + F_2 + F_3 + F_4 \tag{2.218}$$
$$\text{Roll torque:} \quad \tau_\phi = \ell(F_1 - F_2) \tag{2.219}$$
$$\text{Pitch torque:} \quad \tau_\theta = \ell(F_3 - F_4) \tag{2.220}$$
$$\text{Yaw torque:} \quad \tau_\psi = k_\tau(F_1 + F_2 - F_3 - F_4) \tag{2.221}$$

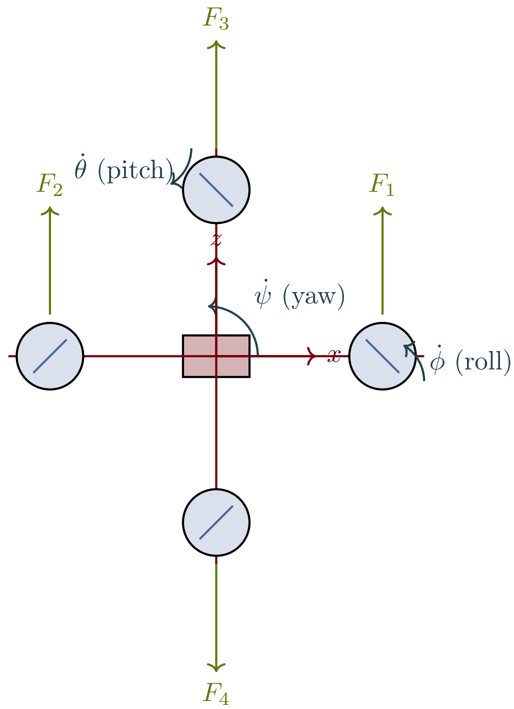
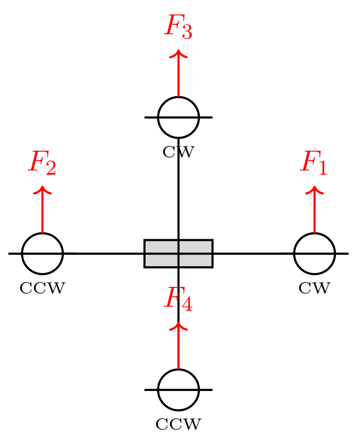


Figure 2.51: Quadrotor free body diagram showing thrust forces at each rotor and angular velocities for roll, pitch, and yaw control.



Roll: $F_1 \neq F_2$, Pitch: $F_3 \neq F_4$, Yaw: CW $\neq$ CCW

Figure 2.52: Quadrotor configuration showing rotor positions and control mixing.

Equations of Motion

Translational (in body frame):

$$m\ddot{x} = -mg\sin\theta \tag{2.222}$$
$$m\ddot{y} = mg\cos\theta\sin\phi \tag{2.223}$$
$$m\ddot{z} = mg\cos\theta\cos\phi - T \tag{2.224}$$

Rotational:

$$I_{xx}\ddot{\phi} = \tau_\phi \tag{2.225}$$
$$I_{yy}\ddot{\theta} = \tau_\theta \tag{2.226}$$
$$I_{zz}\ddot{\psi} = \tau_\psi \tag{2.227}$$

Linearized Model (Hover)

Near hover ($\phi, \theta, \psi \approx 0$):

$$\ddot{\phi} = \frac{\ell}{I_{xx}}(F_1 - F_2), \quad \ddot{\theta} = \frac{\ell}{I_{yy}}(F_3 - F_4), \quad \ddot{z} = g - \frac{T}{m} \tag{2.228}$$

2.15.7 Spacecraft Orbital Mechanics

Hill-Clohessy-Wiltshire Equations

Relative motion near a circular reference orbit:

$$\ddot{x} - 2n\dot{y} - 3n^2x = a_x \tag{2.229}$$
$$\ddot{y} + 2n\dot{x} = a_y \tag{2.230}$$
$$\ddot{z} + n^2z = a_z \tag{2.231}$$

where $n = \sqrt{\mu/a^3}$ is the orbital rate.
The out-of-plane motion (z) is decoupled and is a simple harmonic oscillator.
The in-plane motion (x, y) is coupled, with the solution:

$$x(t) = (4 - 3\cos nt)x_0 + \frac{\sin nt}{n}\dot{x}_0 + \frac{2(1-\cos nt)}{n}\dot{y}_0 \tag{2.232}$$

2.15.8 Exercises: Aerospace Systems

Exercise 2.44. An aircraft has short-period approximation:

$$\frac{\alpha(s)}{\delta_e(s)} = \frac{-2}{s^2 + 1.5s + 4} \tag{2.233}$$

(a) Find the natural frequency and damping ratio

(b) Is this response acceptable for a fighter aircraft?

(c) Sketch the step response

Exercise 2.45. A rocket has pitch moment of inertia $I_{yy} = 10^6$ kg·m^2, thrust $T = 10^6$ N, and gimbal arm $\ell_T = 20$ m.

(a) Find the transfer function from gimbal angle to pitch rate

(b) Design a proportional controller to achieve 1 rad/s^2 pitch acceleration per degree of gimbal

Exercise 2.46. A satellite reaction wheel has inertia $I_w = 0.01$ kg·m^2 and the satellite has inertia $I_s = 100$ kg·m^2 about one axis.

(a) If the wheel accelerates at 100 rad/s^2, find the satellite angular acceleration

(b) Find the maximum satellite rate if the wheel is limited to 6000 rpm

Exercise 2.47. A quadrotor has mass 1.5 kg, arm length 0.25 m, and moment of inertia $I_{xx} = I_{yy} = 0.02$ kg·m^2.

(a) Find the hover thrust per motor

(b) Derive the transfer function from motor force differential to roll angle

(c) If motor response is modeled as first-order with $\tau = 0.05$ s, find the complete roll transfer function

Exercise 2.48. A satellite has inertia $I = 50$ kg·m^2 about the pitch axis. Two reaction wheels are available with inertias $I_w = 0.5$ kg·m^2 each, located 90apart about the body axes.

(a) Write the coupled equations for pitch-yaw control using both wheels

(b) If the pitch wheel accelerates at 200 rad/s^2, find the satellite pitch angular acceleration

(c) Design a control law to achieve zero final angular rate when the pitch wheel reaches 3000 rpm

Exercise 2.49. An aircraft has longitudinal dynamics with short-period natural frequency $\omega_n = 1.2$ rad/s and damping $\zeta = 0.45$. The phugoid has $\omega_p = 0.08$ rad/s and $\zeta_p = 0.06$.

(a) Sketch the pole-zero map for the longitudinal system

(b) Calculate the time constants for each mode

(c) What type of controller (P, PI, or PD) would be best for elevator control? Why?

Exercise 2.50. A rocket undergoes pitch control with gimbal actuation. The rocket parameters are: $I_{yy} = 2 \times 10^6$ kg·m^2, thrust $T = 5 \times 10^6$ N, gimbal arm $\ell = 10$ m, and aerodynamic damping coefficient $M_q = -3 \times 10^5$ N·m·s/rad.

(a) Derive the transfer function from gimbal angle δ to pitch rate q

(b) Find the system poles and determine stability

(c) Design a proportional controller to limit pitch rate to 5 deg/s for 1gimbal command

Exercise 2.51. A quadrotor must perform a roll maneuver from level flight. It has mass $m = 2$ kg, arm length $\ell = 0.3$ m, and $I_{xx} = 0.025$ kg·m^2.

(a) For a rolling maneuver command of 0.5 rad/s^2, calculate the required roll torque

(b) Find the force differential between motors: $\Delta F = F_1 - F_2$

(c) If available motor thrust is 10 N each, what is the maximum roll acceleration?

Exercise 2.52. Consider a satellite attitude control system using a three-axis reaction wheel assembly. The satellite inertia matrix is diagonal: $I_{xx} = 80$, $I_{yy} = 100$, $I_{zz} = 120$ kg·m^2.

(a) Write the decoupled single-axis control law for each axis: $\tau = I\ddot{\theta}$

(b) If maximum wheel torque is 0.2 N·m, find the maximum angular acceleration for each axis

(c) Design a PD controller for the yaw axis: $\tau = K_p e + K_d \dot{e}$, with desired natural frequency 0.1 rad/s and damping ratio 0.7

2.16 Biomedical Systems

Biomedical control systems interface with the human body, requiring special consideration for safety, variability between patients, and the inherent complexity of biological processes.

2.16.1 Pharmacokinetics: Drug Distribution

Pharmacokinetics describes how drugs move through the body.

One-Compartment Model

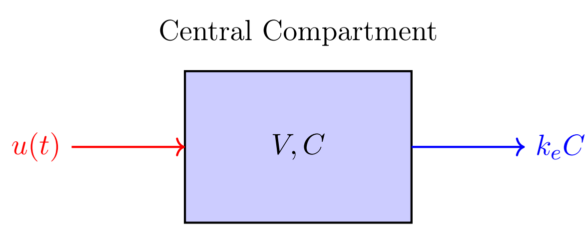


Figure 2.53: One-compartment pharmacokinetic model.

Mass balance:

$$V\frac{\mathrm{d}C}{\mathrm{d}t} = u(t) - k_e V C \tag{2.234}$$

where:

- V = volume of distribution (L)
- C = drug concentration (mg/L)
- $u(t)$ = infusion rate (mg/min)
- k_e = elimination rate constant (1/min)

Transfer function:

$$\frac{C(s)}{U(s)} = \frac{1/V}{s + k_e} \tag{2.235}$$

Half-life: $t_{1/2} = \ln(2)/k_e$

Example 1.15: IV Drug Infusion

A drug has $V = 50$ L and $k_e = 0.1$ min^{-1}. Find:

(a) The half-life

(b) The infusion rate needed for steady-state concentration of 10 mg/L

(c) Time to reach 90% of steady-state

Solution:
(a) Half-life: $t_{1/2} = \ln(2)/0.1 = 6.93$ min
(b) At steady-state: $u_{ss} = k_e V C_{ss} = 0.1 \times 50 \times 10 = 50$ mg/min
(c) Time constant: $\tau = 1/k_e = 10$ min
Time to 90%: $t = -\tau \ln(0.1) = 10 \times 2.3 = 23$ min

Two-Compartment Model

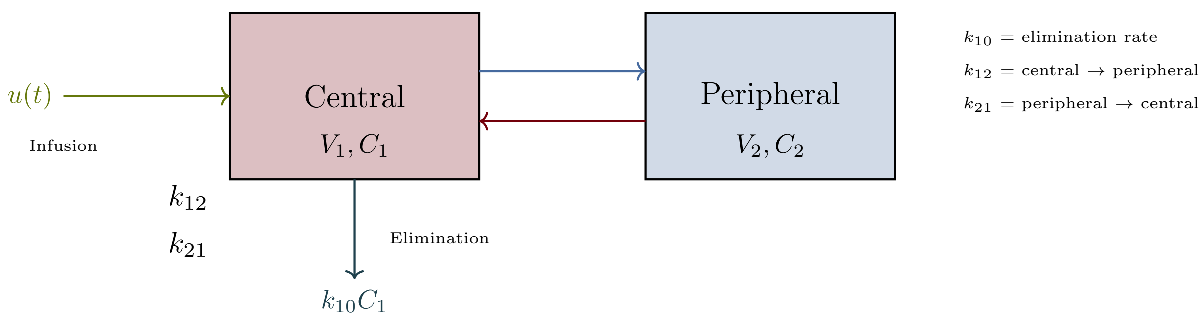


Figure 2.54: Two-compartment pharmacokinetic model diagram showing drug infusion, inter-compartment transfer, and elimination with rate constants labeled.

Mass balances:

$$V_1 \frac{\mathrm{d}C_1}{\mathrm{d}t} = u(t) - (k_{10} + k_{12})V_1 C_1 + k_{21} V_2 C_2 \tag{2.236}$$
$$V_2 \frac{\mathrm{d}C_2}{\mathrm{d}t} = k_{12} V_1 C_1 - k_{21} V_2 C_2 \tag{2.237}$$

This yields a second-order system with two exponential phases:

- **Distribution phase** (α): rapid initial decline
- **Elimination phase** (β): slower terminal decline

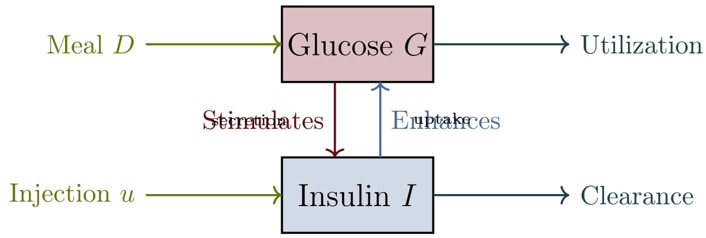


Figure 2.55: Glucose-insulin feedback loop block diagram showing meal and infusion inputs, utilization/clearance outputs, and the bidirectional regulatory feedback.

2.16.2 Glucose-Insulin Dynamics

Minimal Model (Bergman)

$$\frac{\mathrm{d}G}{\mathrm{d}t} = -p_1 G - X(G + G_b) + D(t) \tag{2.238}$$

$$\frac{\mathrm{d}X}{\mathrm{d}t} = -p_2 X + p_3(I - I_b) \tag{2.239}$$

$$\frac{\mathrm{d}I}{\mathrm{d}t} = -n(I - I_b) + \gamma(G - h)^+ + u(t) \tag{2.240}$$

where:

- G = glucose concentration (mg/dL)
- I = insulin concentration (μU/mL)
- X = insulin action (remote compartment)
- $D(t)$ = meal disturbance
- $u(t)$ = insulin infusion (for artificial pancreas)

Linearized Model for Control Design

Near operating point (G_0, I_0):

$$\begin{bmatrix} \Delta\dot{G} \\ \Delta\dot{I} \end{bmatrix} = \begin{bmatrix} -p_1 - X_0 & -K_G \\ K_I & -n \end{bmatrix} \begin{bmatrix} \Delta G \\ \Delta I \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} \Delta D + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \Delta u \tag{2.241}$$

2.16.3 Cardiovascular System

Windkessel Model

The arterial system can be modeled as an electrical analog:

Governing equation:

$$C\frac{\mathrm{d}P}{\mathrm{d}t} + \frac{P}{R} = Q(t) \tag{2.242}$$

where:

- P = arterial pressure (mmHg)

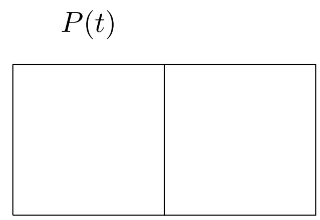


Figure 2.56: Two-element Windkessel model.

- Q = cardiac output (mL/s)
- C = arterial compliance (mL/mmHg)
- R = systemic vascular resistance (mmHg·s/mL)

Blood Pressure Control

The baroreflex regulates blood pressure:

$$\frac{\mathrm{d}P}{\mathrm{d}t} = -\frac{1}{\tau_P}(P - P_{\text{set}}) + K_u u(t) \tag{2.243}$$

where $u(t)$ represents vasoactive drug infusion.

2.16.4 Respiratory System

Mechanical Ventilation Model

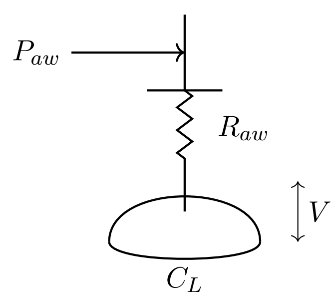


Figure 2.57: Single-compartment lung model.

Equation of motion:

$$P_{aw} = R_{aw}\dot{V} + \frac{V}{C_L} + P_{\text{PEEP}} \tag{2.244}$$

where:

- P_{aw} = airway pressure ($\mathrm{cmH_2O}$)
- R_{aw} = airway resistance ($\mathrm{cmH_2O{\cdot}s/L}$)
- C_L = lung compliance ($\mathrm{L/cmH_2O}$)
- V = lung volume (L)

- P_{PEEP} = positive end-expiratory pressure

Transfer function:

$$\frac{V(s)}{P_{aw}(s)} = \frac{C_L}{R_{aw}C_L s + 1} \tag{2.245}$$

2.16.5 Anesthesia Delivery

Propofol Pharmacokinetics

Three-compartment mammillary model:

$$V_1\dot{C}_1 = u - k_{10}V_1C_1 - k_{12}V_1C_1 + k_{21}V_2C_2 - k_{13}V_1C_1 + k_{31}V_3C_3 \tag{2.246}$$

$$V_2\dot{C}_2 = k_{12}V_1C_1 - k_{21}V_2C_2 \tag{2.247}$$

$$V_3\dot{C}_3 = k_{13}V_1C_1 - k_{31}V_3C_3 \tag{2.248}$$

Pharmacodynamic Effect

The effect-site concentration C_e relates to the measured output (e.g., BIS index):

$$\frac{\mathrm{d}C_e}{\mathrm{d}t} = k_{e0}(C_1 - C_e) \tag{2.249}$$

Hill equation for effect:

$$\text{Effect} = E_0 - E_{\max}\frac{C_e^\gamma}{C_e^\gamma + EC_{50}^\gamma} \tag{2.250}$$

2.16.6 Medical Device Examples

Infusion Pump

Transfer function (actuator dynamics):

$$\frac{Q(s)}{V_{\text{cmd}}(s)} = \frac{K_p}{\tau_p s + 1} \tag{2.251}$$

where Q is flow rate and V_{cmd} is the command voltage.

Pacemaker Rate Response

Activity-based pacing rate:

$$\tau\frac{\mathrm{d}R}{\mathrm{d}t} + R = R_{\text{base}} + K \cdot A(t) \tag{2.252}$$

where $A(t)$ is the activity sensor signal and R is the pacing rate.

Cochlear Implant Signal Processing

Filter bank model:

$$H_k(s) = \frac{\omega_k^2}{s^2 + 2\zeta\omega_k s + \omega_k^2} \tag{2.253}$$

for frequency band k centered at ω_k.

2.16.7 Neural Systems

Hodgkin-Huxley Neuron Model

$$C_m \frac{\mathrm{d}V}{\mathrm{d}t} = I_{\text{ext}} - g_{Na} m^3 h (V - E_{Na}) - g_K n^4 (V - E_K) - g_L (V - E_L) \tag{2.254}$$

with gating variables m, h, n following first-order kinetics.

Simplified Integrate-and-Fire Model

$$\tau_m \frac{\mathrm{d}V}{\mathrm{d}t} = -(V - V_{\text{rest}}) + R_m I_{\text{ext}} \tag{2.255}$$

When V reaches threshold V_{th}, the neuron fires and resets.

2.16.8 Exercises: Biomedical Systems

Exercise 2.53. A drug has one-compartment kinetics with $V = 35$ L and $t_{1/2} = 4$ hours.

(a) Calculate the elimination rate constant

(b) Find the loading dose to achieve 20 mg/L immediately

(c) Find the maintenance infusion rate

(d) Write the transfer function from infusion rate to concentration

Exercise 2.54. A patient's arterial compliance is $C = 2$ mL/mmHg and peripheral resistance is $R = 1$ mmHg·s/mL.

(a) Find the time constant of the arterial system

(b) If cardiac output steps from 80 to 100 mL/s, find the new steady-state pressure

(c) Sketch the pressure response to the step change

Exercise 2.55. A mechanical ventilator delivers pressure to a patient with $R_{aw} = 10$ cmH$_2$O·s/L and $C_L = 0.05$ L/cmH$_2$O.

(a) Find the time constant

(b) If inspiratory pressure is 20 cmH$_2$O above PEEP, find the tidal volume

(c) Find the inspiratory time needed to achieve 95% of the tidal volume

Exercise 2.56. An artificial pancreas controller must regulate glucose. The simplified model has:

$$\frac{G(s)}{U(s)} = \frac{-2}{(10s + 1)(50s + 1)} \tag{2.256}$$

where G is glucose (mg/dL) and U is insulin (U/hr), with time in minutes.

(a) Find the steady-state glucose change for 1 U/hr insulin increase

(b) Find the system poles and time constants

(c) Sketch the glucose response to a step insulin input

Exercise 2.57. A two-compartment drug delivery system has: $V_1 = 10$ L, $V_2 = 40$ L, $k_{10} = 0.2$ hr^{-1}, $k_{12} = 0.15$ hr^{-1}, and $k_{21} = 0.1$ hr^{-1}.

(a) Calculate the drug concentration in both compartments at 4 hours following a 500 mg IV bolus dose

(b) Find the half-lives for the distribution and elimination phases

(c) If the drug is continuously infused at 10 mg/hr, find the steady-state concentrations in both compartments

Exercise 2.58. A patient receiving mechanical ventilation has measured respiratory mechanics: $R_{aw} = 8$ cmH$_2$O·s/L, $C_L = 0.04$ L/cmH$_2$O, $I_r = 0.03$ L/cmH$_2$O (chest wall compliance).

(a) Find the total respiratory system compliance and time constant

(b) For a constant inspiratory pressure of 15 cmH$_2$O (above PEEP), calculate tidal volume after 5 time constants

(c) Design a pressure control strategy to deliver 0.5 L tidal volume in 1 second

Exercise 2.59. A pacemaker rate-response control system regulates heart rate based on activity. The transfer function from activity sensor voltage to pacing rate is:

$$\frac{R(s)}{A(s)} = \frac{K}{0.5s + 1}, \quad K = 100 \text{ bpm/V} \tag{2.257}$$

(a) Calculate the steady-state pacing rate for activity input of 0.5 V

(b) Find the time to reach 90% of steady-state rate

(c) Design a controller to achieve 0 to 120 bpm range with sensor output of 0 to 1 V

Exercise 2.60. An infusion pump delivers IV medications with first-order actuator dynamics: $\frac{Q(s)}{V_{\text{cmd}}(s)} = \frac{K_p}{\tau_p s+1}$, where $K_p = 5$ mL/min/V and $\tau_p = 0.1$ s.

(a) For a step command of 2 V, find the steady-state flow rate

(b) Calculate the actual flow rate at time $t = 0.2$ s

(c) If a patient needs 50 mL of drug infused in exactly 10 minutes, determine the required command voltage accounting for actuator dynamics

Exercise 2.61. A prosthetic limb controller uses proportional myoelectric control, with muscle EMG signal driving joint angle:

$$\theta(s) = \frac{G(s)}{1 + G(s)H(s)} E(s) \tag{2.258}$$

where $G(s) = \frac{K}{\tau s+1}$ (actuator), $H(s) = K_f s$ (feedback with damping), $K = 45$ deg/V, $\tau = 0.2$ s, and $K_f = 0.1$ V·s/deg.

(a) Find the closed-loop transfer function $\theta(s)/E(s)$

(b) For a step EMG input of 0.5 V, calculate steady-state joint angle

(c) Sketch the response and identify if overshoot or oscillation occurs

(d) Design a proportional controller to add to $G(s)$ that achieves 1 deg/V steady-state sensitivity with damping ratio 0.8

Exercise 2.62. A glucose-insulin model for artificial pancreas design is based on the Bergman minimal model. Patient parameters are: $p_1 = 0.03$ min^{-1}, $p_2 = 0.02$ min^{-1}, $p_3 = 1.2 \times 10^{-5}$ mL/(μU·min), $n = 0.1$ min^{-1}, $\gamma = 0.5$ mg/dL$/min/(\mu$U$/mL), with$G$_b = 95$ mg/dL and $I_b = 12$ μU/mL.

(a) Linearize the system about the basal operating point $(G_0, I_0) = (G_b, I_b)$

(b) Find the transfer functions $G(s)/U(s)$ and $G(s)/D(s)$

(c) Design a proportional insulin infusion controller: $u(t) = K_p(G_{\text{ref}} - G(t))$ that regulates glucose to 100 mg/dL with settling time under 3 hours

2.17 Industrial and Computer Systems

This section covers systems from manufacturing, process control, and computer science—domains where control theory meets discrete events, queueing, and information flow.

2.17.1 Industrial Process Control

Conveyor Belt System

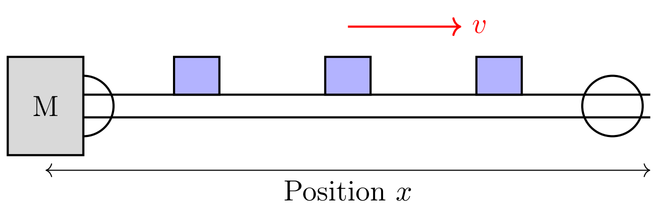


Figure 2.58: Conveyor belt system with motor drive.

Motor-driven conveyor dynamics:

$$J_{\text{eq}}\dot{\omega} + B\omega = K_t i - \tau_L \tag{2.259}$$

where J_{eq} includes reflected inertia of belt and load.
Belt velocity: $v = r\omega$ where r is the drive roller radius.
Transfer function (speed control):

$$\frac{V(s)}{I(s)} = \frac{rK_t}{J_{\text{eq}}s + B} \tag{2.260}$$

Temperature Control Zone

Industrial ovens often have multiple zones:
Each zone:

$$C_i \frac{\mathrm{d}T_i}{\mathrm{d}t} = Q_i - \frac{T_i - T_{i-1}}{R_{i-1,i}} - \frac{T_i - T_{i+1}}{R_{i,i+1}} - \frac{T_i - T_\infty}{R_{i,\infty}} \tag{2.261}$$

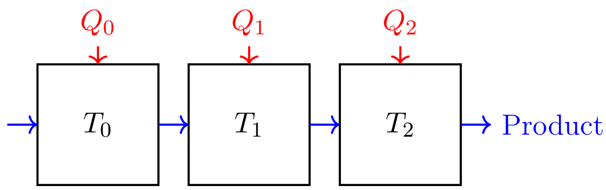


Figure 2.59: Multi-zone temperature control.

2.17.2 Robotic Manipulators

Single-Link Robot Arm

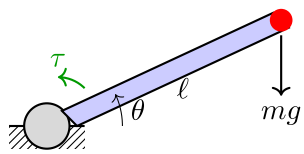


Figure 2.60: Single-link robot arm.

Equation of motion:

$$J\ddot{\theta} + B\dot{\theta} + mg\ell\cos\theta = \tau \tag{2.262}$$

Linearized about horizontal ($\theta_0 = 0$):

$$J\ddot{\theta} + B\dot{\theta} + mg\ell = \tau \tag{2.263}$$

Note: Gravity creates a constant disturbance torque.

Two-Link Planar Arm

The dynamics become coupled and nonlinear:

$$\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{g}(\mathbf{q}) = \boldsymbol{\tau} \tag{2.264}$$

where:

- $\mathbf{M}(\mathbf{q})$ = mass matrix (configuration-dependent)
- $\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})$ = Coriolis and centrifugal terms
- $\mathbf{g}(\mathbf{q})$ = gravity vector
- $\boldsymbol{\tau}$ = joint torques

For a two-link arm:

$$M_{11} = m_1\ell_{c1}^2 + m_2(\ell_1^2 + \ell_{c2}^2 + 2\ell_1\ell_{c2}\cos\theta_2) + I_1 + I_2 \tag{2.265}$$
$$M_{12} = m_2(\ell_{c2}^2 + \ell_1\ell_{c2}\cos\theta_2) + I_2 \tag{2.266}$$
$$M_{22} = m_2\ell_{c2}^2 + I_2 \tag{2.267}$$

2.17.3 CNC Machine Tools

Axis Drive Model

Each axis of a CNC machine:

$$J_{\text{axis}}\ddot{x} + B\dot{x} + F_c \operatorname{sgn}(\dot{x}) = F_{\text{motor}} - F_{\text{cut}} \tag{2.268}$$

where F_c is Coulomb friction and F_{cut} is cutting force disturbance.

Servo System

Position loop transfer function:

$$\frac{X(s)}{X_{\text{cmd}}(s)} = \frac{K_p K_v/s}{1 + K_p K_v/s + s^2/\omega_n^2} \approx \frac{K_p}{\tau_p s + 1} \tag{2.269}$$

for well-tuned systems where $K_v = K_p/\tau_p$.

CNC Machine Control Loop

A complete CNC control architecture uses nested feedback loops for position tracking with real-time disturbance rejection:

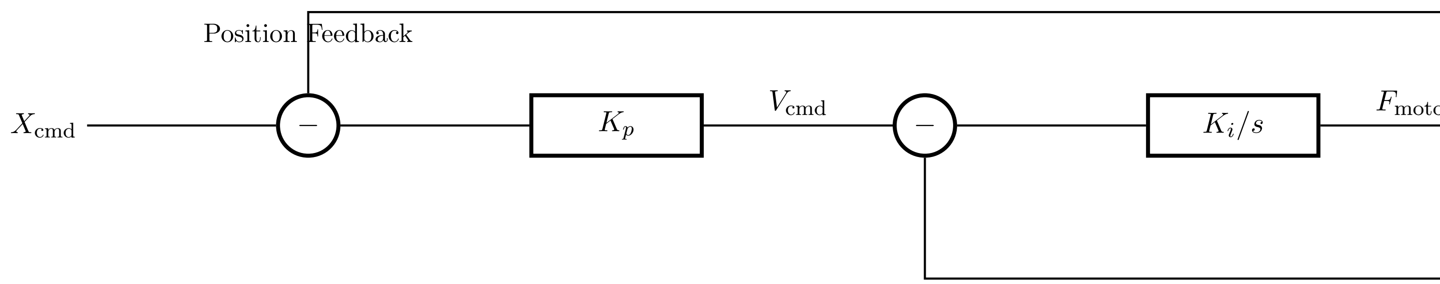


Figure 2.61: CNC machine position control loop with nested feedback and disturbance rejection. The proportional controller commands a velocity set-point, while the integrating controller regulates actual velocity feedback to track the commanded profile.

2.17.4 Power Systems

Synchronous Generator

Swing equation:

$$\frac{2H}{\omega_s}\frac{\mathrm{d}^2\delta}{\mathrm{d}t^2} + D\frac{\mathrm{d}\delta}{\mathrm{d}t} = P_m - P_e \tag{2.270}$$

where:

- H = inertia constant (seconds)
- δ = rotor angle (electrical radians)
- ω_s = synchronous speed
- P_m = mechanical power

- P_e = electrical power

Linearized about operating point:

$$\frac{\Delta\delta(s)}{\Delta P_m(s)} = \frac{\omega_s/2H}{s^2 + (D\omega_s/2H)s + K_s\omega_s/2H} \tag{2.271}$$

where $K_s = \partial P_e/\partial\delta$ is the synchronizing coefficient.

Automatic Generation Control (AGC)

Area control error:

$$\text{ACE} = \Delta P_{\text{tie}} + B\Delta f \tag{2.272}$$

where B is the frequency bias setting.

2.17.5 Queueing Systems

M/M/1 Queue

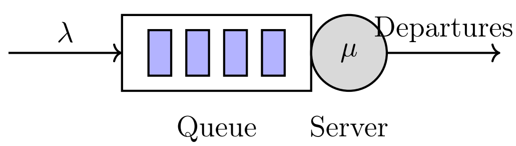


Figure 2.62: M/M/1 queue model.

Steady-state:

$$\text{Utilization:} \quad \rho = \lambda/\mu \tag{2.273}$$

$$\text{Mean queue length:} \quad L = \frac{\rho}{1-\rho} \tag{2.274}$$

$$\text{Mean waiting time:} \quad W = \frac{1}{\mu - \lambda} \tag{2.275}$$

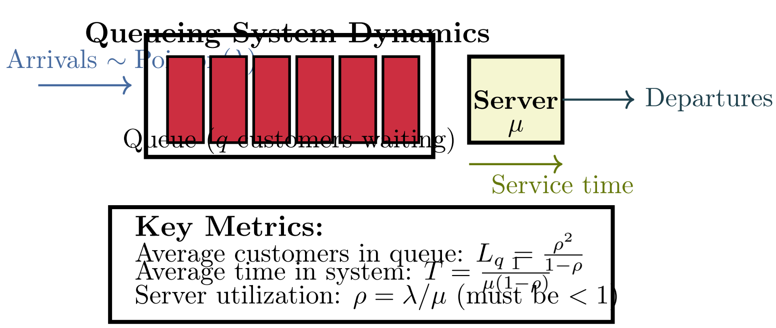


Figure 2.63: Queueing system flow and performance metrics. Customers arrive according to a Poisson process with rate λ, wait in queue, and receive service with exponential service time (mean $1/\mu$). System stability requires $\rho < 1$.

Fluid Flow Model

For control design, queues can be approximated as fluid:

$$\frac{\mathrm{d}q}{\mathrm{d}t} = \lambda(t) - \mu(t) \tag{2.276}$$

where q is queue length, λ is arrival rate, and μ is service rate.
Transfer function:

$$\frac{Q(s)}{\Lambda(s) - M(s)} = \frac{1}{s} \tag{2.277}$$

2.17.6 Computer Systems

CPU Load Control

CPU utilization dynamics:

$$\tau \frac{\mathrm{d}U}{\mathrm{d}t} + U = K_u \cdot R \tag{2.278}$$

where U is utilization, R is admission rate, and τ depends on job service time.

Network Congestion Control

TCP congestion window dynamics (simplified):

$$\frac{\mathrm{d}W}{\mathrm{d}t} = \frac{1}{RTT} - \frac{W}{2} \cdot p \tag{2.279}$$

where W is congestion window, RTT is round-trip time, and p is packet loss probability.

Memory Management

Page fault rate model:

$$f = f_0 \left(\frac{W_s}{M}\right)^{\alpha} \tag{2.280}$$

where M is allocated memory, W_s is working set size, and α is a workload parameter.

2.17.7 Data Center Thermal Management

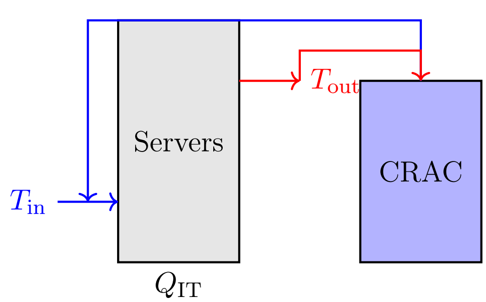


Figure 2.64: Data center thermal model.

Thermal dynamics:

$$C_{\text{room}} \frac{\mathrm{d}T}{\mathrm{d}t} = Q_{\text{IT}} - \dot{m} c_p (T - T_{\text{supply}}) \tag{2.281}$$

where $\dot{m}$ is air mass flow rate (controlled by CRAC fans).

2.17.8 Building Automation

HVAC Zone Model

$$C_z \frac{\mathrm{d}T_z}{\mathrm{d}t} = Q_{\mathrm{HVAC}} + Q_{\mathrm{internal}} + Q_{\mathrm{solar}} - UA(T_z - T_{\mathrm{out}}) \tag{2.282}$$

Transfer function from HVAC power to zone temperature:

$$\frac{T_z(s)}{Q_{\mathrm{HVAC}}(s)} = \frac{1/UA}{\tau s + 1} \tag{2.283}$$

where $\tau = C_z/UA$.

Lighting Control

Dimming dynamics:

$$\tau_L \frac{\mathrm{d}L}{\mathrm{d}t} + L = K_L \cdot d \tag{2.284}$$

where L is light level, d is dimmer setting, and τ_L is lamp response time.

2.17.9 Traffic Flow

Cell Transmission Model

For a road segment:

$$\frac{\mathrm{d}n_i}{\mathrm{d}t} = q_{i-1} - q_i \tag{2.285}$$

where n_i is vehicle count in cell i and q_i is flow rate.

Flow-density relationship:

$$q = v_f n \left(1 - \frac{n}{n_{\max}}\right) \tag{2.286}$$

Traffic Signal Timing

Green time allocation affects queue discharge:

$$\frac{\mathrm{d}q}{\mathrm{d}t} = \lambda - s \cdot g(t) \tag{2.287}$$

where s is saturation flow rate and $g(t) \in \{0, 1\}$ indicates green signal.

2.17.10 Exercises: Industrial and Computer Systems

Exercise 2.63. A conveyor belt has equivalent inertia $J = 2$ kg·m^2, friction $B = 0.5$ N·m·s/rad, drive roller radius $r = 0.1$ m, and motor constant $K_t = 0.5$ N·m/A.

(a) Find the transfer function from motor current to belt velocity

(b) Calculate the current needed for 0.5 m/s belt speed

(c) Find the time constant

Exercise 2.64. A single-link robot arm has $J = 0.5$ kg·m^2, $B = 0.1$ N·m·s/rad, $m = 2$ kg, and $\ell = 0.5$ m.

(a) Write the nonlinear equation of motion

(b) Linearize about $\theta_0 = 0$ (horizontal)

(c) Find the gravity compensation torque

(d) Derive the transfer function from torque to angle (with gravity compensated)

Exercise 2.65. An M/M/1 queue has arrival rate $\lambda = 80$ jobs/s and service rate $\mu = 100$ jobs/s.

(a) Calculate the utilization

(b) Find the mean queue length

(c) Find the mean response time (waiting + service)

(d) If the arrival rate increases to 90 jobs/s, how do the metrics change?

Exercise 2.66. A data center has thermal capacitance $C = 10^6$ J/K, heat load $Q_{\text{IT}} = 100$ kW, and air flow rate that removes heat according to $\dot{Q}_{\text{cool}} = 5000(T - 18)$ W where T is in °C.

(a) Find the steady-state temperature

(b) Calculate the thermal time constant

(c) If IT load suddenly increases to 120 kW, find the new steady-state and time to reach it

Exercise 2.67. A traffic intersection has arrival rate 600 vehicles/hour and saturation flow 1800 vehicles/hour. The cycle length is 90 seconds.

(a) Find the minimum green time to prevent queue buildup

(b) If green time is 40 seconds, calculate the average queue at the start of green

(c) Model the queue dynamics as a hybrid system

Exercise 2.68 (CNC Positioning Control)**.** A CNC machine has proportional position controller gain $K_p = 50$ V/mm and velocity controller gain $K_i = 10$ A/(mm/s). The motor has torque constant $K_t = 0.8$ N·m/A, axis inertia $J = 0.2$ kg·m^2, and viscous damping $B = 2$ N·m·s/rad. The lead screw couples the motor to the axis with a 5 mm pitch.

(a) Convert the velocity command from the position controller into screw rotational velocity (rad/s)

(b) Write the motor dynamics in state-space form with velocity feedback

(c) Find the steady-state position error for a constant velocity ramp command of 10 mm/s

(d) Design the proportional gain K_p to achieve a settling time of 0.5 s for a 1 mm step command

(e) Explain how cutting force disturbance $F_{\text{cut}} = 500$ N affects steady-state accuracy

Exercise 2.69 (Queueing System Design)**.** A server farm handles web requests with M/M/1 queue dynamics. Requests arrive at $\lambda = 50$ req/s and each server processes requests at $\mu = 60$ req/s.

(a) Calculate the server utilization and assess stability

(b) Compute the average number of requests in the system

(c) Find the average response time (waiting + processing)

(d) If we want to maintain $W < 50$ ms, how many additional servers are needed?

(e) Model the queue length as a first-order system: $\dot{q} = \lambda - \mu(t)$, and find the characteristic time constant

Exercise 2.70 (Data Center Thermal Control)**.** A data center has thermal dynamics:

$$C\frac{\mathrm{d}T}{\mathrm{d}t} = Q_{\mathrm{IT}} - Q_{\mathrm{cool}}(T, \dot{m}) \tag{2.288}$$

where $C = 5 \times 10^6$ J/K is the room thermal capacitance, $Q_{\mathrm{IT}} = 500$ kW is the IT heat load, and cooling is $Q_{\mathrm{cool}} = \dot{m}c_p(T - T_{\mathrm{supply}})$ with $c_p = 1005$ J/(kg·K), $T_{\mathrm{supply}} = 15$C, and controlled mass flow $\dot{m}(t)$ in kg/s.

(a) Find the steady-state room temperature when $\dot{m} = 100$ kg/s

(b) Linearize the system about this operating point

(c) Design a proportional controller for mass flow: $\dot{m}(t) = K_p(T_{\mathrm{ref}} - T(t))$ with $T_{\mathrm{ref}} = 20$C

(d) Find the time constant of the closed-loop system with $K_p = 5$ kg/(s·°C)

(e) Analyze stability: what maximum K_p prevents oscillation (critical damping)?

Exercise 2.71 (Network Congestion Window Dynamics)**.** TCP congestion control uses a window-based mechanism. The congestion window W (in packets) evolves according to:

$$\frac{\mathrm{d}W}{\mathrm{d}t} = \frac{A}{W} - B \cdot p(W) \tag{2.289}$$

where A and B are constants, and $p(W)$ is the packet loss probability, modeled as $p(W) = 0.3 \cdot (W/100)^2$ when $W > 50$ and $p(W) = 0$ otherwise (no loss during congestion avoidance).

(a) Find the equilibrium window size when $A = 2$, $B = 1$

(b) Analyze stability of the equilibrium: is it stable or unstable?

(c) If the RTT suddenly increases by 20%, how does the equilibrium window change?

(d) Model a step increase in competing traffic (increase in B) and sketch the expected transient response

(e) Propose a controller to maintain target window size $W_{\mathrm{ref}} = 80$ packets

Exercise 2.72 (Two-Layer Queueing Network)**.** Consider a tandem queueing network where jobs arrive at Queue 1 (servers rate $\mu_1 = 100$ jobs/s), then proceed to Queue 2 (servers rate $\mu_2 = 80$ jobs/s). The arrival rate is $\lambda = 70$ jobs/s.

(a) Are both queues stable? Justify your answer

(b) Calculate the average delay at Queue 1 and Queue 2 separately

(c) Find the total average time a job spends in the network

(d) If Queue 2 becomes a bottleneck (μ_2 reduced to 65 jobs/s), what is the new stability boundary?

(e) Propose a feedback control strategy (admission control) to maintain both queues stable despite temporary traffic spikes

(f) Sketch a block diagram showing fluid-flow models of both queues with control feedback

Appendix A

Laplace Transform Tables and Properties

The Laplace transform is fundamental to classical control systems analysis. This appendix provides comprehensive tables of transform pairs, properties, and useful techniques for solving control problems. All tables use the unilateral Laplace transform defined as:

$$F(s) = \mathcal{L}\{f(t)\} = \int_0^\infty f(t)e^{-st}\,\mathrm{d}t \tag{A.1}$$

where $s = \sigma + j\omega$ is the complex frequency variable, with region of convergence (ROC) $\Re(s) > \sigma_0$ for appropriate σ_0.

A.1 Common Laplace Transform Pairs

Table A.1 presents the most common Laplace transform pairs essential for control systems analysis. These pairs are used repeatedly throughout classical control design.

Table A.1: Common Laplace Transform Pairs

Signal $f(t)$	**Laplace Transform** $F(s)$	**Constraints / Notes**
$\delta(t)$ (unit impulse)	1	Dirac delta function
$u(t)$ (unit step)	$\frac{1}{s}$	$s > 0$
$u(t) - u(t-a)$ (rectangular pulse)	$\frac{1 - e^{-as}}{s}$	$s > 0$
t (ramp)	$\frac{1}{s^2}$	$s > 0$
t^2	$\frac{2}{s^3}$	$s > 0$
$t^n,\ n \in \mathbb{Z}^+$	$\frac{n!}{s^{n+1}}$	$s > 0$
$t^\alpha,\ \alpha > -1$	$\frac{\Gamma(\alpha+1)}{s^{\alpha+1}}$	$s > 0$; generalizes to real powers
e^{-at}	$\frac{1}{s+a}$	$s > -a$
te^{-at}	$\frac{1}{(s+a)^2}$	$s > -a$
$t^n e^{-at}$	$\frac{n!}{(s+a)^{n+1}}$	$s > -a$

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Signal $f(t)$	**Laplace Transform** $F(s)$	**Constraints / Notes**
$1 - e^{-at}$	$\frac{a}{s(s+a)}$	$s > 0$
$\sin(\omega t)$	$\frac{\omega}{s^2+\omega^2}$	$s > 0$
$\cos(\omega t)$	$\frac{s}{s^2+\omega^2}$	$s > 0$
$\tan(\omega t)$	$\frac{\omega}{s^2+\omega^2}$	Not standard; see sine form
$e^{-at}\sin(\omega t)$	$\frac{\omega}{(s+a)^2+\omega^2}$	$s > -a$
$e^{-at}\cos(\omega t)$	$\frac{s+a}{(s+a)^2+\omega^2}$	$s > -a$
$t\sin(\omega t)$	$\frac{2\omega s}{(s^2+\omega^2)^2}$	$s > 0$
$t\cos(\omega t)$	$\frac{s^2-\omega^2}{(s^2+\omega^2)^2}$	$s > 0$
$te^{-at}\sin(\omega t)$	$\frac{2\omega(s+a)}{[(s+a)^2+\omega^2]^2}$	$s > -a$
$te^{-at}\cos(\omega t)$	$\frac{(s+a)^2-\omega^2}{[(s+a)^2+\omega^2]^2}$	$s > -a$
$\sinh(at)$	$\frac{a}{s^2-a^2}$	$s > a$
$\cosh(at)$	$\frac{s}{s^2-a^2}$	$s > a$
$e^{-at}\sinh(\omega t)$	$\frac{\omega}{(s+a)^2-\omega^2}$	$s > -a+\omega$
$e^{-at}\cosh(\omega t)$	$\frac{s+a}{(s+a)^2-\omega^2}$	$s > -a+\omega$
$\delta(t-a)$ (shifted impulse)	e^{-as}	$a \geq 0$
$f(t-a)u(t-a)$ (shifted function)	$e^{-as}F(s)$	Time shift property
$\int_0^t f(\tau)\,\mathrm{d}\tau$ (integration)	$\frac{F(s)}{s}$	Integration property

ROC and Stability

The Region of Convergence (ROC) specifies where the Laplace transform is valid. For causal signals (defined for $t \geq 0$), the ROC is always of the form $\Re(s) > \sigma_0$ for some constant σ_0. Physical signals with exponential growth rates determine this boundary.

A.2 Laplace Transform Properties

Table A.2 summarizes the fundamental properties of the Laplace transform. These properties are essential for manipulating transforms algebraically and solving differential equations.

Table A.2: Laplace Transform Properties

Property Name	Time Domain $f(t)$	Frequency Domain $F(s)$
Linearity	$af(t) + bg(t)$ (constants a, b)	$aF(s) + bG(s)$
Time Shift	$f(t-a)u(t-a)$ $(a > 0)$	$e^{-as}F(s)$
Frequency Shift	$e^{-at}f(t)$	$F(s+a)$
Time Scaling	$f(at)$ $(a > 0)$	$\frac{1}{a}F\left(\frac{s}{a}\right)$
Differentiation	$\frac{\mathrm{d}f}{\mathrm{d}t}$	$sF(s) - f(0^+)$
n-th Derivative	$\frac{\mathrm{d}^n f}{\mathrm{d}t^n}$	$s^n F(s) - \sum_{k=0}^{n-1} s^{n-1-k} f^{(k)}(0^+)$
Integration	$\int_0^t f(\tau)\,\mathrm{d}\tau$	$\frac{F(s)}{s}$
Multiplication by t	$tf(t)$	$-\frac{\mathrm{d}F}{\mathrm{d}s}$
Division by t	$\frac{f(t)}{t}$	$\int_s^\infty F(\sigma)\,\mathrm{d}\sigma$
Convolution	$f(t) * g(t)$ $\int_0^t f(\tau)g(t-\tau)\mathrm{d}\tau$	$F(s) \cdot G(s)$
Initial Value	$f(0^+)$	$\lim_{s\to\infty} sF(s)$ (if limit exists)
Final Value	$\lim_{t\to\infty} f(t)$	$\lim_{s\to 0} sF(s)$ (if limit exists and poles in $\Re(s) > 0$)
Parseval's Theorem	$\int_0^\infty \lvert f(t)\rvert^2\,\mathrm{d}t$	$\frac{1}{2\pi j}\int_{\sigma-j\infty}^{\sigma+j\infty} F(s)F(-s)\,\mathrm{d}s$

Important Theorems

Initial Value Theorem: Provides the instantaneous behavior of $f(t)$ at $t = 0$ without finding $f(t)$ explicitly. Used to verify initial conditions in differential equations.
Final Value Theorem: Determines the steady-state behavior of $f(t)$ as $t \to \infty$. Critical for stability analysis and design specification. *Caution:* The FVT only applies if all poles of $sF(s)$ lie in the left-half plane or on the imaginary axis.

A.3 Partial Fraction Expansion

Partial fraction expansion is the primary technique for finding inverse Laplace transforms. This section provides the standard forms and methods for decomposing rational transfer functions.

A.3.1 Distinct Real Poles

For a rational function with distinct real poles:

$$F(s) = \frac{N(s)}{D(s)} = \frac{N(s)}{(s-p_1)(s-p_2)\cdots(s-p_n)} \tag{A.2}$$

the partial fraction expansion is:

$$F(s) = \frac{K_1}{s-p_1} + \frac{K_2}{s-p_2} + \cdots + \frac{K_n}{s-p_n} \tag{A.3}$$

where the residues are computed as:

$$K_i = \lim_{s \to p_i} (s-p_i)F(s) = \frac{N(p_i)}{D'(p_i)} \tag{A.4}$$

The inverse transform is:

$$f(t) = \sum_{i=1}^{n} K_i e^{p_i t} u(t) \tag{A.5}$$

Example: Distinct Real Poles

Consider:

$$F(s) = \frac{10}{(s+1)(s+2)}$$

Expand as:

$$F(s) = \frac{K_1}{s+1} + \frac{K_2}{s+2}$$

Compute residues:

$$\begin{aligned} K_1 &= \lim_{s \to -1} (s+1)\frac{10}{(s+1)(s+2)} = \frac{10}{-1+2} = 10 \\ K_2 &= \lim_{s \to -2} (s+2)\frac{10}{(s+1)(s+2)} = \frac{10}{-2+1} = -10 \end{aligned}$$

Therefore:

$$F(s) = \frac{10}{s+1} - \frac{10}{s+2}$$

Inverse transform:

$$f(t) = (10e^{-t} - 10e^{-2t})u(t)$$

A.3.2 Repeated Real Poles

For repeated poles of multiplicity m:

$$F(s) = \frac{N(s)}{(s-p)^m \cdot D_r(s)} \tag{A.6}$$

the expansion includes terms:

$$\frac{K_1}{s-p} + \frac{K_2}{(s-p)^2} + \cdots + \frac{K_m}{(s-p)^m} + \text{(other poles)} \tag{A.7}$$

The residues for the repeated pole are:

$$K_k = \frac{1}{(m-k)!} \lim_{s \to p} \frac{\mathrm{d}^{m-k}}{\mathrm{d}s^{m-k}} \left[(s-p)^m F(s)\right] \tag{A.8}$$

The inverse transform contribution from these terms is:

$$\mathcal{L}^{-1}\left\{\frac{K}{(s-p)^k}\right\} = \frac{Kt^{k-1}}{(k-1)!} e^{pt}\, u(t) \tag{A.9}$$

Example: Repeated Real Poles

Consider:

$$F(s) = \frac{5}{(s+1)^2}$$

This is a repeated pole of multiplicity $m = 2$ at $p = -1$. The residues are:

$$K_2 = \lim_{s \to -1} (s+1)^2 \cdot \frac{5}{(s+1)^2} = 5$$
$$K_1 = \frac{\mathrm{d}}{\mathrm{d}s}\left[(s+1)^2 \cdot \frac{5}{(s+1)^2}\right]_{s=-1} = 0$$

Expansion:

$$F(s) = \frac{5}{(s+1)^2}$$

Inverse transform:

$$f(t) = 5te^{-t}\, u(t)$$

A.3.3 Complex Conjugate Poles

For complex conjugate poles $p = -\alpha \pm j\omega$:

$$F(s) = \frac{K}{(s - (-\alpha + j\omega))(s - (-\alpha - j\omega))} = \frac{K}{(s+\alpha)^2 + \omega^2} \tag{A.10}$$

This can be written as:

$$F(s) = \frac{As + B}{(s+\alpha)^2 + \omega^2} \tag{A.11}$$

and inverted using:

$$\mathcal{L}^{-1}\left\{\frac{A(s+\alpha) + (B - A\alpha)}{(s+\alpha)^2 + \omega^2}\right\} = Ae^{-\alpha t}\cos(\omega t) + \frac{B - A\alpha}{\omega} e^{-\alpha t}\sin(\omega t) \tag{A.12}$$

Alternatively, express as:

$$F(s) = \frac{K}{(s+\alpha)^2 + \omega^2} \quad \Rightarrow \quad f(t) = \frac{K}{\omega} e^{-\alpha t}\sin(\omega t)\, u(t) \tag{A.13}$$

Example: Complex Conjugate Poles

Consider:

$$F(s) = \frac{10}{s^2 + 2s + 5}$$

Factor the denominator:

$$s^2 + 2s + 5 = (s+1)^2 + 4 = (s+1)^2 + 2^2$$

Identify $\alpha = 1$, $\omega = 2$, and rewrite:

$$F(s) = \frac{10}{(s+1)^2 + 4}$$

Using the standard pair with $K = 10$ and $\omega = 2$:

$$f(t) = \frac{10}{2} e^{-t} \sin(2t)\, u(t) = 5e^{-t} \sin(2t)\, u(t)$$

Strategy for Partial Fractions

1. If degree of numerator $\geq$ degree of denominator, perform polynomial long division first.
2. Factor the denominator completely (real and complex factors).
3. Set up the partial fraction form based on pole types:
   - Distinct real pole p: contribute $\dfrac{K}{s-p}$
   - Repeated real pole p (multiplicity m): contribute $\dfrac{K_1}{s-p} + \dfrac{K_2}{(s-p)^2} + \cdots + \dfrac{K_m}{(s-p)^m}$
   - Complex conjugate pole pair: contribute $\dfrac{As+B}{(s+\alpha)^2 + \omega^2}$
4. Solve for coefficients using cover-up method, residue formula, or algebraic matching.
5. Use Table A.1 to invert each term.

A.4 Common Transfer Function Forms

Transfer functions are the primary tool for classical control analysis. This section presents standard forms frequently encountered in control system design.

A.4.1 First-Order System

The standard first-order transfer function is:

$$G(s) = \frac{K}{s+a} = \frac{K/a}{s/a + 1} = \frac{K}{\tau s + 1} \tag{A.14}$$

where:

- K is the DC gain (or steady-state gain)

- a is the pole location (for stability, $a > 0$)
- $\tau = 1/a$ is the time constant

Key characteristics:

- **Step response:** $y(t) = K(1 - e^{-t/\tau})u(t)$, settling to $y(\infty) = K$
- **Time to 63%:** $t_{63\%} = \tau$
- **DC gain:** $G(0) = K$
- **Frequency bandwidth:** $\omega_{BW} = 1/\tau = a$

A.4.2 Second-Order System

The standard second-order transfer function is:

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \tag{A.15}$$

where:

- ω_n is the natural frequency (rad/s)
- ζ is the damping ratio (unitless)

Pole locations: $s = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$

Response Characteristics by Damping

1. **Underdamped** $(0 < \zeta < 1)$: Oscillatory response with exponential decay

$$y(t) = 1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}} \sin\left(\omega_n\sqrt{1-\zeta^2}\, t + \arccos(\zeta)\right) \tag{A.16}$$

   - **Overshoot:** $M_p = e^{-\zeta\pi/\sqrt{1-\zeta^2}} \times 100\%$
   - **Peak time:** $t_p = \dfrac{\pi}{\omega_n\sqrt{1-\zeta^2}}$
   - **Settling time (2%):** $t_s \approx \dfrac{4}{\zeta\omega_n}$

2. **Critically damped** $(\zeta = 1)$: Fastest response without overshoot

$$y(t) = 1 - (1 + \omega_n t)e^{-\omega_n t} \tag{A.17}$$

3. **Overdamped** $(\zeta > 1)$: Slow, no-overshoot response

$$y(t) = 1 + \frac{\zeta e^{-\omega_n(\zeta-\sqrt{\zeta^2-1})t}}{2\sqrt{\zeta^2-1}(\zeta-\sqrt{\zeta^2-1})} - \frac{\zeta e^{-\omega_n(\zeta+\sqrt{\zeta^2-1})t}}{2\sqrt{\zeta^2-1}(\zeta+\sqrt{\zeta^2-1})} \tag{A.18}$$

4. **Undamped** $(\zeta = 0)$: Perpetual oscillation

$$y(t) = 1 - \cos(\omega_n t) \tag{A.19}$$

Design Trade-offs

For step input to a second-order system, the damping ratio ζ controls the trade-off between:

- **Overshoot and settling time:** As ζ decreases, overshoot increases but settling is faster
- **Stability margin:** Higher ζ provides greater robustness to parameter variations
- **Practical choice:** $\zeta \in [0.5, 0.8]$ balances performance and robustness

A.4.3 PID Controller

The PID (Proportional-Integral-Derivative) controller is the most common feedback controller:

$$G_c(s) = K_p + \frac{K_i}{s} + K_d s = K_p\left(1 + \frac{1}{T_i s} + T_d s\right) \tag{A.20}$$

where:

- K_p is the proportional gain
- K_i is the integral gain (or $T_i = K_p/K_i$ is the integral time constant)
- K_d is the derivative gain (or $T_d = K_d/K_p$ is the derivative time constant)

Rewritten in standard form:

$$G_c(s) = K_p \frac{(T_i T_d s^2 + T_i s + 1)}{T_i s} \tag{A.21}$$

Component Roles

1. **Proportional (P):** Immediate response proportional to error
   - Reduces rise time and steady-state error
   - May introduce steady-state error for step inputs
   - Can destabilize if gain too high
2. **Integral (I):** Eliminates steady-state error
   - Integrates error over time to drive steady-state error to zero
   - Slower response, increased lag
   - Can cause overshoot and oscillation
3. **Derivative (D):** Anticipatory action, improves transient response
   - Responds to rate of change of error
   - Improves damping and reduces overshoot
   - Amplifies high-frequency noise
   - Typically not used alone; combined with P and I

Practical PID Forms

Ideal PID (series form):

$$G_c(s) = K_c \left(1 + \frac{1}{T_i s}\right)(1 + T_d s) \tag{A.22}$$

Parallel PID:

$$G_c(s) = K_p + \frac{K_i}{s} + K_d s \tag{A.23}$$

Filtered derivative (with derivative filter):

$$G_c(s) = K_p + \frac{K_i}{s} + \frac{K_d s}{s/N + 1} \tag{A.24}$$

where N (typically $N = 5$ to 20) filters high-frequency noise.

A.4.4 Lead Compensator

A lead compensator (lead-lag network) improves transient response and stability margins:

$$G_c(s) = K_c \frac{1 + sT_d}{1 + sT_d/\alpha} = K_c \frac{\tau s + 1}{\alpha \tau s + 1} \tag{A.25}$$

where:

- K_c is the steady-state gain
- T_d (or $\tau = T_d$) is the lead time constant
- $\alpha < 1$ is the separation factor
- Maximum phase lead: $\phi_{\max} = \sin^{-1}\left(\frac{1-\alpha}{1+\alpha}\right)$
- Frequency at maximum phase: $\omega_m = \dfrac{1}{T_d\sqrt{\alpha}}$

Characteristics:

- Increases bandwidth and improves transient response
- Adds phase lead in mid-frequency range
- Increases high-frequency gain (noise amplification)
- Used to improve stability margins (gain and phase margins)

A.4.5 Lag Compensator

A lag compensator improves steady-state accuracy:

$$G_c(s) = K_c \frac{1 + sT_i}{\beta + sT_i} = K_c \frac{\tau s + 1}{\beta \tau s + 1} \tag{A.26}$$

where:

- K_c is the high-frequency gain

- T_i (or $\tau = T_i$) is the lag time constant
- $\beta > 1$ is the separation factor
- DC gain increase: βK_c (improves error)
- Phase lag is present but minimal in control bandwidth

Characteristics:

- Increases low-frequency gain to reduce steady-state error
- Minimal effect on transient response if designed properly
- Reduces high-frequency noise amplification
- Slows system response slightly

Compensator Design Guidelines

- **Lead compensator:** Use when transient response (damping, settling time) needs improvement
- **Lag compensator:** Use when steady-state error must be reduced without major transient changes
- **Lead-lag compensator:** Combines both; lead for transient improvement, lag for steady-state
- **PID:** Versatile choice; proportional and derivative for transient, integral for steady-state

A.5 Useful Identities and Manipulations

This section collects useful identities and algebraic manipulations frequently needed in control analysis.

A.5.1 Completing the Square

For quadratic denominators, completing the square is essential for identifying natural frequency and damping:

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = (s + \zeta\omega_n)^2 + \omega_n^2(1 - \zeta^2) \tag{A.27}$$

This reveals the pole locations: $s = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$

General case:

$$s^2 + as + b = \left(s + \frac{a}{2}\right)^2 + \left(b - \frac{a^2}{4}\right) \tag{A.28}$$

A.5.2 Numerator/Denominator Relationships

For rational functions:

DC gain:

$$\lim_{s\to 0} F(s) = \frac{\text{numerator at } s=0}{\text{denominator at } s=0} \tag{A.29}$$

High-frequency behavior:

$$\lim_{s\to\infty} s^n F(s) = \begin{cases} \text{nonzero constant} & \text{if numerator degree} = n + \text{denominator degree} \\ 0 & \text{if numerator degree} < n + \text{denominator degree} \end{cases} \tag{A.30}$$

A.5.3 Convolution Simplifications

For common control system cascades:

$$\frac{1}{s}\cdot\frac{1}{s+a} = \frac{1}{a}\left(\frac{1}{s} - \frac{1}{s+a}\right) \tag{A.31}$$

$$\frac{1}{(s+a)(s+b)} = \frac{1}{b-a}\left(\frac{1}{s+a} - \frac{1}{s+b}\right) \quad (a \neq b) \tag{A.32}$$

$$\frac{s}{(s+a)(s+b)} = \frac{a}{a-b}\cdot\frac{1}{s+a} + \frac{b}{b-a}\cdot\frac{1}{s+b} \quad (a \neq b) \tag{A.33}$$

A.6 Quick Reference Checklist

When Working with Laplace Transforms

To transform a differential equation:

1. Replace $\frac{d}{dt}$ with s (assuming zero initial conditions)
2. Replace $\int$ with $\frac{1}{s}$
3. Solve algebraically for $Y(s)$

To find $f(t)$ from $F(s)$:

1. Check if $F(s)$ appears in Table A.1; if so, read off $f(t)$ directly
2. If not, use partial fraction expansion to decompose into standard forms
3. Invert each piece using table
4. For repeated poles, use $t^{n-1}e^{pt}$ terms
5. For complex poles, use damped sinusoid forms

To use initial/final value theorems:

1. **Initial value:** $f(0^+) = \lim_{s\to\infty} sF(s)$ (if limit exists)
2. **Final value:** $f(\infty) = \lim_{s\to 0} sF(s)$ (if poles in left-half plane)

For transfer functions and stability:

1. Poles in left-half plane: stable
2. Poles on imaginary axis: marginally stable
3. Poles in right-half plane: unstable
4. Repeated pole on imaginary axis: unstable

Appendix Notes

The Laplace transform tables and properties presented in this appendix are standard results found in most advanced mathematics and control systems texts. Key references include:

- Kamen, E. W., & Heck, B. S. (2014). *Fundamentals of Signals and Systems Using MATLAB*. Pearson Education.
- Oppenheim, A. V., Willsky, A. S., & Nawab, S. H. (1996). *Signals and Systems* (2nd ed.). Prentice Hall.
- Franklin, G. F., Powell, J. D., & Emami-Naeini, A. (2014). *Feedback Control of Dynamic Systems* (7th ed.). Pearson Education.

This appendix is designed to be a comprehensive reference for studying and applying Laplace transforms in control systems analysis and design. Proficiency with these tables and properties is fundamental to classical control theory.